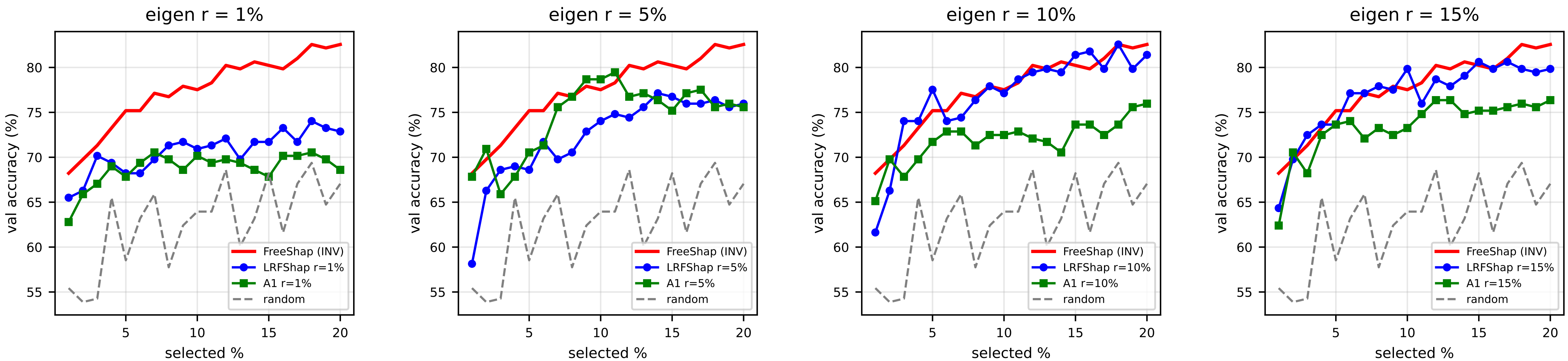
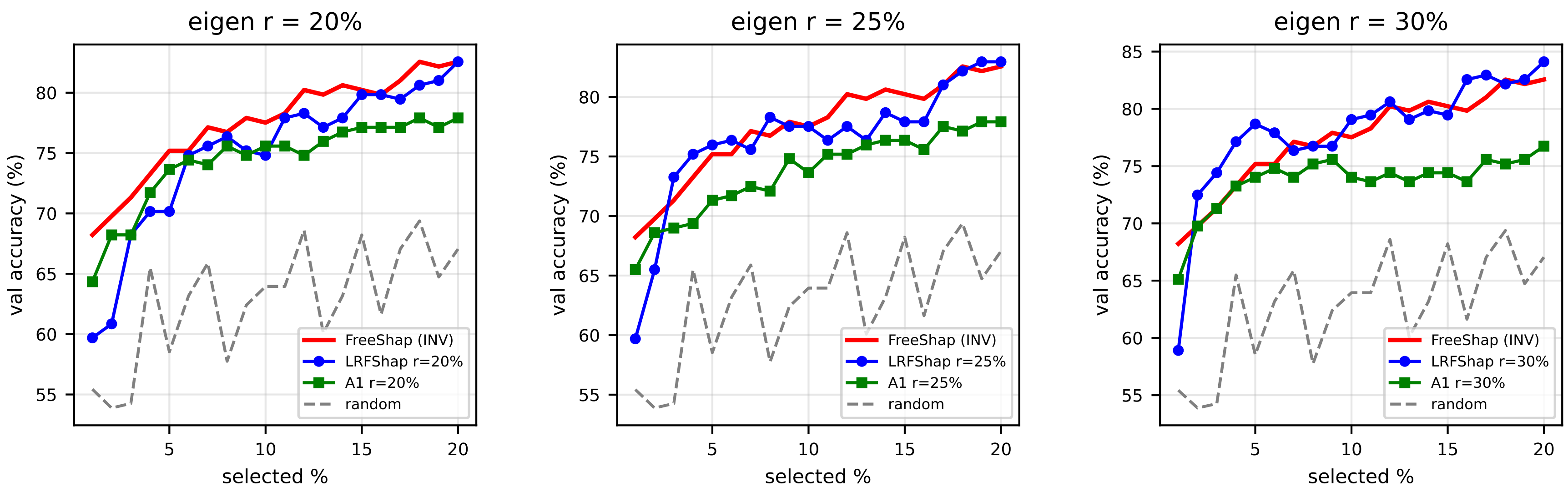
MRPC forced 50/50 (n=2300, valbal=258) | n_train=2300, valbal=258



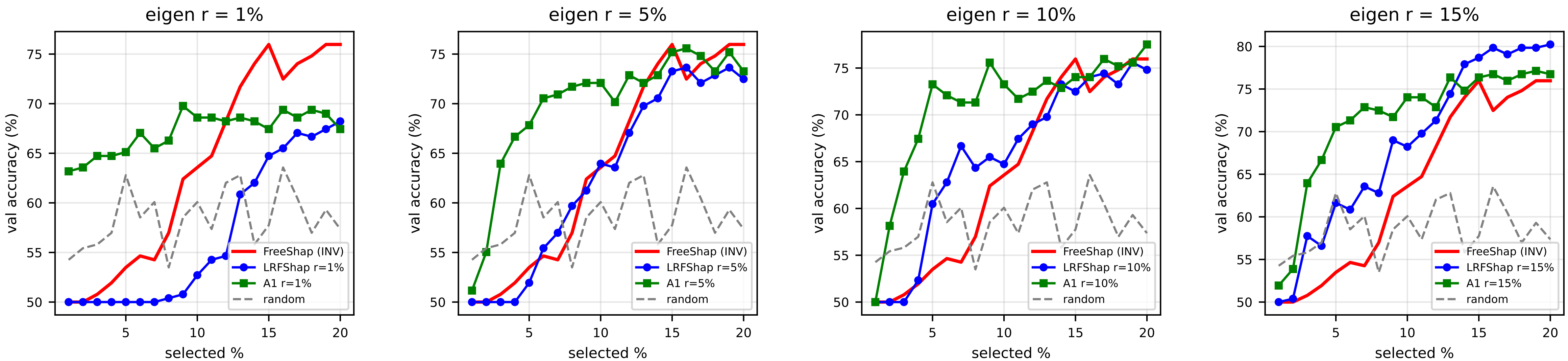


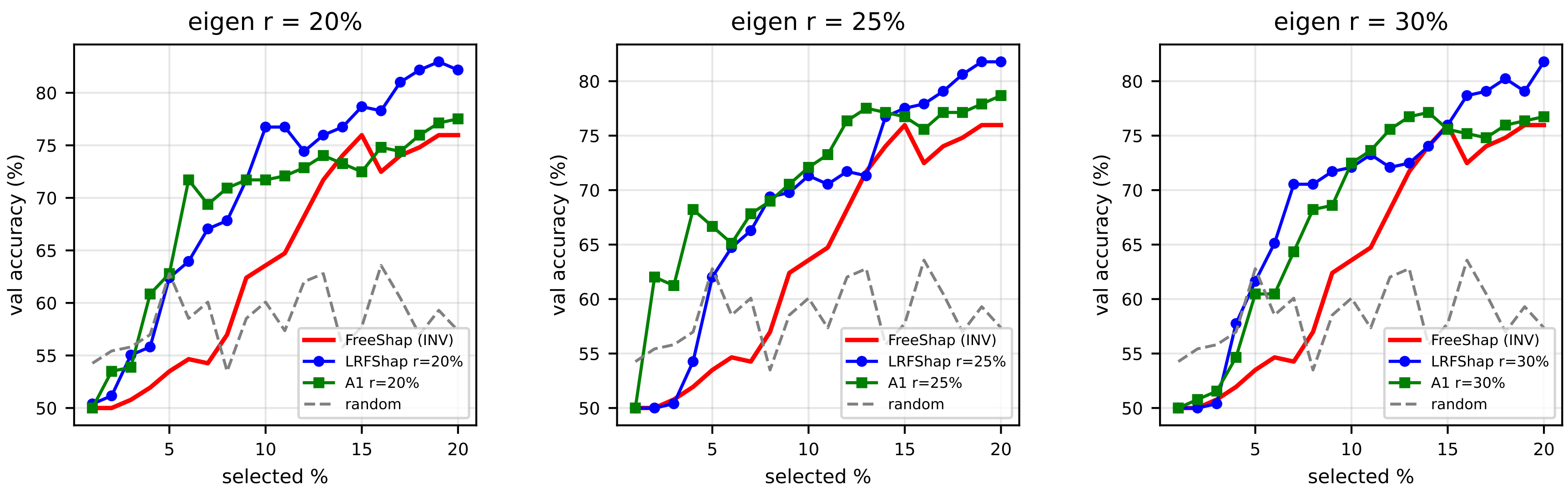

```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	68.22	69.77	71.32	73.26	75.19	77.52	82.56
r=1%	X	X	X	X	X	X	X
r=5%	O	O	X	X	O	O	X
r=10%	O	O	X	X	X	X	X
r=15%	X	O	X	X	-	X	X
r=20%	O	O	-	O	O	O	X
r=25%	O	O	X	X	X	X	X
r=30%	O	X	X	X	X	X	X

MRPC forced 70/30 label1-majority (n=2300, valbal=258) | n_train=2300, valbal=258



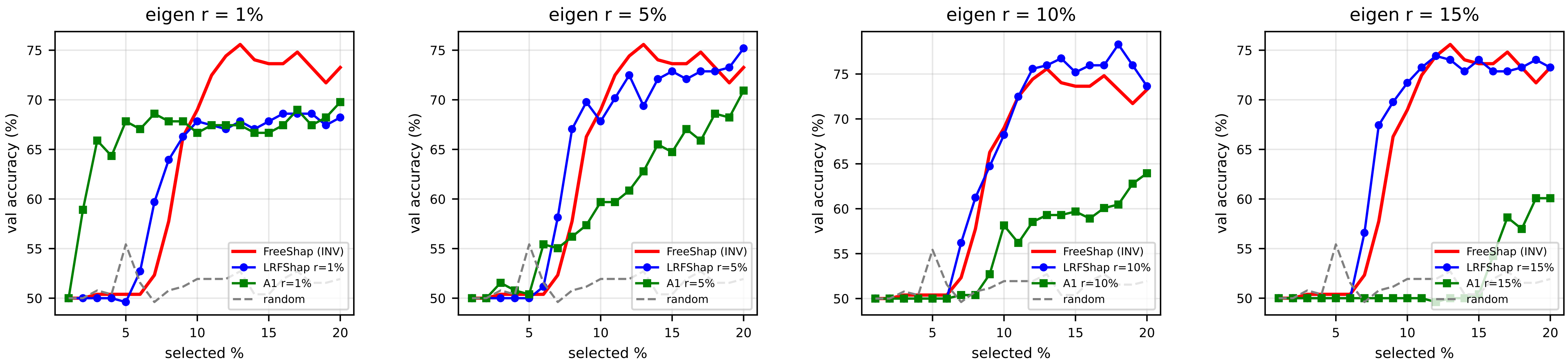


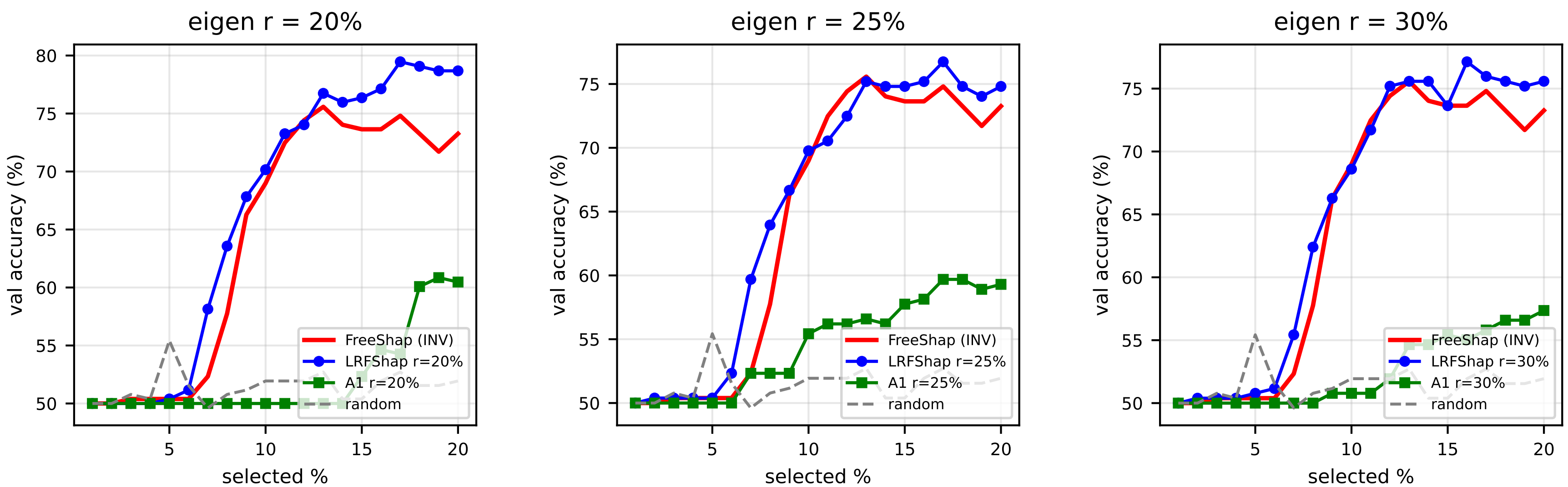

```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	**sel2%**	**sel3%**	**sel4%**	**sel5%**	**sel10%**	**sel20%**
INV (acc)	50.00 [!]	50.00 [!]	50.78 [!]	51.94 [!]	53.49 [!]	63.57	75.97
r=1%	O†	O†	O†	O†	O†	O†	X
r=5%	!†	!†	O†	O†	O†	O	O
r=10%	!†	O†	O†	O†	O†	O	O
r=15%	!†	!†	O	O†	O†	O	X
r=20%	!†	!†	!†	O†	!†	X	X
r=25%	!†	O†	O†	O†	O†	O	X
r=30%	!†	!†	!†	!	!†	O	X

MRPC forced 90/10 label1-majority (n=2300, valbal=258) | n_train=2300, valbal=258



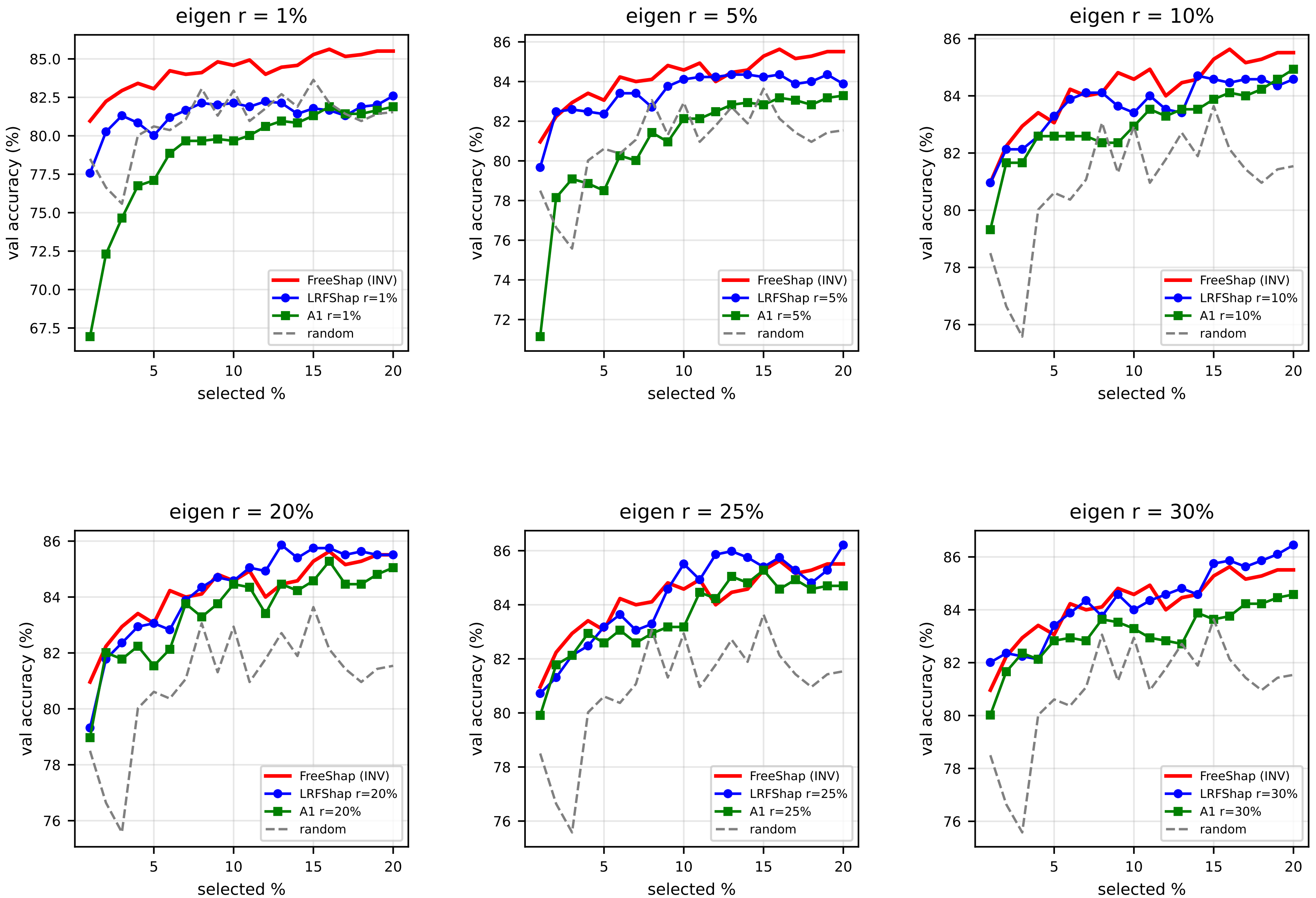



```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	50.00 [!]	50.00 [!]	50.39 [!]	50.39 [!]	50.39 [!]	68.99	73.26
r=1%	!†	O†	O†	O†	O†	X	O
r=5%	!†	!†	O†	O†	!†	X	X
r=10%	!†	!†	!†	!†	!†	X	X
r=15%	!†	!†	!†	!†	!†	!	X
r=20%	!†	!†	!†	!†	!†	!	X
r=25%	!†	!	!†	!†	!†	X	X
r=30%	!†	!	!†	!†	!†	!	X

SST-2 forced 50/50 (n=5000, valbal=856) | n_train=5000, valbal=856

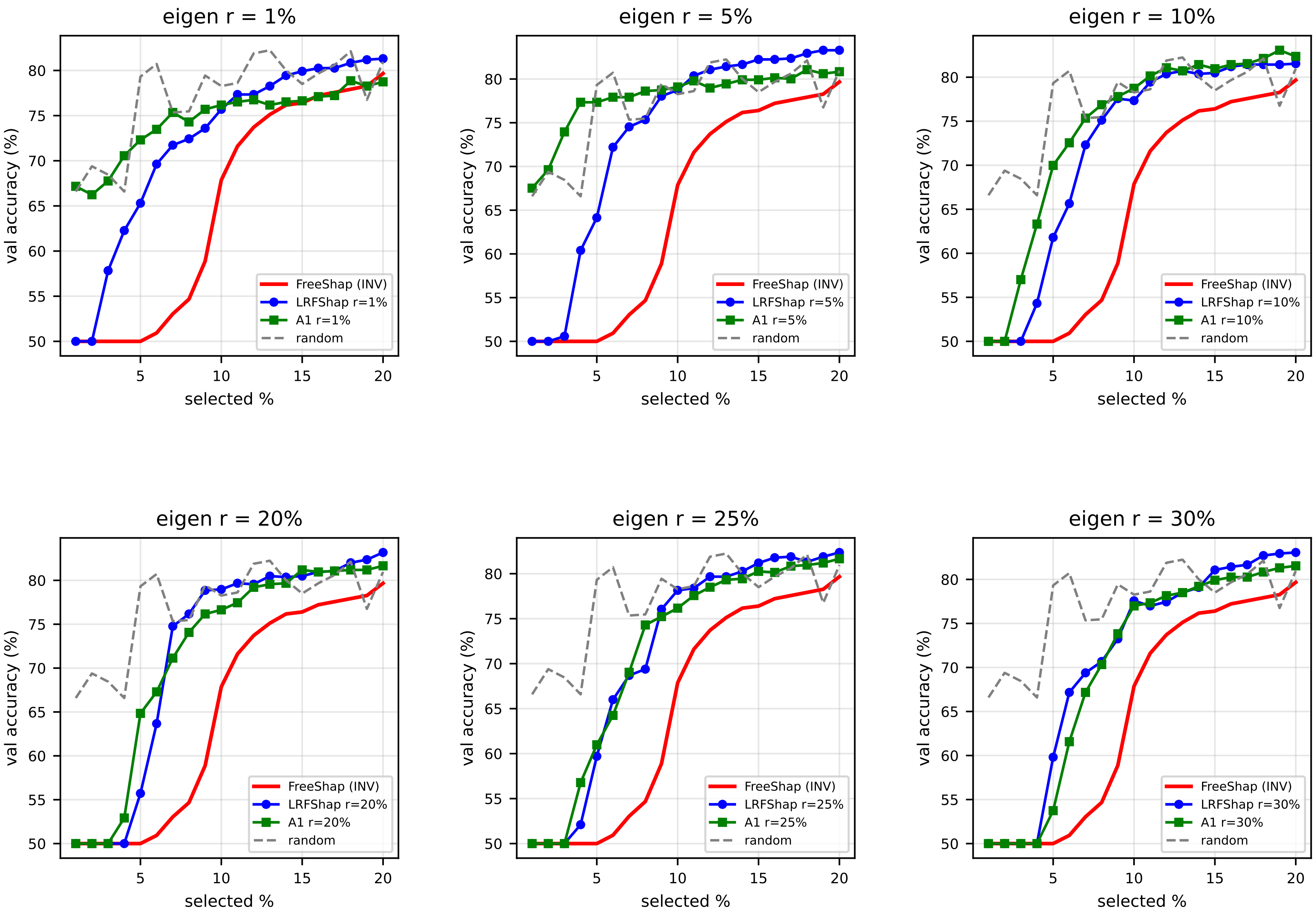



```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	**sel2%**	**sel3%**	**sel4%**	**sel5%**	**sel10%**	**sel20%**
INV (acc)	80.96	82.24	82.94	83.41	83.06	84.58	85.51
r=1%	!†	!	!	!	!†	!†	X
r=5%	!	X	X	!	!	!	X
r=10%	X	X	X	-	X	!	O
r=15%	X	O	O	O	X	X	X
r=20%	X	O	X	X	X	X	X
r=25%	X	O	-	O	X	X	X
r=30%	X	X	O	-	X	X	X

SST-2 forced 70/30 label1-majority (n=5000, valbal=856) | n_train=5000, valbal=856

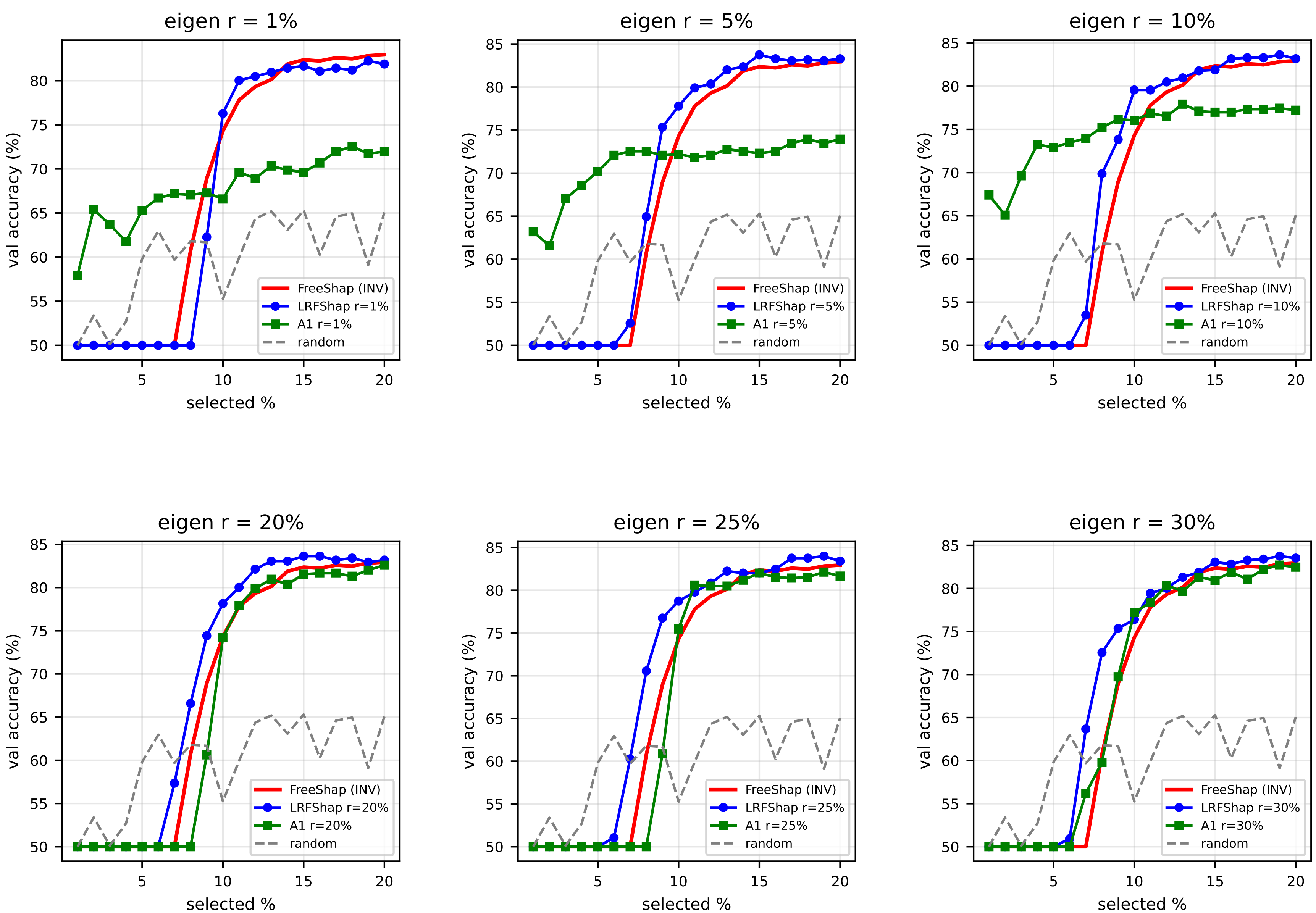



```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	67.87 [!]	79.67 [!]
r=1%	O†	!†	!†	O†	!†	!†	!
r=5%	O†	O†	O†	O†	!†	O	!
r=10%	!†	!†	!†	!†	!†	O†	O
r=15%	!†	!†	!†	!†	!†	!†	X
r=20%	!†	!†	!†	!†	!†	!	X
r=25%	!†	!†	!†	!†	!†	!†	X
r=30%	!†	!†	!†	!†	!†	!†	X

SST-2 forced 90/10 label1-majority (n=5000, valbal=856) | n_train=5000, valbal=856

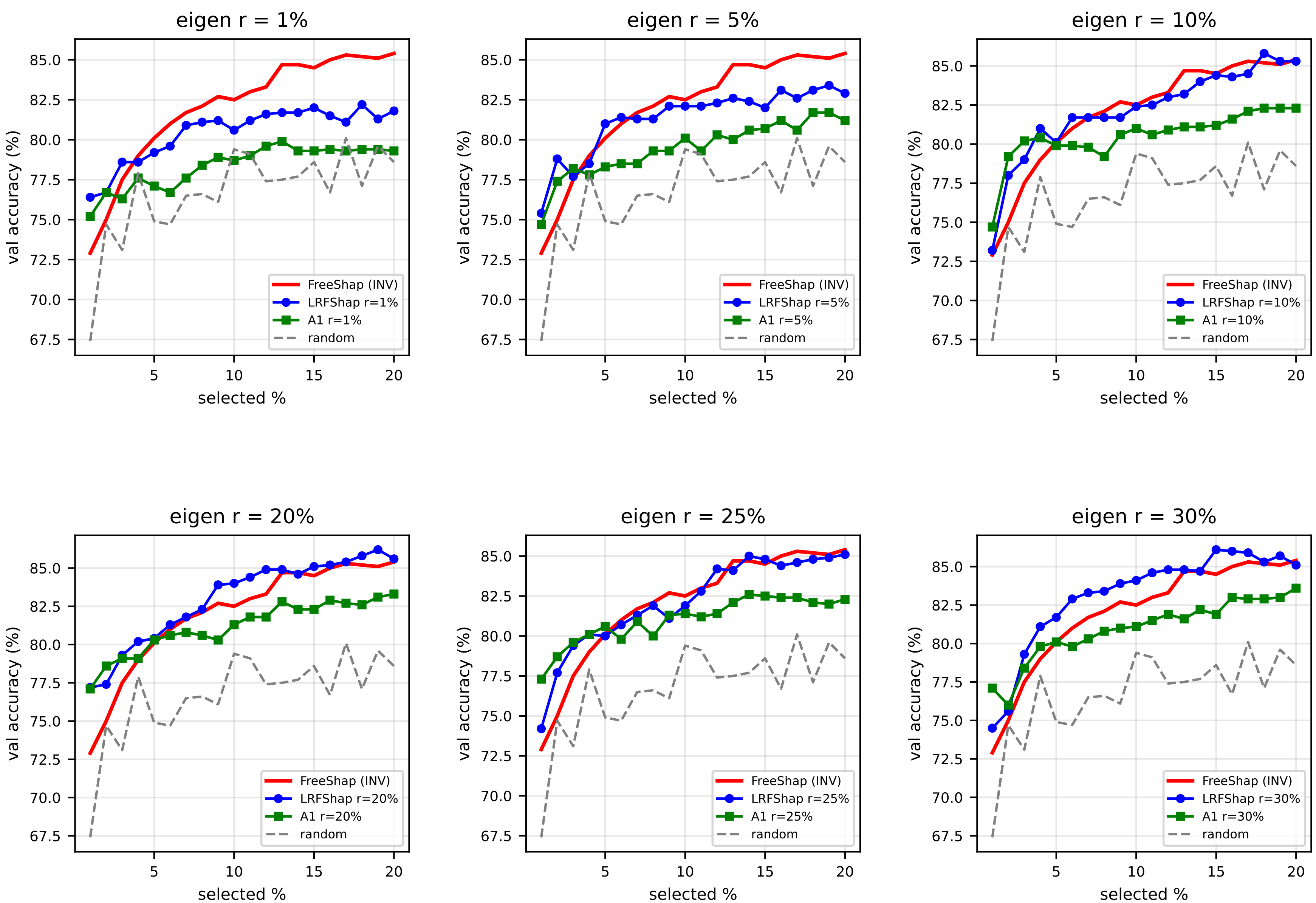



```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	74.30	82.94
r=1%	O†	O†	O†	O†	O†	X	X
r=5%	O†	O†	O†	O†	O†	X	X
r=10%	O†	O†	O†	O†	O†	X	X
r=15%	!†	!†	!†	!†	!†	X	X
r=20%	!†	!†	!†	!†	!†	X	X
r=25%	!†	!†	!†	!†	!†	X	X
r=30%	!†	!†	!†	!†	!†	O	X

MR forced 50/50 (n=4500, valbal=1000) | n_train=4500, valbal=1000

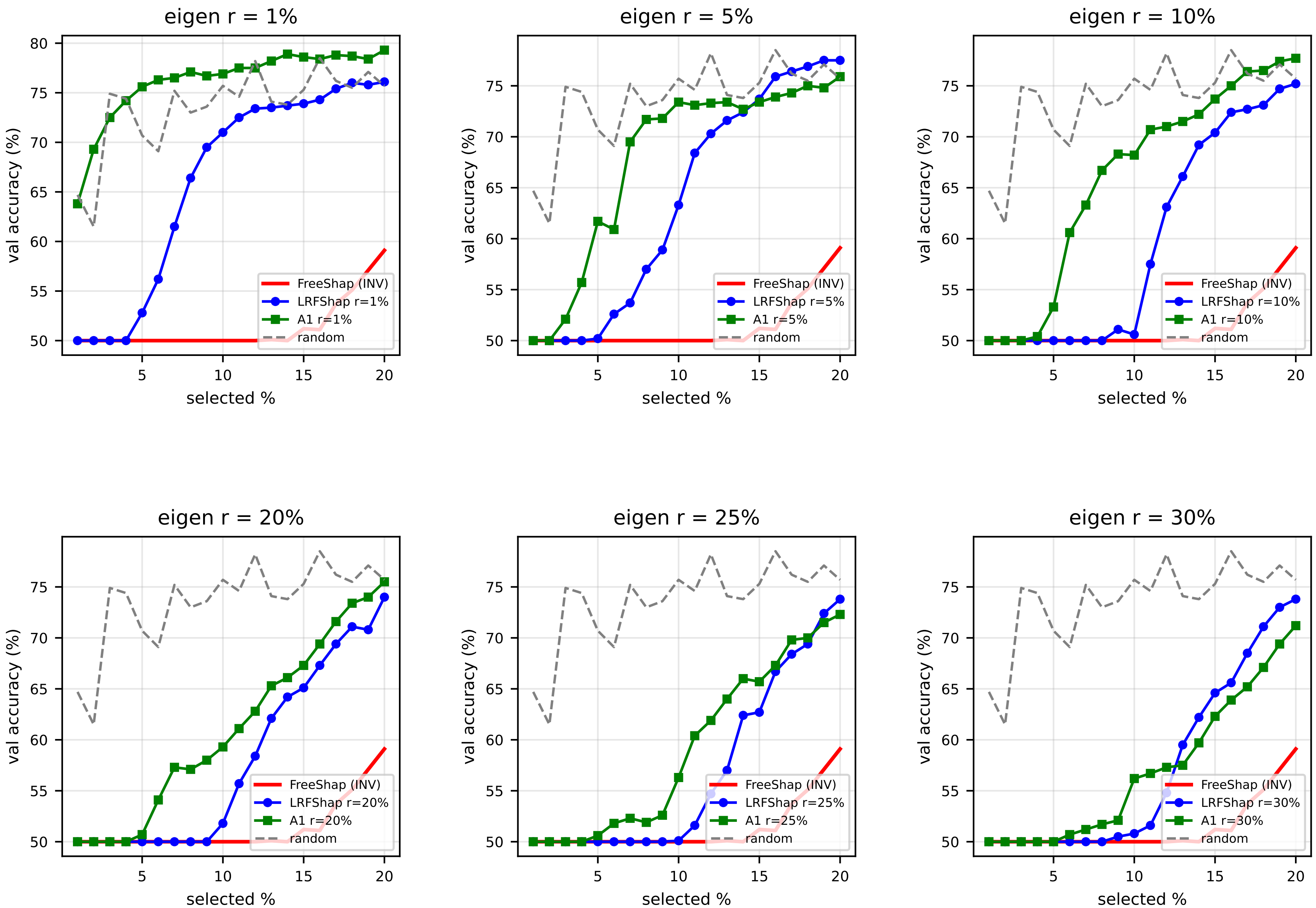



```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	**sel2%**	**sel3%**	**sel4%**	**sel5%**	**sel10%**	**sel20%**
INV (acc)	72.90	75.00	77.50	79.00	80.10	82.50	85.40
r=1%	X	-	X	!	X	!	X
r=5%	X	X	O	!	X	X	X
r=10%	O	O	O	X	X	X	X
r=15%	X	O	O	X	X	X	X
r=20%	X	O	X	X	X	X	X
r=25%	O	O	O	-	O	X	X
r=30%	O	O	X	X	X	X	X

MR forced 70/30 label1-majority (n=4500, valbal=1000) | n_train=4500, valbal=1000

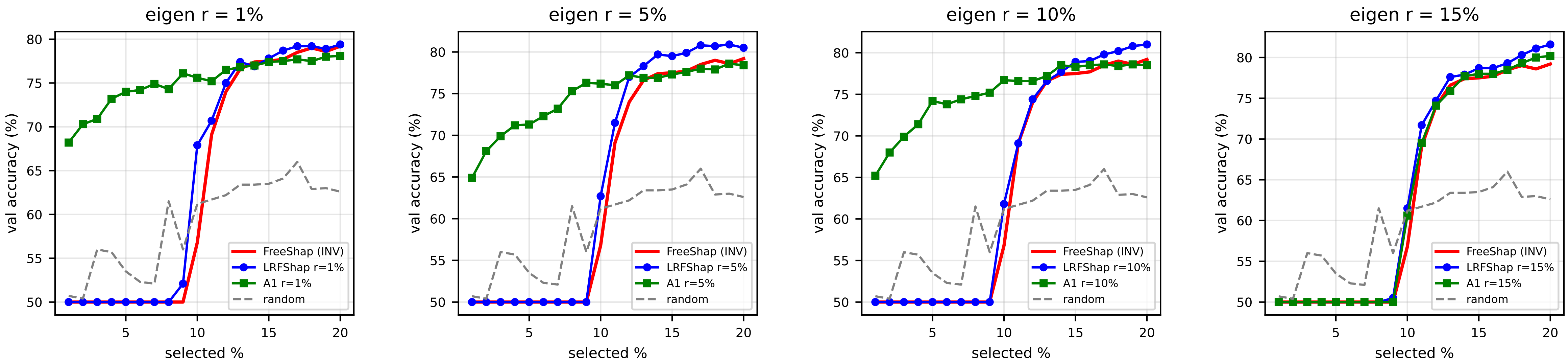


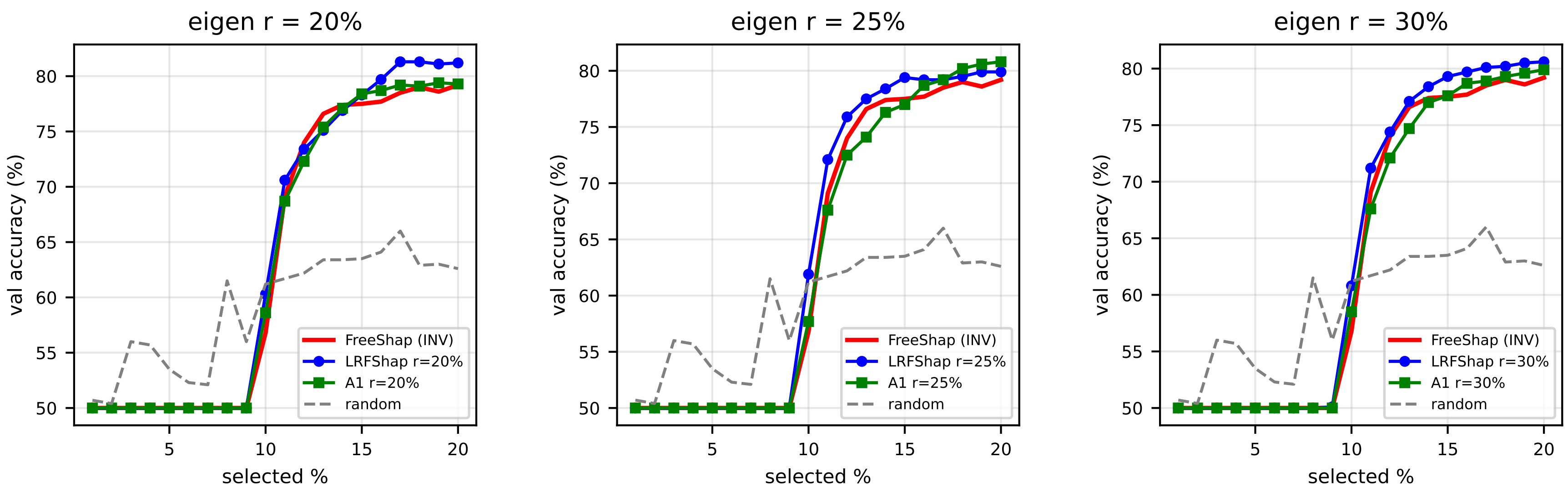

```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	59.10 [!]
r=1%	!†	O†	!†	!†	O†	O†	O
r=5%	!†	!†	!†	!†	!†	!†	X
r=10%	!†	!†	!†	!†	!†	!†	O†
r=15%	!†	!†	!†	!†	!†	!†	!†
r=20%	!†	!†	!†	!†	!†	!†	!†
r=25%	!†	!†	!†	!†	!†	!†	!†
r=30%	!†	!†	!†	!†	!†	!†	!†

MR forced 90/10 label1-majority (n=4500, valbal=1000) | n_train=4500, valbal=1000



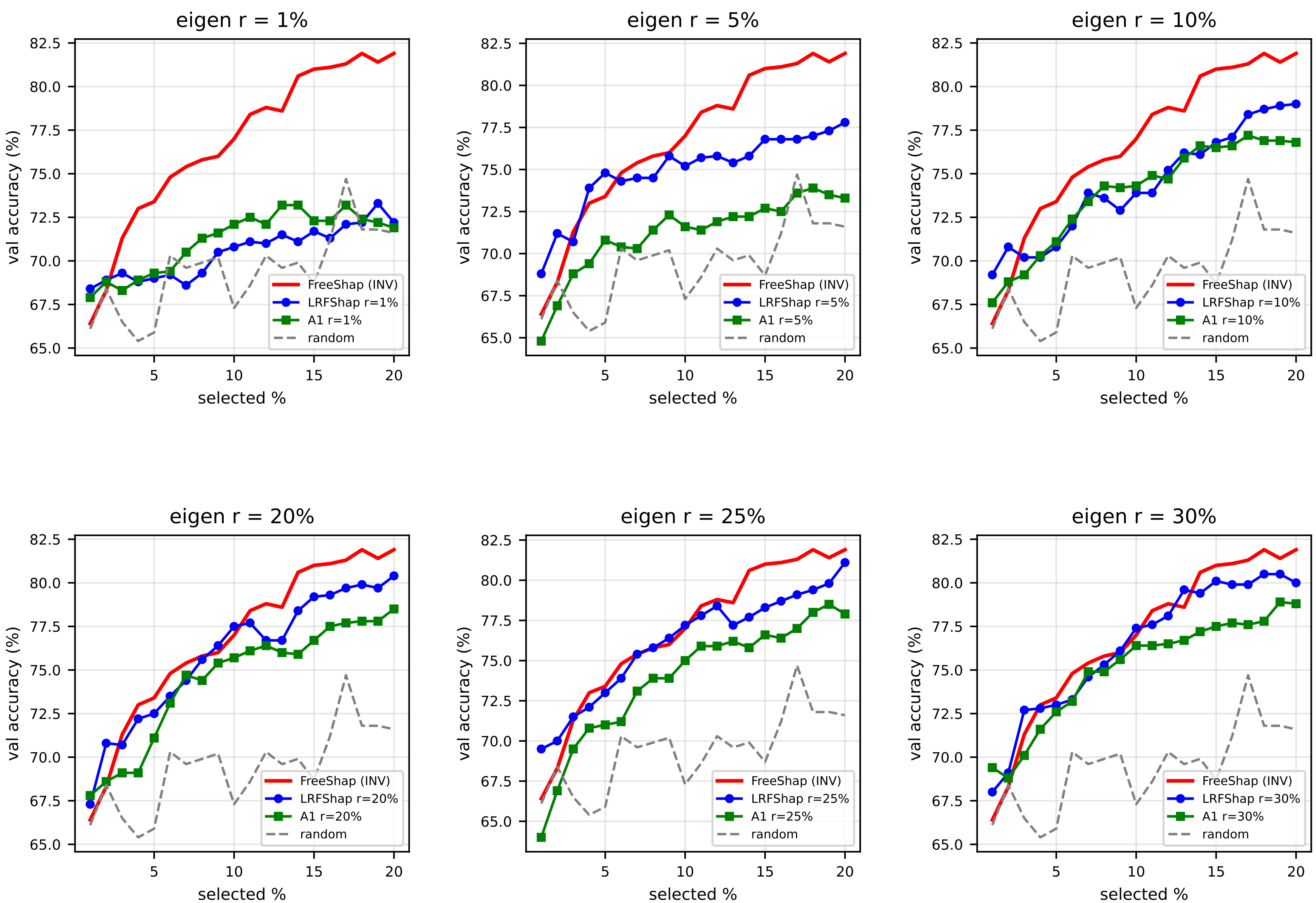



```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	**sel2%**	**sel3%**	**sel4%**	**sel5%**	**sel10%**	**sel20%**
INV (acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	56.80 [!]	79.20
r=1%	O†	O†	O†	O†	O†	O	X
r=5%	O†	O†	O†	O†	O†	O	X
r=10%	O†	O†	O†	O†	O†	O	X
r=15%	!†	!†	!†	!†	!†	!	X
r=20%	!†	!†	!†	!†	!†	!†	X
r=25%	!†	!†	!†	!†	!†	!	O
r=30%	!†	!†	!†	!†	!†	!†	X

QQP forced 50/50 (n=5000, valbal=1000) | n_train=5000, valbal=1000

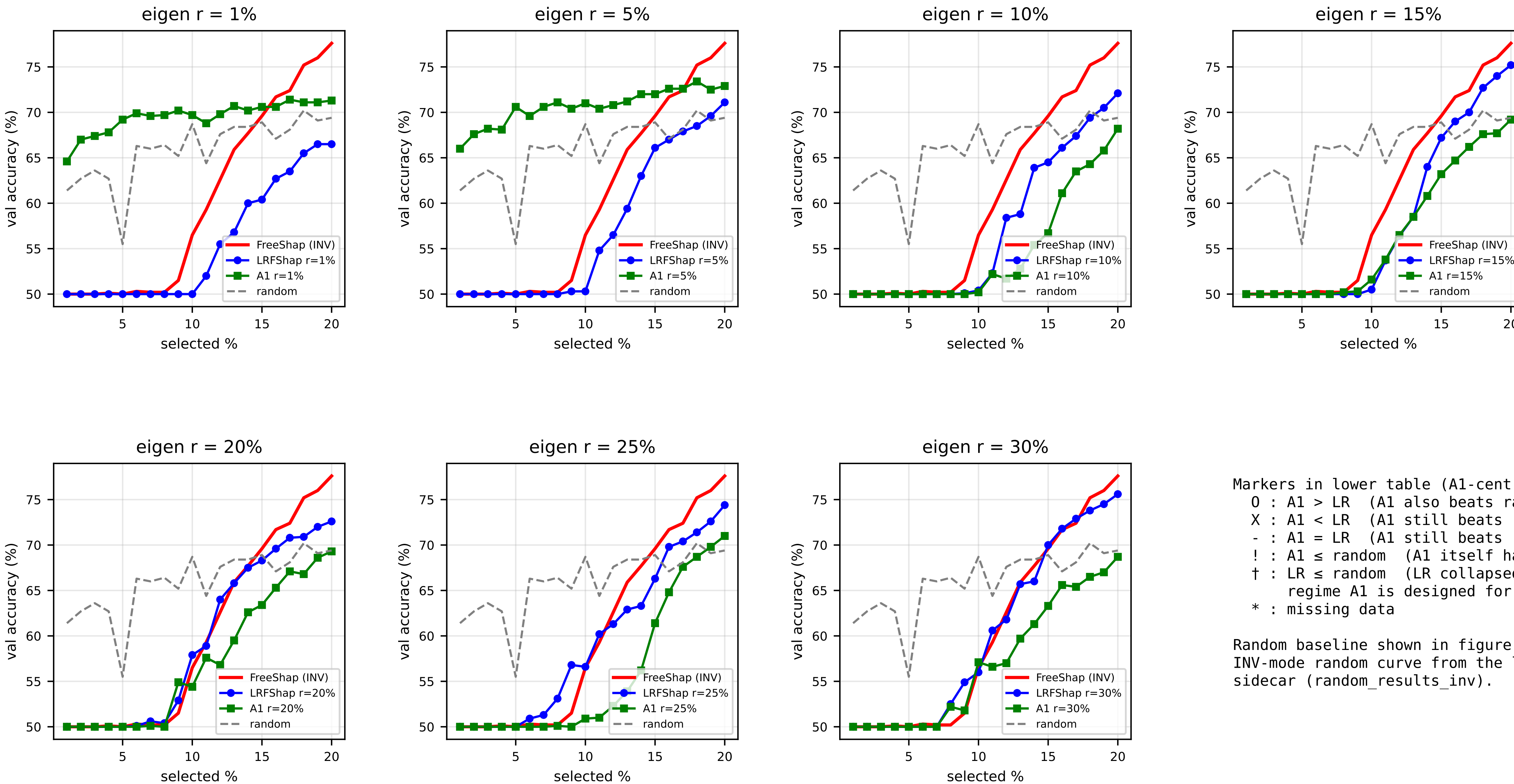



```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	66.40	68.30 [!]	71.30	73.00	73.40	77.00	81.90
r=1%	X	X	X	O	O	O	X
r=5%	!	!	X	X	X	X	X
r=10%	X	X	X	O	O	O	X
r=15%	X	!	X	X	X	X	X
r=20%	O	X	X	X	X	X	X
r=25%	!	!	X	X	X	X	X
r=30%	O	X	X	X	X	X	X

QQP forced 70/30 label0-majority (n=5000, valbal=1000) | n_train=5000, valbal=1000

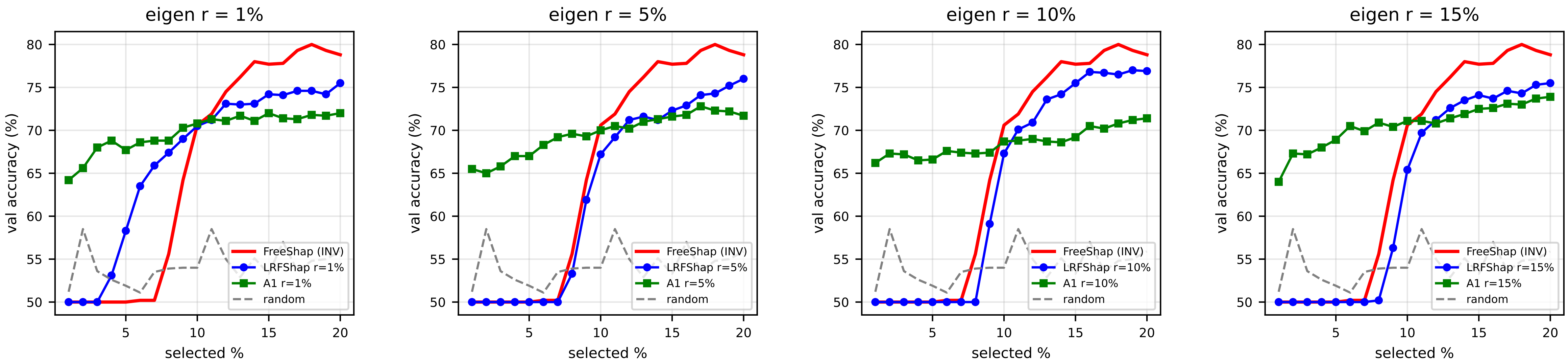


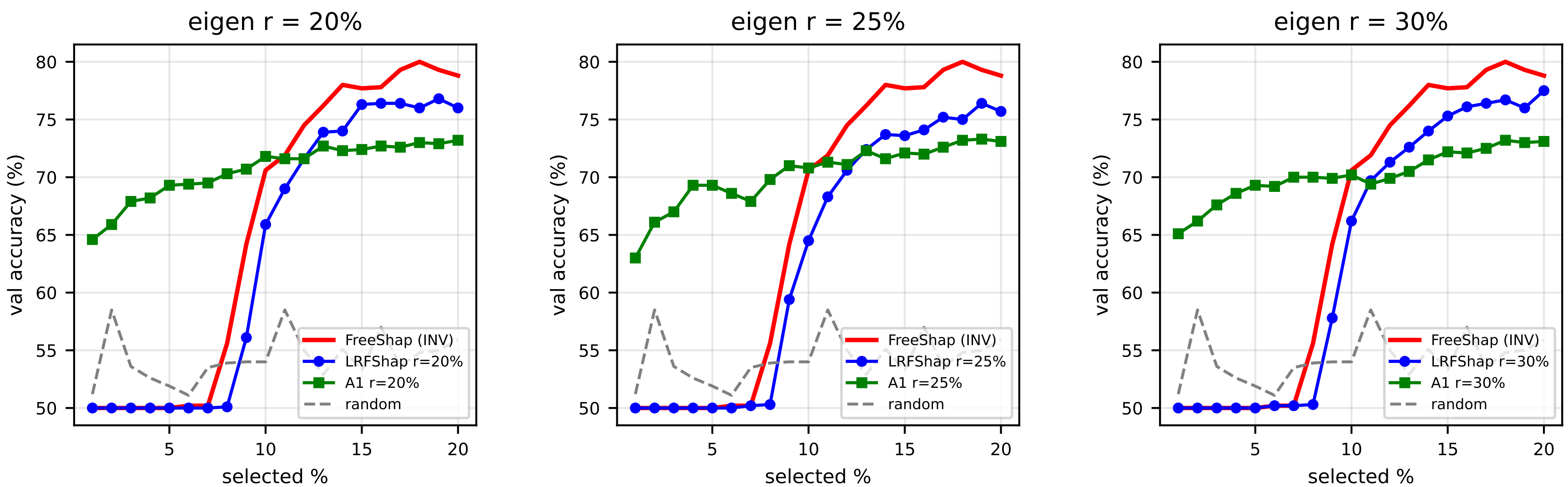

```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	**sel2%**	**sel3%**	**sel4%**	**sel5%**	**sel10%**	**sel20%**
INV (acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.10 [!]	50.00 [!]	56.50 [!]	77.60
r=1%	O†	O†	O†	O†	O†	O†	O†
r=5%	O†	O†	O†	O†	O†	O†	O
r=10%	!†	!†	!†	!†	!†	!†	!
r=15%	!†	!†	!†	!†	!†	!†	!
r=20%	!†	!†	!†	!†	!†	!†	!
r=25%	!†	!†	!†	!†	!†	!†	X
r=30%	!†	!†	!†	!†	!†	!†	!

QQP forced 90/10 label0-majority (n=5000, valbal=1000) | n_train=5000, valbal=1000



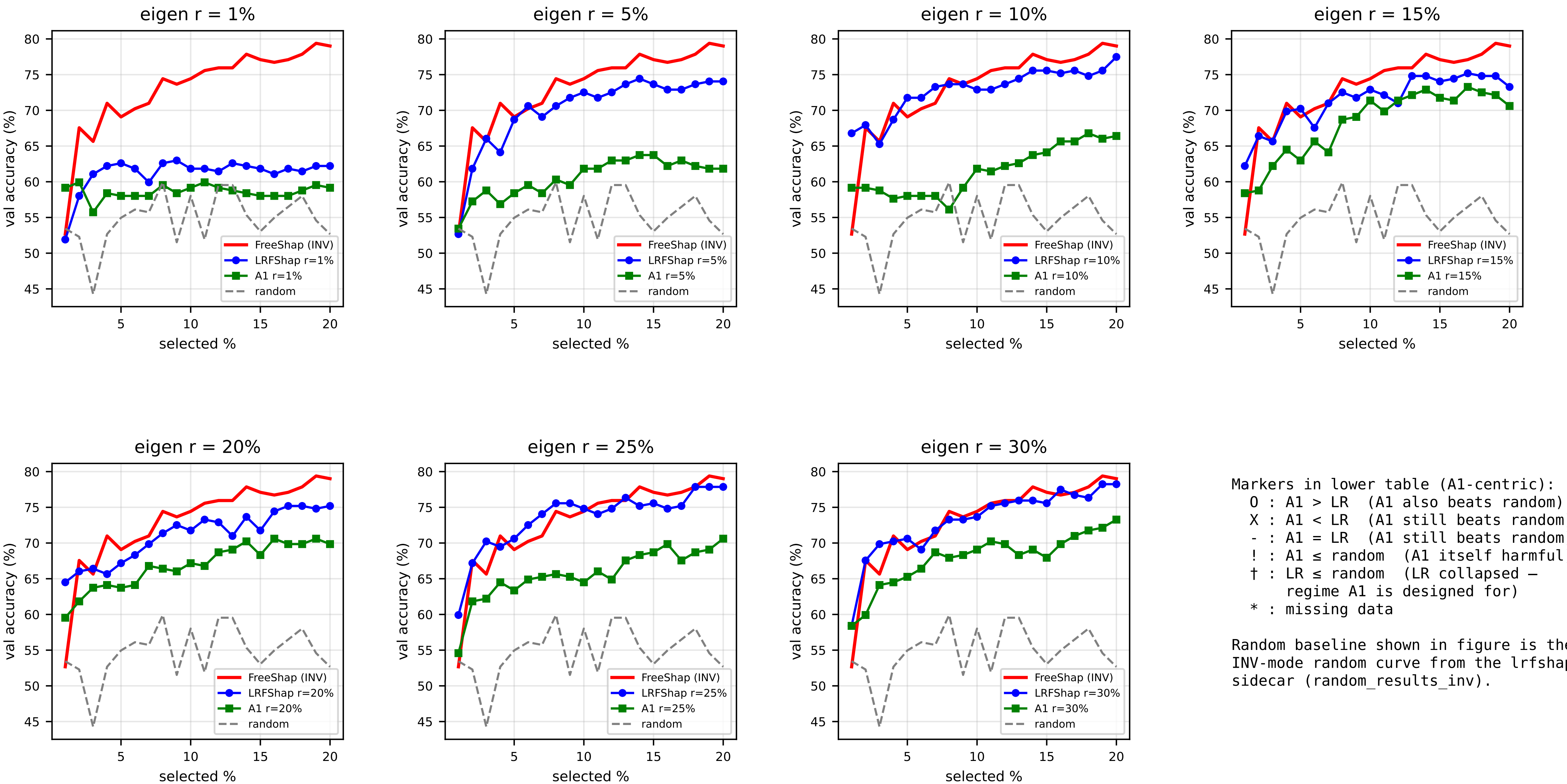



```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

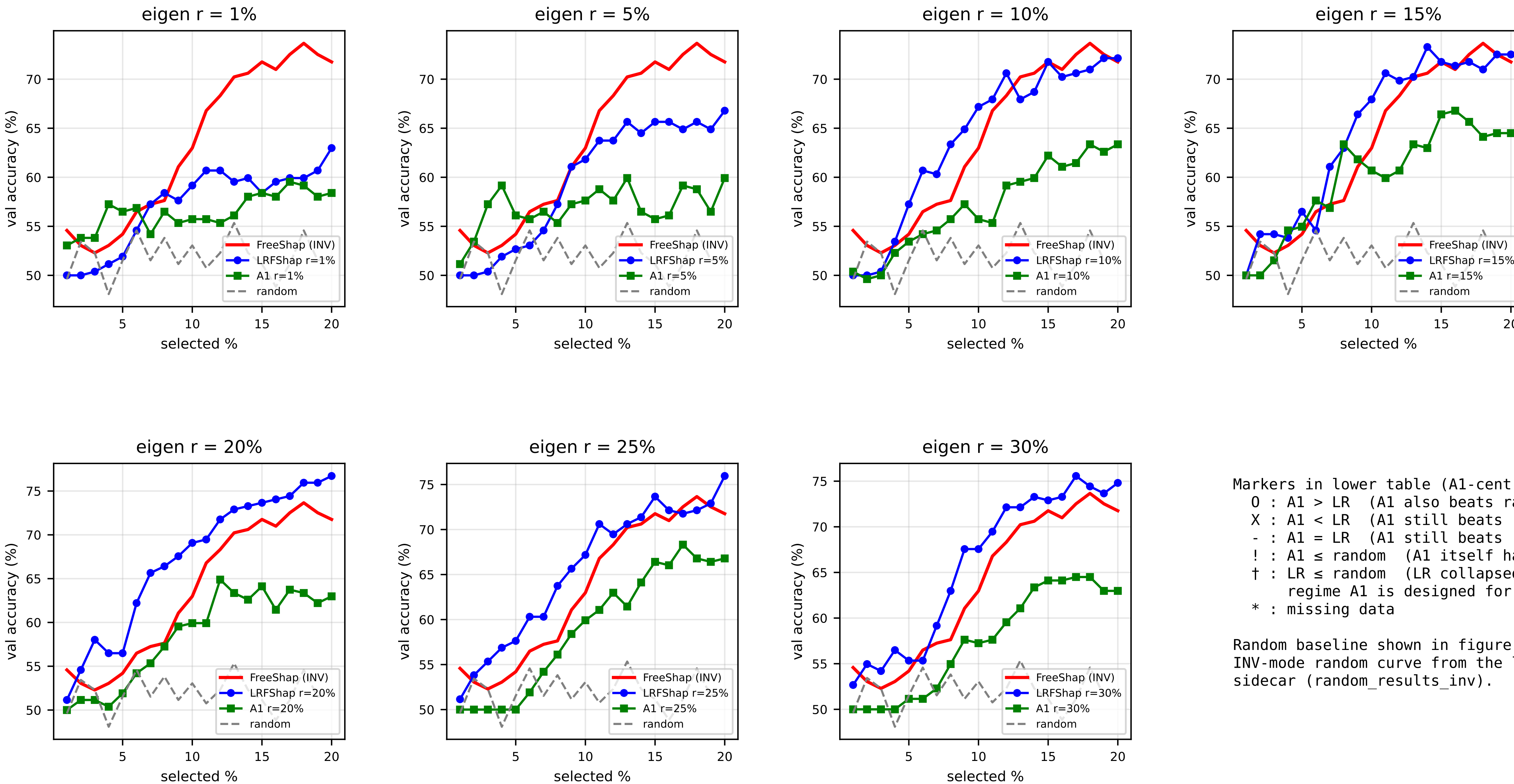
	sel1%	**sel2%**	**sel3%**	**sel4%**	**sel5%**	**sel10%**	**sel20%**
INV (acc)	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	50.00 [!]	70.60	78.80
r=1%	O†	O†	O†	O	O	O	X
r=5%	O†	O†	O†	O†	O†	O	X
r=10%	O†	O†	O†	O†	O†	O	X
r=15%	O†	O†	O†	O†	O†	O	X
r=20%	O†	O†	O†	O†	O†	O	X
r=25%	O†	O†	O†	O†	O†	O	X
r=30%	O†	O†	O†	O†	O†	O	X

RTE forced 50/50 (n=1300, valbal=262) | n_train=1300, valbal=262



	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	52.67 [!]	67.56	65.65	70.99	69.08	74.43	79.01
r=1%	O†	O	X	X	X	X	X
r=5%	!†	X	X	X	X	X	X
r=10%	X	X	X	X	X	X	X
r=15%	X	X	X	X	X	X	X
r=20%	X	X	X	X	X	X	X
r=25%	X	X	X	X	X	X	X
r=30%	-	X	X	X	X	X	X

RTE forced 70/30 label1-majority (n=1300, valbal=262) | n_train=1300, valbal=262

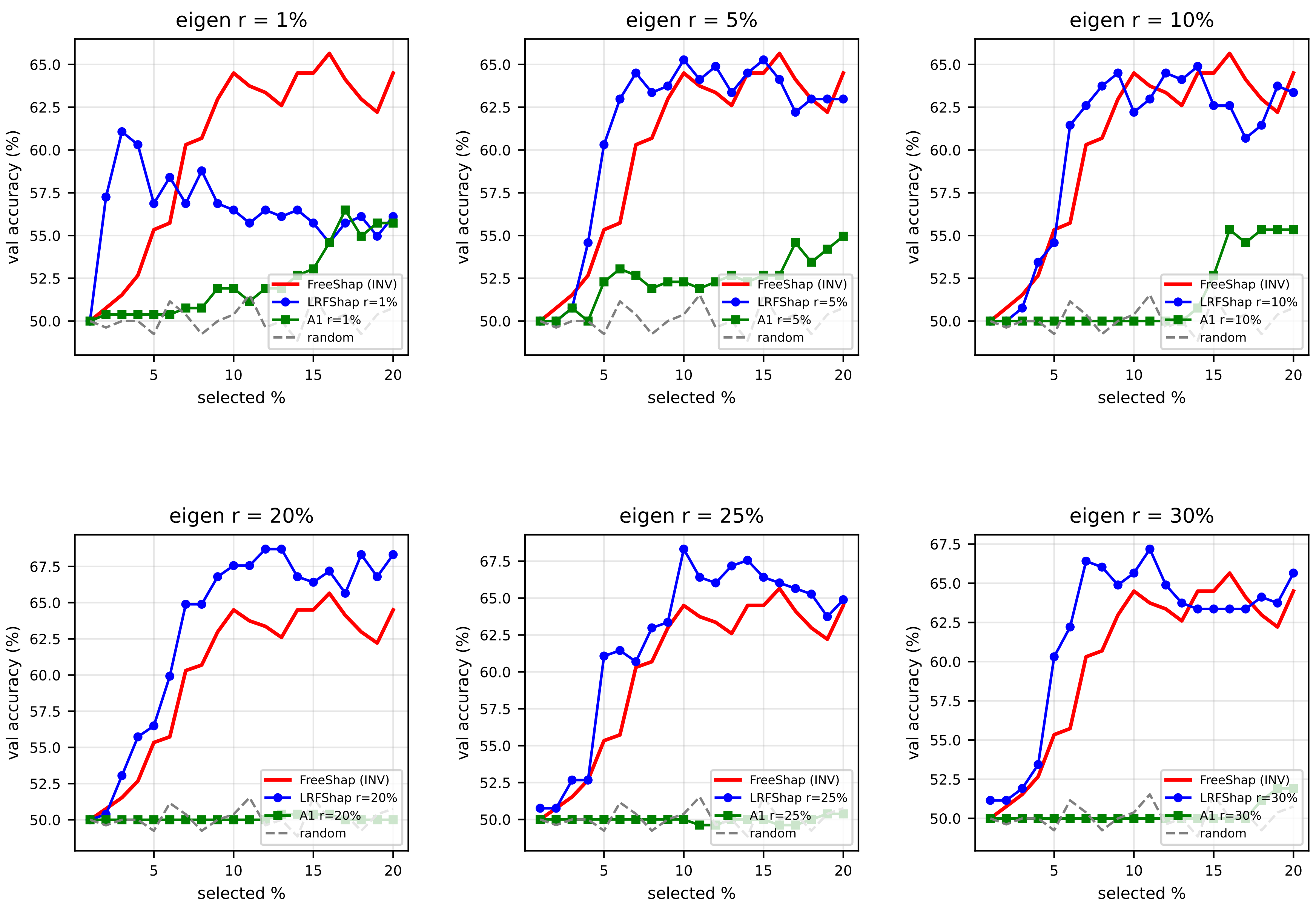



```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	54.58	53.05 [!]	52.29 [!]	53.05	54.20	62.98	71.76
r=1%	O	O†	O†	O	O	X	X
r=5%	O	!†	O†	O	O	X	X
r=10%	O	!†	!†	X	X	X	X
r=15%	-	!	!	O	X	X	X
r=20%	X	!	!	X	X	X	X
r=25%	X	!	!	X	!	X	X
r=30%	X	!	!	X	!	X	X

RTE forced 90/10 label1-majority (n=1300, valbal=262) | n_train=1300, valbal=262

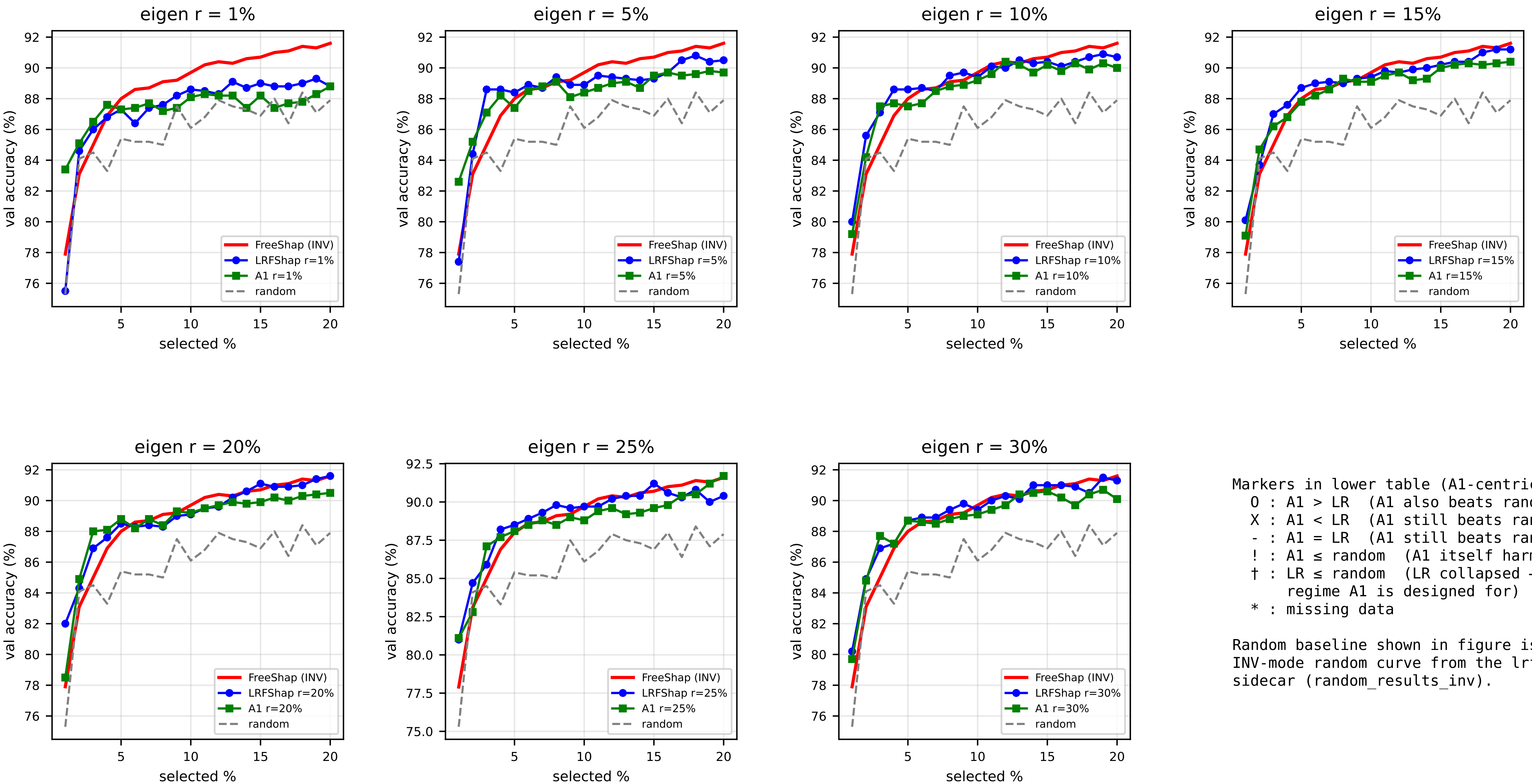



```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	50.00 [!]	50.76	51.53	52.67	55.34	64.50	64.50
r=1%	!†	X	X	X	X	X	X
r=5%	!†	-	-	!	X	X	X
r=10%	!†	-	!	!	X	!	X
r=15%	!†	X	!	!	X	!	!
r=20%	!†	X	!	!	X	!	!
r=25%	!	X	!	!	X	!	!
r=30%	!	X	!	!	X	!	X

AG News balanced 25/25/25/25 (n=5000, valbal=1000) | n_train=5000, valbal=1000

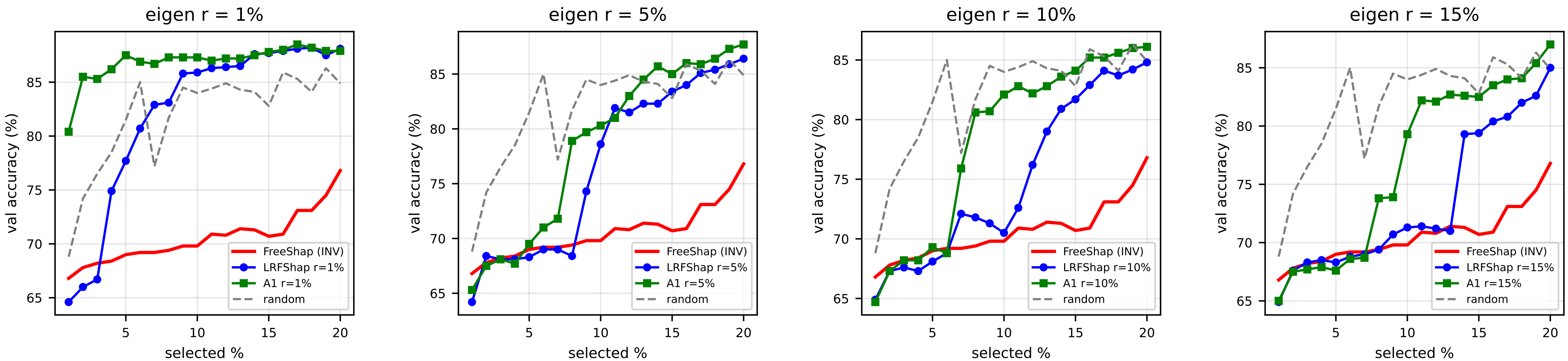


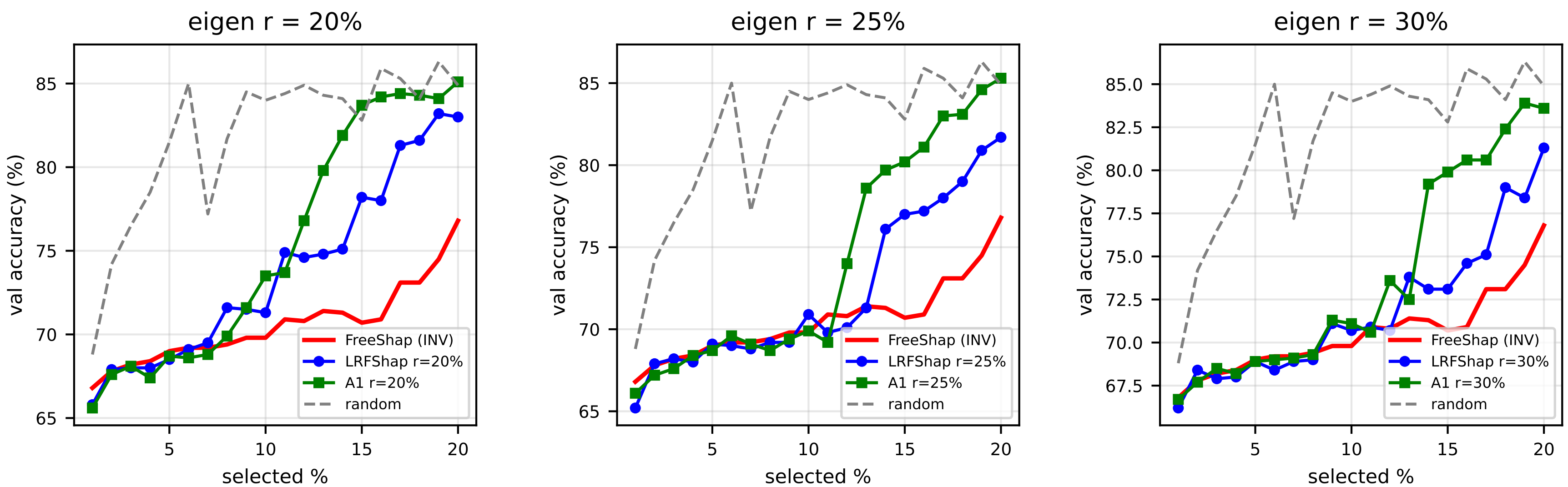

```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	**sel2%**	**sel3%**	**sel4%**	**sel5%**	**sel10%**	**sel20%**
INV (acc)	77.90	83.10 [!]	85.00	86.90	88.00	89.70	91.60
r=1%	O	O	O	O	-	X	-
r=5%	O	O	X	X	X	X	X
r=10%	X	X	O	X	X	X	X
r=15%	X	O†	X	X	X	X	X
r=20%	X	O	O	O	O	O	X
r=25%	O	!	O	X	X	X	O
r=30%	X	X	O	-	-	X	X

AG News forced 55/15/15/15 label0-majority (n=5000, valbal=1000) | n_train=5000, valbal=1000



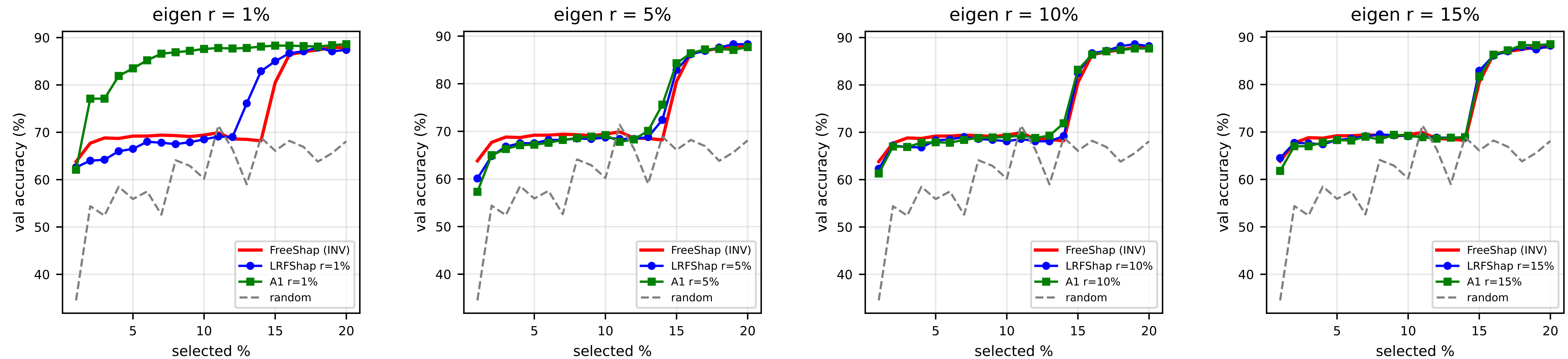


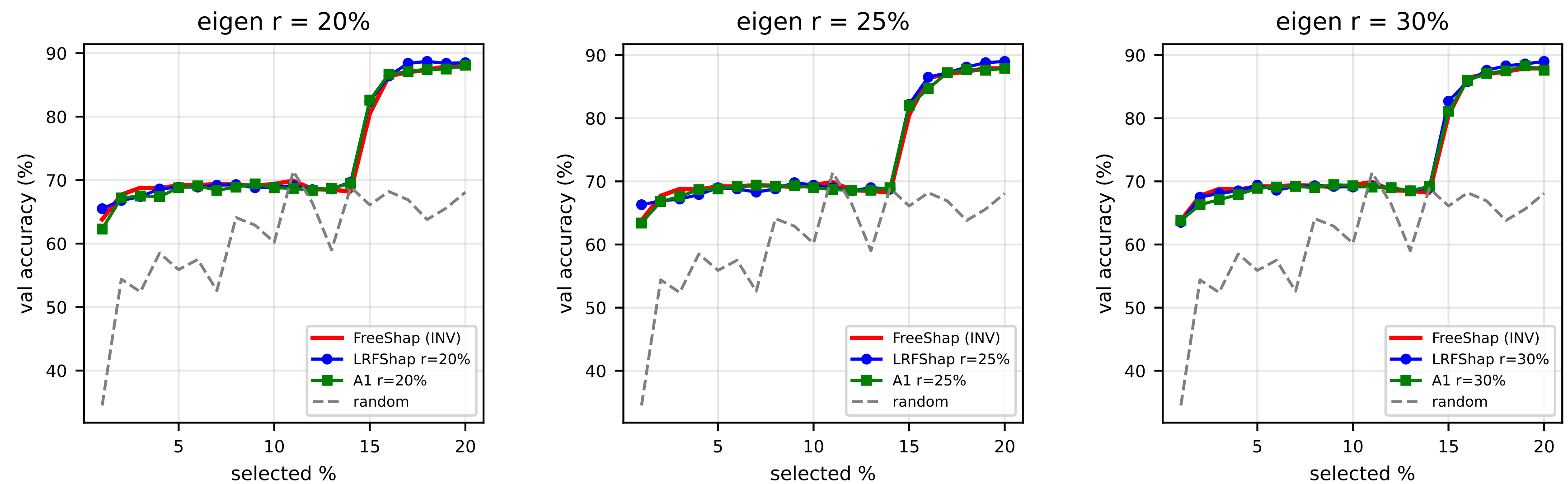

```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	66.80 [!]	67.80 [!]	68.20 [!]	68.40 [!]	69.00 [!]	69.80 [!]	76.80 [!]
r=1%	O†	O†	O†	O†	O†	O	X
r=5%	!†	!†	!†	!†	!†	!†	O
r=10%	!†	!†	!†	!†	!†	!†	O†
r=15%	!†	!†	!†	!†	!†	!†	O
r=20%	!†	!†	!†	!†	!†	!†	O†
r=25%	!†	!†	!†	!†	!†	!†	O†
r=30%	!†	!†	!†	!†	!†	!†	!†

AG News forced 85/5/5/5 label0-majority (n=5000, valbal=1000) | n_train=5000, valbal=1000



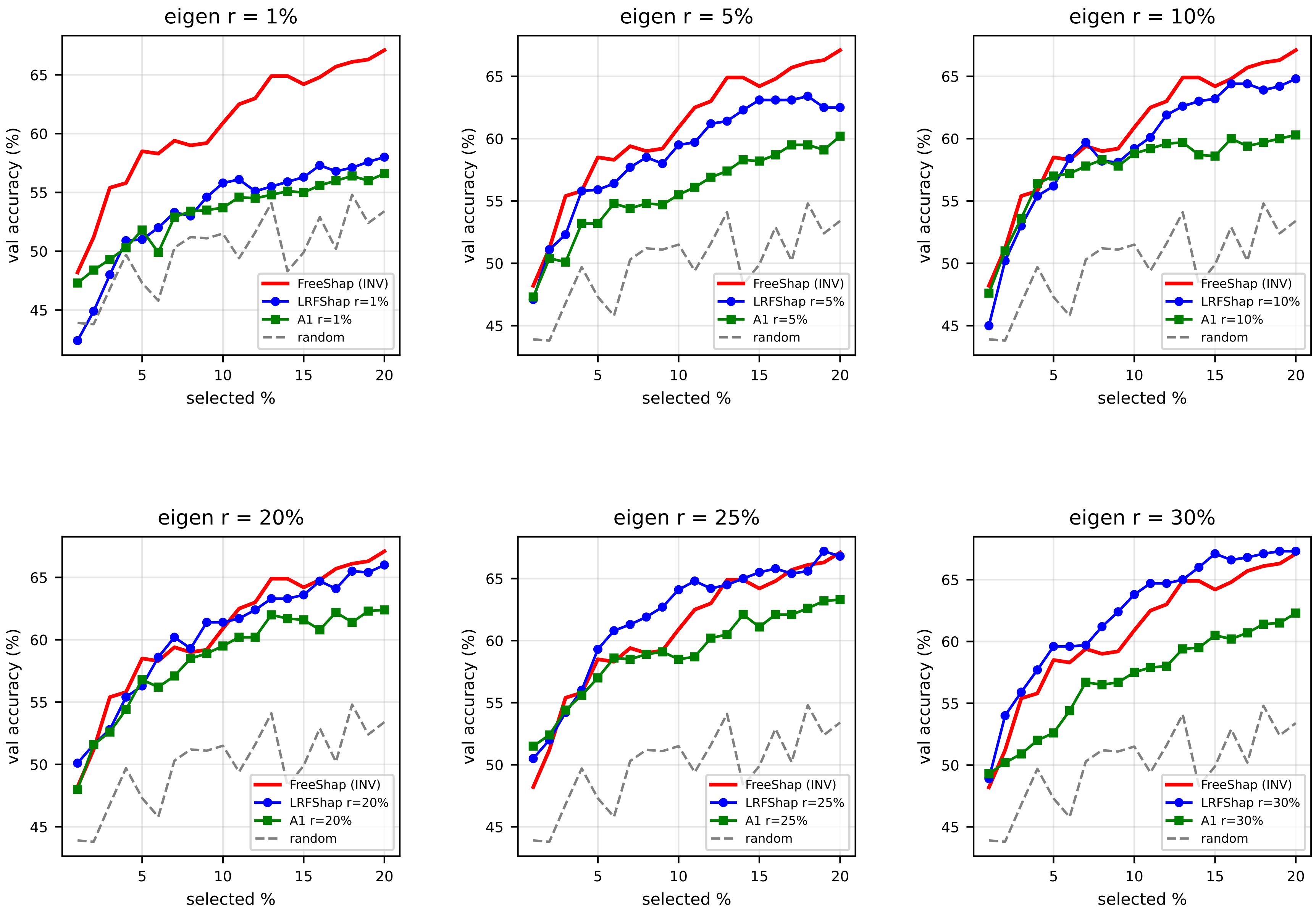



```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	**sel2%**	**sel3%**	**sel4%**	**sel5%**	**sel10%**	**sel20%**
INV (acc)	63.80	67.70	68.80	68.70	69.20	69.40	87.90
r=1%	X	O	O	O	O	O	O
r=5%	X	O	X	X	X	O	X
r=10%	X	X	-	O	X	O	X
r=15%	X	X	X	O	-	O	O
r=20%	X	O	O	X	X	X	X
r=25%	X	X	O	O	X	X	X
r=30%	O	X	X	X	X	O	X

MNLI balanced 33/33/33 (n=5000, valbal=1000) | n_train=5000, valbal=1000

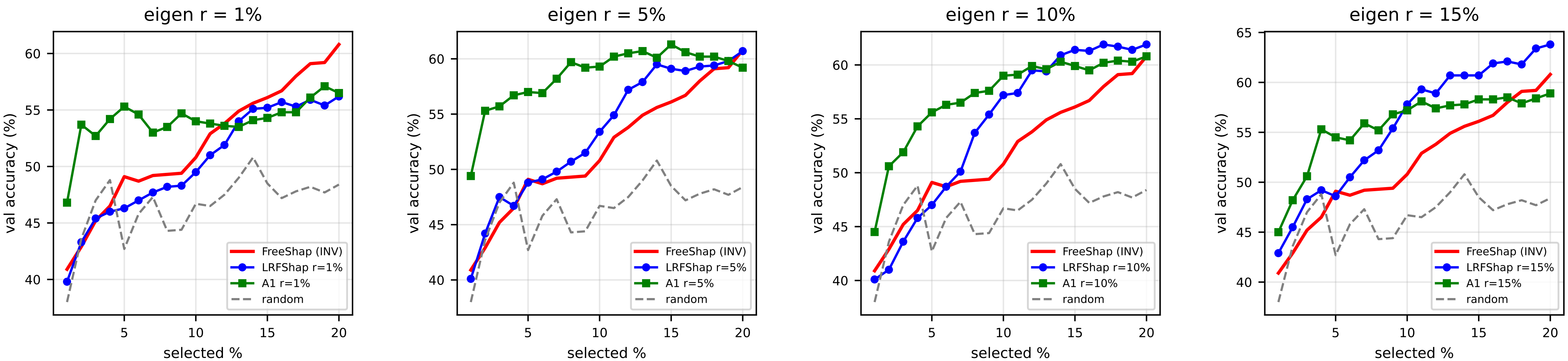


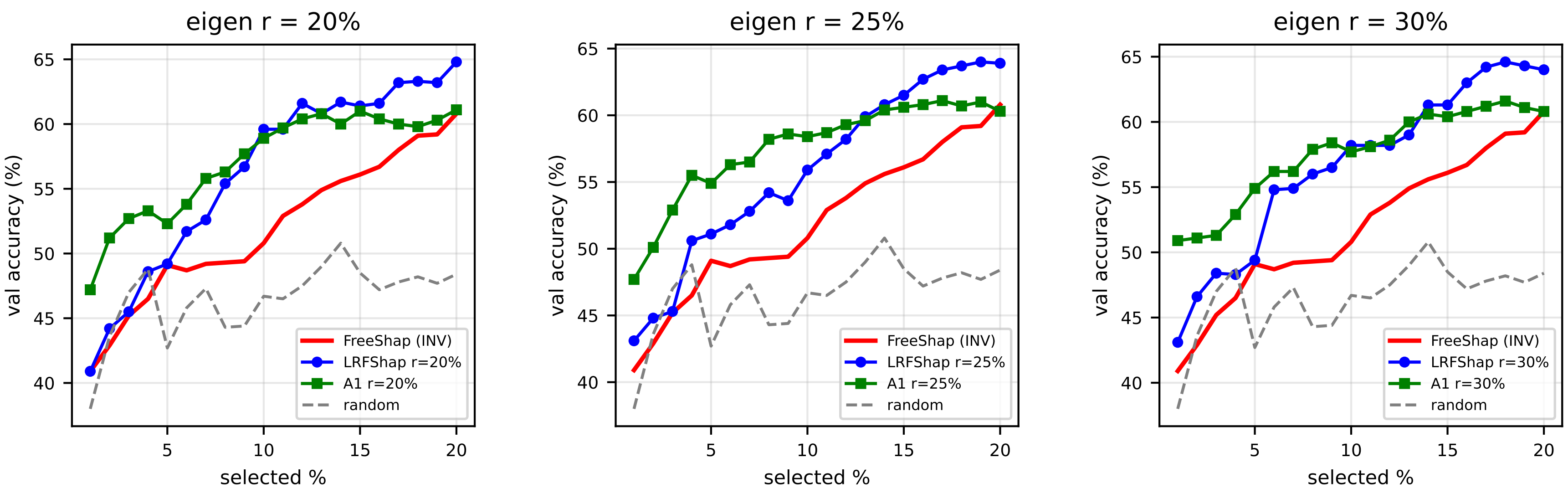

```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	**sel2%**	**sel3%**	**sel4%**	**sel5%**	**sel10%**	**sel20%**
INV (acc)	48.20	51.20	55.40	55.80	58.50	60.90	67.10
r=1%	O†	O	O	X	O	X	X
r=5%	O	X	X	X	X	X	X
r=10%	O	O	O	O	O	X	X
r=15%	X	O	X	X	X	X	X
r=20%	X	-	X	X	O	X	X
r=25%	O	O	O	X	X	X	X
r=30%	O	X	X	X	X	X	X

MNLI forced 60/20/20 label0-majority (n=5000, valbal=1000) | n_train=5000, valbal=1000



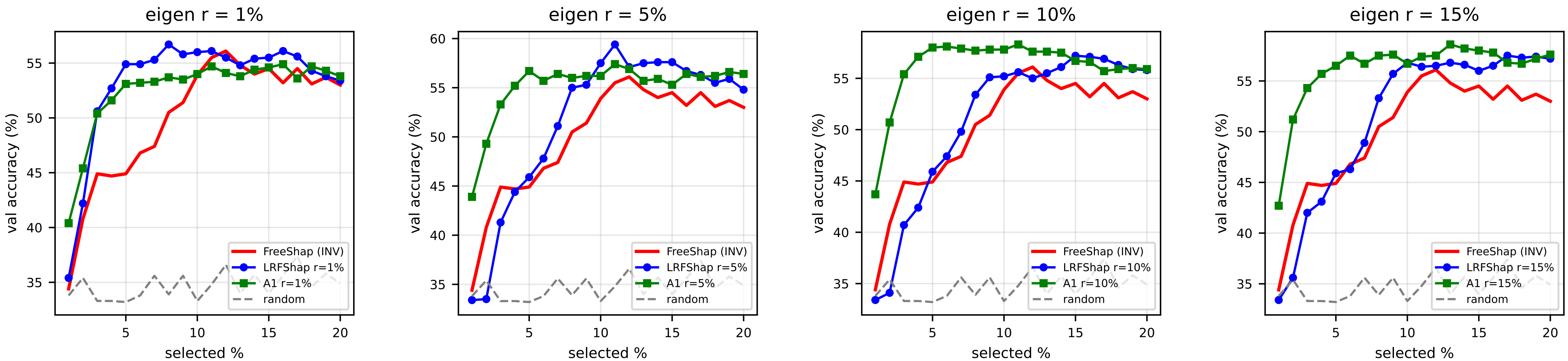


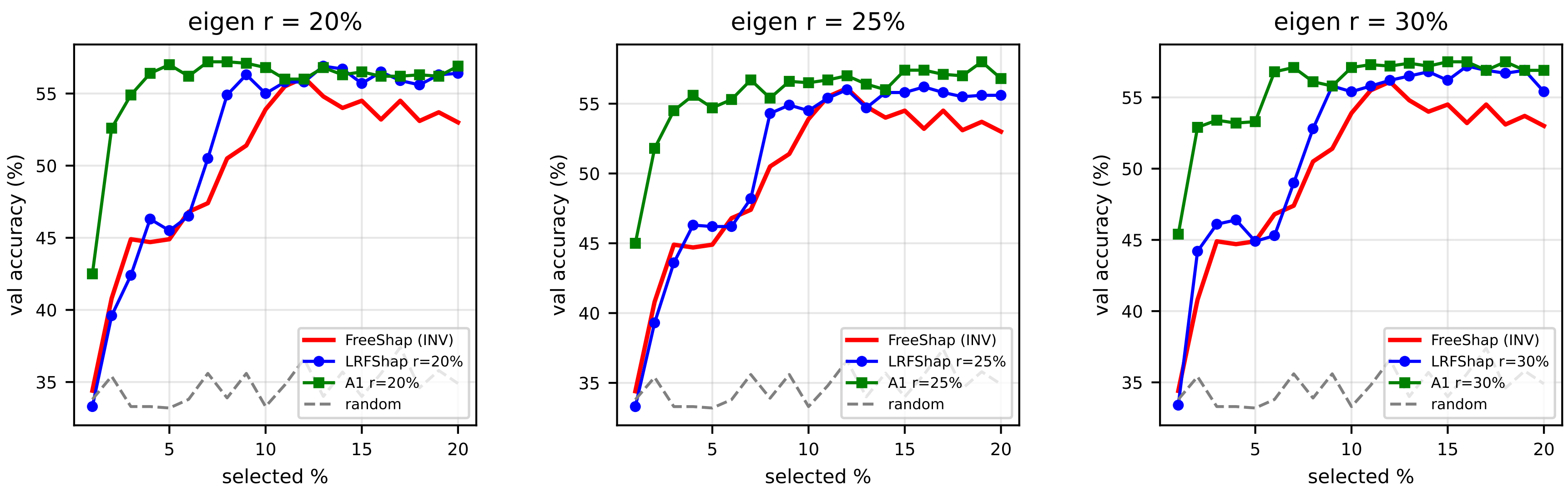

```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%
INV (acc)	40.90	42.90 [!]	45.20 [!]	46.50 [!]	49.10	50.80	60.80
r=1%	O	O†	O†	O†	O	O	O
r=5%	O	O	O	O†	O	O	X
r=10%	O	O†	O†	O†	O	O	X
r=15%	O	O	O	O	O	X	X
r=20%	O	O	O†	O†	O	X	X
r=25%	O	O	O†	O	O	O	X
r=30%	O	O	O	O†	O	X	X

MNLI forced 90/5/5 label0-majority (n=5000, valbal=1000) | n_train=5000, valbal=1000






```
Markers in lower table (A1-centric):
  O : A1 > LR  (A1 also beats random)
  X : A1 < LR  (A1 still beats random)
  - : A1 = LR  (A1 still beats random)
  ! : A1 ≤ random  (A1 itself harmful)
  † : LR ≤ random  (LR collapsed —
      regime A1 is designed for)
  * : missing data

Random baseline shown in figure is the
INV-mode random curve from the lrfshap
sidecar (random_results_inv).
```

	sel1%	**sel2%**	**sel3%**	**sel4%**	**sel5%**	**sel10%**	**sel20%**
INV (acc)	34.40	40.80	44.90	44.70	44.90	53.90	53.00
r=1%	O	O	X	X	X	X	O
r=5%	O†	O†	O	O	O	X	O
r=10%	O†	O†	O	O	O	O	O
r=15%	O†	O	O	O	O	X	O
r=20%	O†	O	O	O	O	O	O
r=25%	O†	O	O	O	O	O	O
r=30%	O†	O	O	O	O	O	O

Summary: A1 vs LRFShap vs random (per-cell breakdown)

Breakdown by setting (rows = dataset/ratio; columns aggregate over rank × sel%)

group	cells	A1 > LR	A1 < LR	A1 = LR	LR ≤ rnd (harm	A1 ≤ rnd (harm	A1 win %	LR-safe %	A1-safe %
mrpc/pos50	49	15	32	2	0	0	30.6	100.0	100.0
mrpc/pos70	49	28	6	0	34	15	57.1	30.6	69.4
mrpc/pos90	49	7	10	0	33	32	14.3	32.7	34.7
sst2/pos50	49	8	27	3	3	11	16.3	93.9	77.6
sst2/pos70	49	9	4	0	40	36	18.4	18.4	26.5
sst2/pos90	49	16	13	0	35	20	32.7	28.6	59.2
mr/pos50	49	13	31	2	0	3	26.5	100.0	93.9
mr/pos70	49	5	1	0	47	43	10.2	4.1	12.2
mr/pos90	49	19	6	0	37	24	38.8	24.5	51.0
qqp/pos50	49	8	36	0	0	5	16.3	100.0	89.8
qqp/pos30	49	14	1	0	43	34	28.6	12.2	30.6
qqp/pos10	49	42	7	0	33	0	85.7	32.7	100.0
rte/pos50	49	2	45	1	2	1	4.1	95.9	98.0
rte/pos70	49	11	24	1	6	13	22.4	87.8	73.5
rte/pos90	49	0	20	3	5	26	0.0	89.8	46.9
ag_news/cls25_25_25_25	49	17	27	4	1	1	34.7	98.0	98.0
ag_news/cls55_15_15_15	49	11	1	0	45	37	22.4	8.2	24.5
ag_news/cls85_05_05_05	49	19	28	2	0	0	38.8	100.0	100.0
mnli/cls33_33_33	49	16	32	1	1	0	32.7	98.0	100.0
mnli/cls60_20_20	49	40	9	0	11	0	81.6	77.6	100.0
mnli/cls90_05_05	49	43	6	0	8	0	87.8	83.7	100.0
ALL (21 settings)	**1029**	**343**	**366**	**19**	**384**	**301**	**33.3**	**62.7**	**70.7**

Breakdown by rank (rows = eigen rank r%; aggregated across all settings × sel%)

group	cells	A1 > LR	A1 < LR	A1 = LR	LR ≤ rnd (harm	A1 ≤ rnd (harm	A1 win %	LR-safe %	A1-safe %
rank r=1%	147	81	45	3	59	18	55.1	59.9	87.8
rank r=5%	147	63	52	2	55	30	42.9	62.6	79.6
rank r=10%	147	58	49	3	59	37	39.5	59.9	74.8
rank r=15%	147	34	57	3	52	53	23.1	64.6	63.9
rank r=20%	147	36	55	2	55	54	24.5	62.6	63.3
rank r=25%	147	42	50	2	52	53	28.6	64.6	63.9
rank r=30%	147	29	58	4	52	56	19.7	64.6	61.9
ALL ranks (21 settings)	**1029**	**343**	**366**	**19**	**384**	**301**	**33.3**	**62.7**	**70.7**

```
Notes:
  - n_cells = 7 ranks × 7 sel% = 49 per setting (when all data present);
    per-rank rows aggregate sel% × all settings.
  - 'A1 win %' = fraction of cells where A1 strictly beats LR (and beats random — '!' cells excluded).
  - 'LR-safe %' = fraction of cells where LR > random.
  - 'A1-safe %' = fraction of cells where A1 > random.
  - Per-cell random baseline = INV (FS) random (random_results_inv from the lrfshap inv sidecar).
    Same baseline used for the figure curves on each setting page.
```

Per-rank breakdown grouped by imbalance level (balanced / mild / extreme)

BALANCED imbalance — settings: mrpc/pos50, sst2/pos50, mr/pos50, qqp/pos50, rte/pos50, ag_news/cls25_25_25_25, mnli/cls33_33_33

group	cells	A1 > LR	A1 < LR	A1 = LR	LR ≤ rnd (harm	A1 ≤ rnd (harm	A1 win %	LR-safe %	A1-safe %
r=1%	49	13	25	3	5	8	26.5	89.8	83.7
r=5%	49	8	33	0	1	8	16.3	98.0	83.7
r=10%	49	15	32	1	0	1	30.6	100.0	98.0
r=15%	49	8	39	1	1	1	16.3	98.0	98.0
r=20%	49	14	33	2	0	0	28.6	100.0	100.0
r=25%	49	14	30	2	0	3	28.6	100.0	93.9
r=30%	49	7	38	4	0	0	14.3	100.0	100.0
ALL ranks (7 settings)	**343**	**79**	**230**	**13**	**7**	**21**	**23.0**	**98.0**	**93.9**

MILD imbalance — settings: mrpc/pos70, sst2/pos70, mr/pos70, qqp/pos30, rte/pos70, ag_news/cls55_15_15_15, mnli/cls60_20_20

group	cells	A1 > LR	A1 < LR	A1 = LR	LR ≤ rnd (harm	A1 ≤ rnd (harm	A1 win %	LR-safe %	A1-safe %
r=1%	49	37	4	0	35	8	75.5	28.6	83.7
r=5%	49	28	4	0	31	17	57.1	36.7	65.3
r=10%	49	17	5	0	36	27	34.7	26.5	44.9
r=15%	49	11	7	1	29	30	22.4	40.8	38.8
r=20%	49	7	10	0	32	32	14.3	34.7	34.7
r=25%	49	12	8	0	32	29	24.5	34.7	40.8
r=30%	49	6	8	0	31	35	12.2	36.7	28.6
ALL ranks (7 settings)	**343**	**118**	**46**	**1**	**226**	**178**	**34.4**	**34.1**	**48.1**

EXTREME imbalance — settings: mrpc/pos90, sst2/pos90, mr/pos90, qqp/pos10, rte/pos90, ag_news/cls85_05_05_05, mnli/cls90_05_05

group	cells	A1 > LR	A1 < LR	A1 = LR	LR ≤ rnd (harm	A1 ≤ rnd (harm	A1 win %	LR-safe %	A1-safe %
r=1%	49	31	16	0	19	2	63.3	61.2	95.9
r=5%	49	27	15	2	23	5	55.1	53.1	89.8
r=10%	49	26	12	2	23	9	53.1	53.1	81.6
r=15%	49	15	11	1	22	22	30.6	55.1	55.1
r=20%	49	15	12	0	23	22	30.6	53.1	55.1
r=25%	49	16	12	0	20	21	32.7	59.2	57.1
r=30%	49	16	12	0	21	21	32.7	57.1	57.1
ALL ranks (7 settings)	**343**	**146**	**90**	**5**	**151**	**102**	**42.6**	**56.0**	**70.3**

Group definitions (class-count agnostic):
- BALANCED: pos50 (binary majority 50/50), cls25 (AG News 4-class even), cls33 (MNLI 3-class even).
- MILD: pos70 (binary 70/30), qqp pos30 (label0 70%), cls55 (AG News label0 55%), cls60 (MNLI label0 60%).
- EXTREME: pos90 (binary 90/10), qqp pos10 (label0 90%), cls85 (AG News label0 85%), cls90 (MNLI label0 90%).

Each row aggregates 7 sel% × all settings in that group at the given rank.
Random baseline = INV (FS) random (random_results_inv from the lrfshap inv sidecar).

Indicator: A1 > LR   (cell = #settings satisfying / #valid settings)

group / rank	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%	ALL sel%
r=1%	3/7	3/7	2/7	2/7	2/7	1/7	0/7	13/49
r=5%	4/7	2/7	1/7	0/7	1/7	1/7	0/7	9/49
r=10%	3/7	3/7	3/7	2/7	2/7	1/7	1/7	15/49
r=15%	0/7	5/7	2/7	1/7	0/7	0/7	0/7	8/49
r=20%	2/7	4/7	1/7	2/7	3/7	2/7	0/7	14/49
r=25%	4/7	4/7	3/7	1/7	1/7	0/7	1/7	14/49
r=30%	4/7	1/7	2/7	0/7	0/7	0/7	0/7	7/49
BALANCED ALL (n=7)	20/49	22/49	14/49	8/49	9/49	5/49	2/49	80/343
r=1%	7/7	7/7	7/7	7/7	7/7	6/7	3/7	44/49
r=5%	6/7	5/7	6/7	6/7	7/7	6/7	3/7	39/49
r=10%	2/7	2/7	4/7	5/7	5/7	5/7	4/7	27/49
r=15%	3/7	2/7	3/7	5/7	4/7	4/7	2/7	23/49
r=20%	1/7	2/7	2/7	3/7	5/7	2/7	2/7	17/49
r=25%	2/7	2/7	2/7	4/7	4/7	3/7	1/7	18/49
r=30%	2/7	2/7	3/7	2/7	1/7	4/7	1/7	15/49
MILD ALL (n=7)	23/49	22/49	27/49	32/49	33/49	30/49	16/49	183/343
r=1%	4/7	6/7	5/7	5/7	5/7	3/7	3/7	31/49
r=5%	4/7	5/7	5/7	5/7	5/7	3/7	1/7	28/49
r=10%	4/7	4/7	4/7	5/7	4/7	4/7	1/7	26/49
r=15%	2/7	2/7	2/7	3/7	2/7	2/7	2/7	15/49
r=20%	2/7	3/7	3/7	2/7	2/7	2/7	1/7	15/49
r=25%	2/7	2/7	3/7	3/7	2/7	2/7	2/7	16/49
r=30%	3/7	2/7	2/7	2/7	2/7	4/7	1/7	16/49
EXTREME ALL (n=7)	21/49	24/49	24/49	25/49	22/49	20/49	11/49	147/343

```
Reading guide:
  - Each cell is '<numerator>/<denominator>' where the numerator counts how
    many settings (out of the group's 7) satisfy `A1 > LR` at that
    (rank, sel%).  Denominator drops any settings whose sidecar is missing
    at that cell (so a partial run reduces denominator, never inflates it).
  - 'ALL sel%' column sums numerators and denominators across the 7 sel%
    columns (so each row 's ALL = sum of its 7 numerators / sum of its 7
    denominators; not an average of percentages).
  - The bolded '<GROUP> ALL (n=7)' row sums over the group's 7 settings
    and the 7 ranks for each sel%.
```

Indicator: A1 > INV (full kernel)   (cell = #settings satisfying / #valid settings)

group / rank	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%	ALL sel%
r=1%	4/7	3/7	1/7	1/7	0/7	0/7	0/7	9/49
r=5%	3/7	3/7	2/7	1/7	0/7	1/7	0/7	10/49
r=10%	4/7	3/7	2/7	3/7	0/7	0/7	0/7	12/49
r=15%	4/7	4/7	3/7	1/7	0/7	0/7	1/7	13/49
r=20%	4/7	4/7	2/7	2/7	2/7	0/7	0/7	14/49
r=25%	4/7	2/7	2/7	2/7	2/7	0/7	1/7	13/49
r=30%	5/7	3/7	2/7	2/7	1/7	0/7	0/7	13/49
BALANCED ALL (n=7)	28/49	22/49	14/49	12/49	5/49	1/49	2/49	84/343
r=1%	6/7	7/7	7/7	7/7	7/7	6/7	2/7	42/49
r=5%	4/7	5/7	6/7	6/7	7/7	6/7	3/7	37/49
r=10%	1/7	2/7	3/7	4/7	5/7	5/7	4/7	24/49
r=15%	2/7	2/7	3/7	5/7	5/7	5/7	4/7	26/49
r=20%	1/7	2/7	2/7	3/7	4/7	5/7	5/7	22/49
r=25%	1/7	2/7	2/7	3/7	4/7	5/7	4/7	21/49
r=30%	1/7	2/7	3/7	2/7	3/7	6/7	4/7	21/49
MILD ALL (n=7)	16/49	22/49	26/49	30/49	35/49	38/49	26/49	193/343
r=1%	4/7	6/7	6/7	6/7	6/7	4/7	2/7	34/49
r=5%	4/7	4/7	5/7	5/7	4/7	2/7	1/7	25/49
r=10%	4/7	4/7	4/7	4/7	4/7	3/7	1/7	24/49
r=15%	2/7	2/7	2/7	2/7	2/7	3/7	3/7	16/49
r=20%	2/7	2/7	2/7	2/7	2/7	3/7	3/7	16/49
r=25%	2/7	2/7	2/7	2/7	2/7	4/7	2/7	16/49
r=30%	2/7	2/7	2/7	2/7	2/7	3/7	2/7	15/49
EXTREME ALL (n=7)	20/49	22/49	23/49	23/49	22/49	22/49	14/49	146/343

```
Reading guide:
  - Each cell is '<numerator>/<denominator>' where the numerator counts how
    many settings (out of the group's 7) satisfy `A1 > INV (full kernel)` at that
    (rank, sel%).  Denominator drops any settings whose sidecar is missing
    at that cell (so a partial run reduces denominator, never inflates it).
  - 'ALL sel%' column sums numerators and denominators across the 7 sel%
    columns (so each row 's ALL = sum of its 7 numerators / sum of its 7
    denominators; not an average of percentages).
  - The bolded '<GROUP> ALL (n=7)' row sums over the group's 7 settings
    and the 7 ranks for each sel%.
```

Indicator: A1 > random (cell = #settings satisfying / #valid settings)

group / rank	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%	ALL sel%
r=1%	6/7	6/7	6/7	5/7	6/7	5/7	7/7	41/49
r=5%	4/7	6/7	7/7	5/7	6/7	6/7	7/7	41/49
r=10%	7/7	7/7	7/7	7/7	7/7	6/7	7/7	48/49
r=15%	7/7	6/7	7/7	7/7	7/7	7/7	7/7	48/49
r=20%	7/7	7/7	7/7	7/7	7/7	7/7	7/7	49/49
r=25%	6/7	5/7	7/7	7/7	7/7	7/7	7/7	46/49
r=30%	7/7	7/7	7/7	7/7	7/7	7/7	7/7	49/49
BALANCED ALL (n=7)	44/49	44/49	48/49	45/49	47/49	45/49	49/49	322/343
r=1%	6/7	6/7	5/7	6/7	6/7	6/7	6/7	41/49
r=5%	4/7	3/7	5/7	5/7	4/7	5/7	6/7	32/49
r=10%	2/7	2/7	2/7	3/7	3/7	4/7	6/7	22/49
r=15%	2/7	1/7	2/7	3/7	3/7	3/7	5/7	19/49
r=20%	2/7	1/7	1/7	3/7	2/7	3/7	5/7	17/49
r=25%	2/7	2/7	2/7	3/7	2/7	3/7	6/7	20/49
r=30%	2/7	1/7	1/7	2/7	1/7	3/7	4/7	14/49
MILD ALL (n=7)	20/49	16/49	18/49	25/49	21/49	27/49	38/49	165/343
r=1%	5/7	7/7	7/7	7/7	7/7	7/7	7/7	47/49
r=5%	5/7	6/7	7/7	6/7	6/7	7/7	7/7	44/49
r=10%	5/7	6/7	5/7	5/7	6/7	6/7	7/7	40/49
r=15%	3/7	4/7	3/7	3/7	4/7	4/7	6/7	27/49
r=20%	3/7	4/7	3/7	3/7	4/7	4/7	6/7	27/49
r=25%	3/7	4/7	3/7	3/7	4/7	5/7	6/7	28/49
r=30%	3/7	4/7	3/7	3/7	4/7	4/7	7/7	28/49
EXTREME ALL (n=7)	27/49	35/49	31/49	30/49	35/49	37/49	46/49	241/343

Reading guide:
- Each cell is '<numerator>/<denominator>' where the numerator counts how many settings (out of the group's 7) satisfy `A1 > random` at that (rank, sel%). Denominator drops any settings whose sidecar is missing at that cell (so a partial run reduces denominator, never inflates it).
- 'ALL sel%' column sums numerators and denominators across the 7 sel% columns (so each row 's ALL = sum of its 7 numerators / sum of its 7 denominators; not an average of percentages).
- The bolded '<GROUP> ALL (n=7)' row sums over the group's 7 settings and the 7 ranks for each sel%.

Indicator: A1 > max(LR, INV, random) (cell = #settings satisfying / #valid settings)

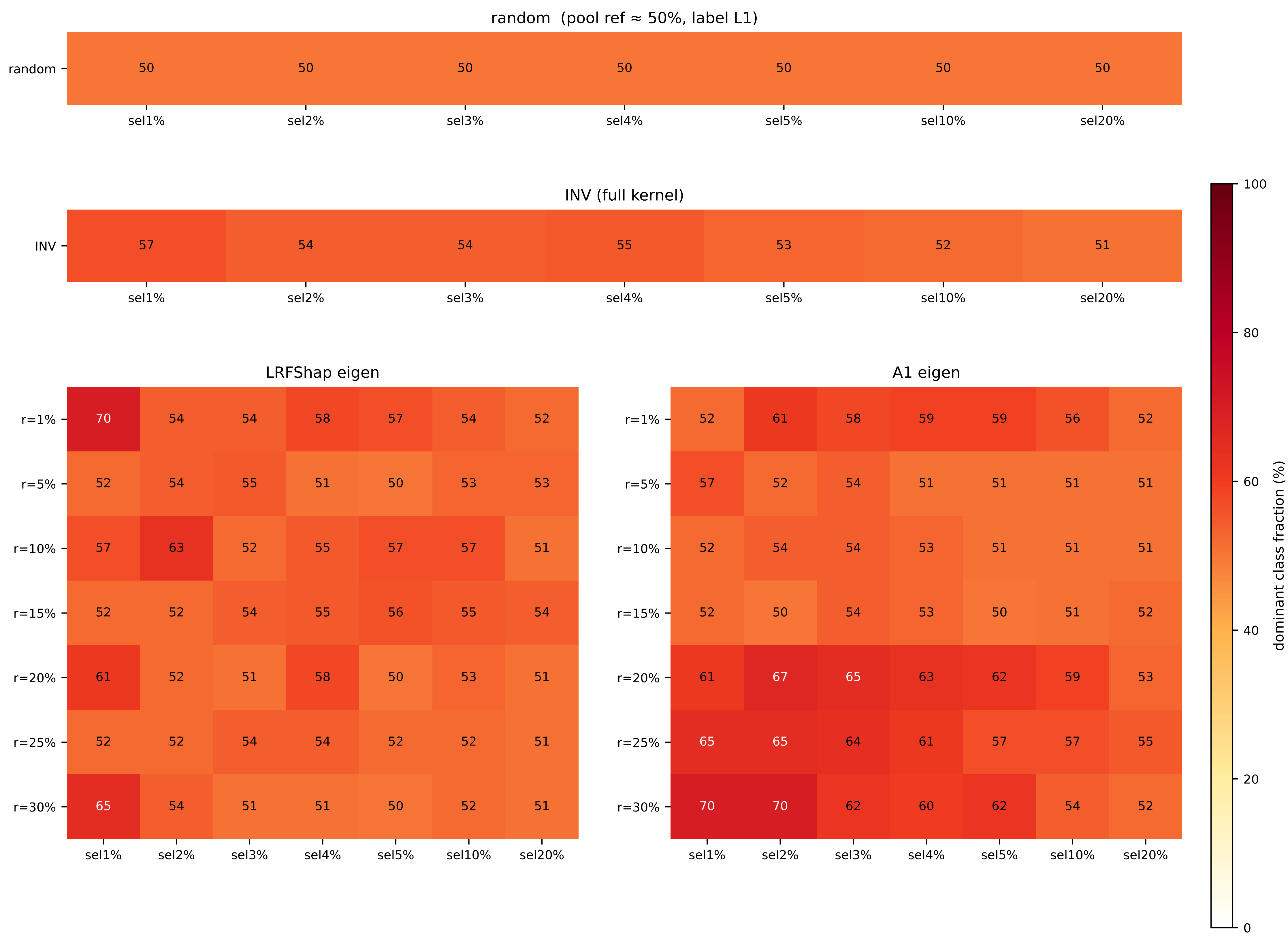
group / rank	sel1%	sel2%	sel3%	sel4%	sel5%	sel10%	sel20%	ALL sel%
r=1%	2/7	1/7	1/7	1/7	0/7	0/7	0/7	5/49
r=5%	1/7	2/7	1/7	0/7	0/7	1/7	0/7	5/49
r=10%	1/7	1/7	2/7	1/7	0/7	0/7	0/7	5/49
r=15%	0/7	4/7	2/7	0/7	0/7	0/7	0/7	6/49
r=20%	1/7	2/7	1/7	1/7	1/7	0/7	0/7	6/49
r=25%	3/7	2/7	2/7	0/7	1/7	0/7	1/7	9/49
r=30%	3/7	1/7	1/7	0/7	0/7	0/7	0/7	5/49
BALANCED ALL (n=7)	11/49	13/49	10/49	3/49	2/49	1/49	1/49	41/343
r=1%	5/7	6/7	5/7	6/7	6/7	5/7	1/7	34/49
r=5%	3/7	3/7	5/7	5/7	4/7	4/7	1/7	25/49
r=10%	1/7	2/7	2/7	2/7	2/7	3/7	4/7	16/49
r=15%	1/7	1/7	2/7	3/7	2/7	1/7	1/7	11/49
r=20%	1/7	1/7	1/7	2/7	1/7	0/7	1/7	7/49
r=25%	1/7	2/7	2/7	2/7	2/7	2/7	1/7	12/49
r=30%	1/7	1/7	1/7	1/7	1/7	1/7	0/7	6/49
MILD ALL (n=7)	13/49	16/49	18/49	21/49	18/49	16/49	9/49	111/343
r=1%	4/7	6/7	5/7	5/7	5/7	3/7	2/7	30/49
r=5%	4/7	4/7	5/7	5/7	4/7	1/7	1/7	24/49
r=10%	4/7	4/7	4/7	4/7	4/7	2/7	1/7	23/49
r=15%	2/7	2/7	2/7	2/7	2/7	1/7	2/7	13/49
r=20%	2/7	2/7	2/7	2/7	2/7	2/7	1/7	13/49
r=25%	2/7	2/7	2/7	2/7	2/7	2/7	2/7	14/49
r=30%	2/7	2/7	2/7	2/7	2/7	2/7	1/7	13/49
EXTREME ALL (n=7)	20/49	22/49	22/49	22/49	21/49	13/49	10/49	130/343

```
Reading guide:
  - Each cell is '<numerator>/<denominator>' where the numerator counts how
    many settings (out of the group's 7) satisfy `A1 > max(LR, INV, random)` at that
    (rank, sel%).  Denominator drops any settings whose sidecar is missing
    at that cell (so a partial run reduces denominator, never inflates it).
  - 'ALL sel%' column sums numerators and denominators across the 7 sel%
    columns (so each row 's ALL = sum of its 7 numerators / sum of its 7
    denominators; not an average of percentages).
  - The bolded '<GROUP> ALL (n=7)' row sums over the group's 7 settings
    and the 7 ranks for each sel%.
```

mrpc pos50 valbal — top-Shapley class composition

MRPC forced 50/50 (n=2300, valbal=258)

	sel 1%	**sel 2%**	**sel 3%**	**sel 4%**	**sel 5%**	**sel 10%**	**sel 20%**
INV / random	L1=57 / L1=50	L1=54 / L1=50	L1=54 / L1=50	L1=55 / L1=50	L1=53 / L1=50	L1=52 / L1=50	L1=51 / L1=50
LR / A1							
r=1%	L0=70 / L1=52	L0=54 / L0=61	L0=54 / L0=58	L0=58 / L0=59	L0=57 / L0=59	L0=54 / L0=56	L0=52 / L0=52
r=5%	L0=52 / L0=57	L0=54 / L1=52	L0=55 / L1=54	L0=51 / L1=51	L0=50 / L1=51	L1=53 / L0=51	L1=53 / L1=51
r=10%	L1=57 / L0=52	L1=63 / L1=54	L1=52 / L1=54	L1=55 / L1=53	L1=57 / L1=51	L1=57 / L0=51	L1=51 / L1=51
r=15%	L0=52 / L0=52	L1=52 / L0=50	L1=54 / L1=54	L1=55 / L1=53	L1=56 / L1=50	L1=55 / L0=51	L1=54 / L1=52
r=20%	L1=61 / L1=61	L1=52 / L1=67	L1=51 / L1=65	L1=58 / L1=63	L1=50 / L1=62	L0=53 / L1=59	L0=51 / L1=53
r=25%	L1=52 / L1=65	L0=52 / L1=65	L0=54 / L1=64	L0=54 / L1=61	L0=52 / L1=57	L0=52 / L1=57	L1=51 / L1=55
r=30%	L1=65 / L1=70	L0=54 / L1=70	L0=51 / L1=62	L1=51 / L1=60	L0=50 / L1=62	L0=52 / L1=54	L1=51 / L1=52

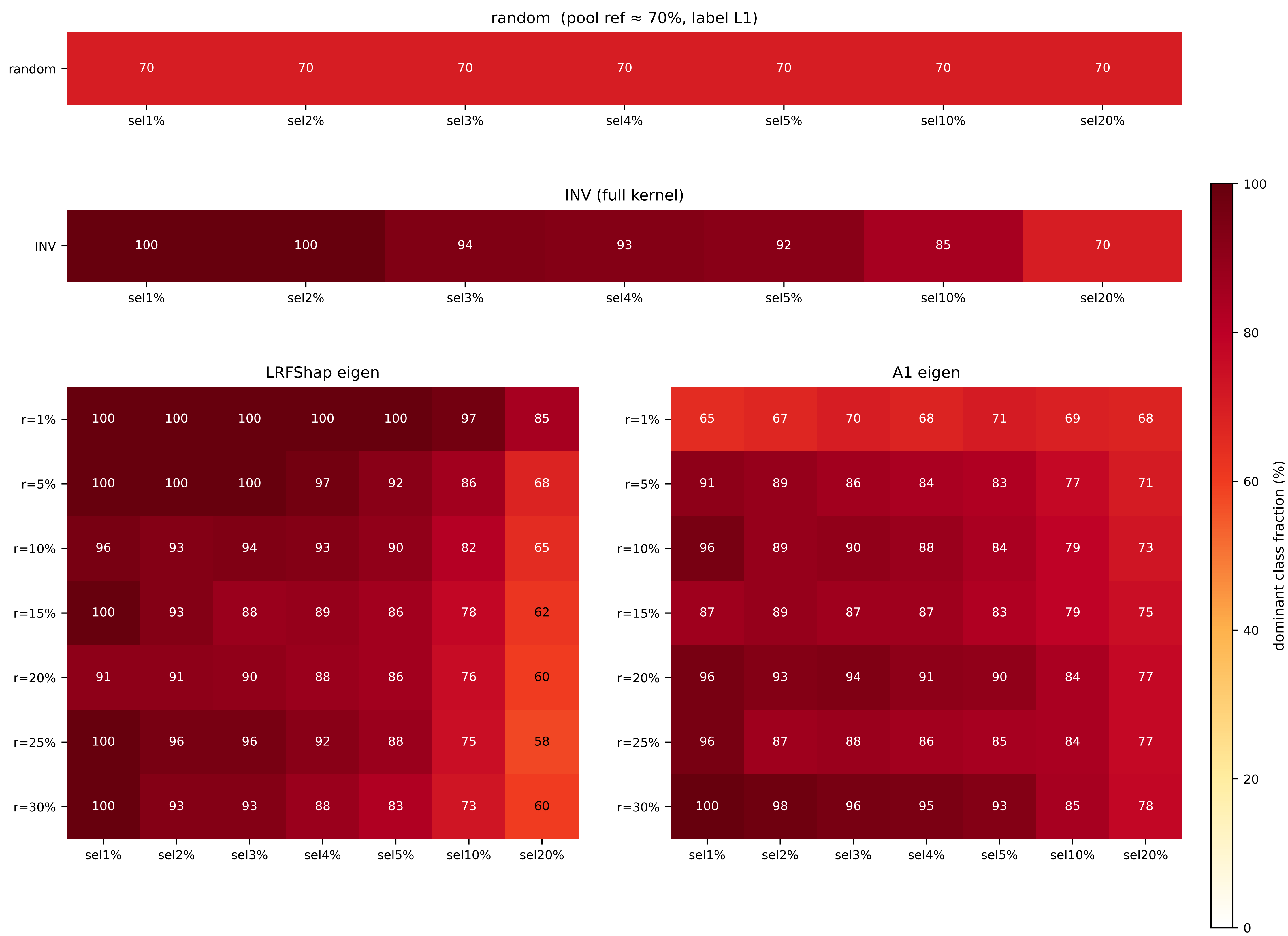



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

mrpc pos70 valbal — top-Shapley class composition

MRPC forced 70/30 label1-majority (n=2300, valbal=258)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L0=100 / L1=70	L0=100 / L1=70	L0=94 / L1=70	L0=93 / L1=70	L0=92 / L1=70	L0=85 / L1=70	L0=70 / L1=70
LR / A1							
r=1%	L0=100 / L1=6	L0=100 / L1=6	L0=100 / L1=70	L0=100 / L1=68	L0=100 / L1=71	L0=97 / L1=69	L0=85 / L1=68
r=5%	L0=100 / L1=91	L0=100 / L1=89	L0=100 / L1=86	L0=97 / L1=84	L0=92 / L1=83	L0=86 / L1=77	L0=68 / L1=71
r=10%	L0=96 / L1=96	L0=93 / L1=89	L0=94 / L1=90	L0=93 / L1=88	L0=90 / L1=84	L0=82 / L1=79	L0=65 / L1=73
r=15%	L0=100 / L1=87	L0=93 / L1=89	L0=88 / L1=87	L0=89 / L1=87	L0=86 / L1=83	L0=78 / L1=79	L0=62 / L1=75
r=20%	L0=91 / L1=96	L0=91 / L1=93	L0=90 / L1=94	L0=88 / L1=91	L0=86 / L1=90	L0=76 / L1=84	L0=60 / L1=77
r=25%	L0=100 / L1=96	L0=96 / L1=87	L0=96 / L1=88	L0=92 / L1=86	L0=88 / L1=85	L0=75 / L1=84	L0=58 / L1=77
r=30%	L0=100 / L1=10	L0=93 / L1=98	L0=93 / L1=96	L0=88 / L1=95	L0=83 / L1=93	L0=73 / L1=85	L0=60 / L1=78

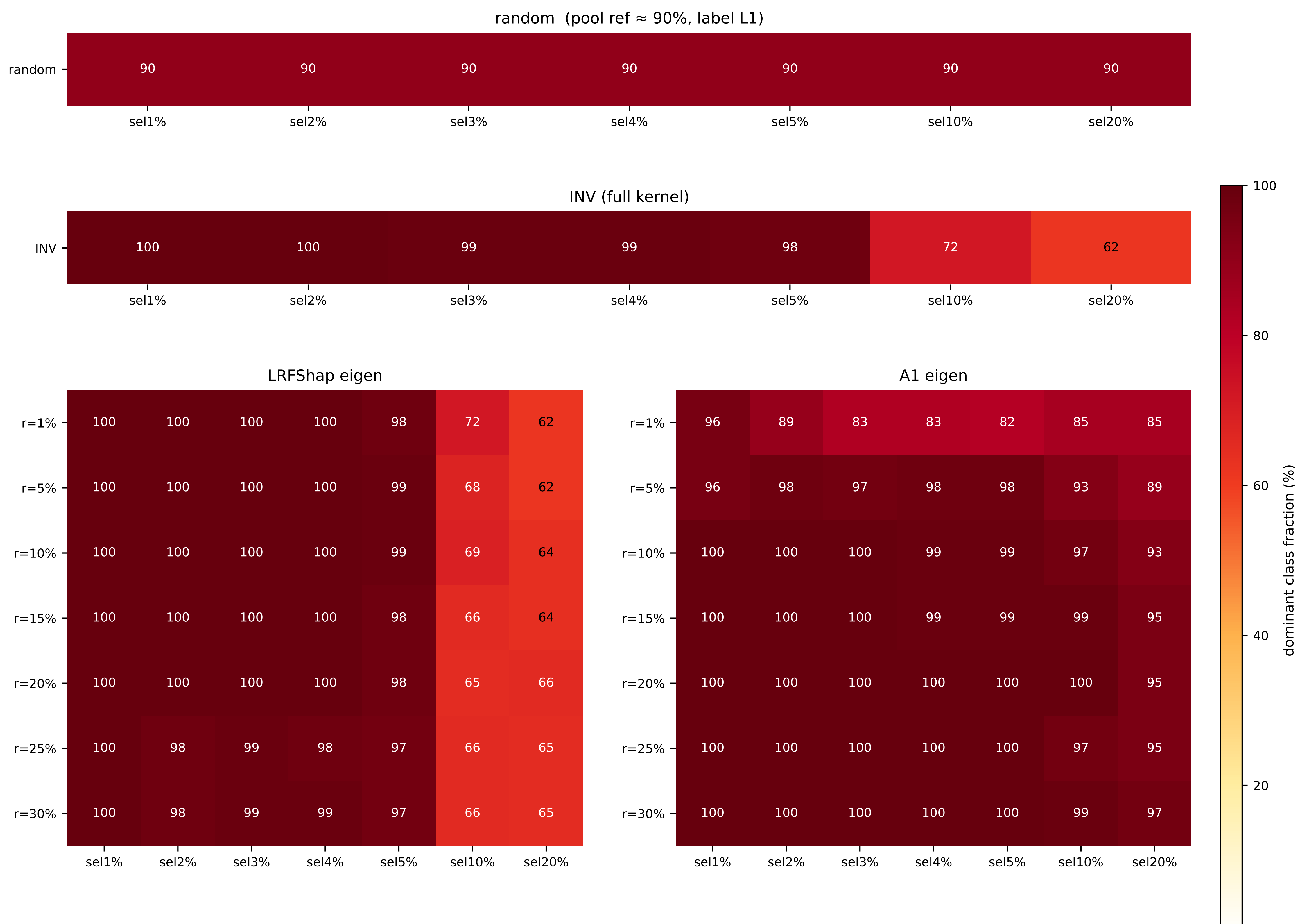



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

mrpc pos90 valbal — top-Shapley class composition

MRPC forced 90/10 label1-majority (n=2300, valbal=258)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L0=100 / L1=90	L0=100 / L1=90	L0=99 / L1=90	L0=99 / L1=90	L0=98 / L1=90	L0=72 / L1=90	L1=62 / L1=90
LR / A1							
r=1%	L0=100 / L1=96	L0=100 / L1=89	L0=100 / L1=83	L0=100 / L1=83	L0=98 / L1=82	L0=72 / L1=85	L1=62 / L1=85
r=5%	L0=100 / L1=96	L0=100 / L1=98	L0=100 / L1=97	L0=100 / L1=98	L0=99 / L1=98	L0=68 / L1=93	L1=62 / L1=89
r=10%	L0=100 / L1=10	L0=100 / L1=10	L0=100 / L1=10	L0=100 / L1=99	L0=99 / L1=99	L0=69 / L1=97	L1=64 / L1=93
r=15%	L0=100 / L1=10	L0=100 / L1=10	L0=100 / L1=10	L0=100 / L1=99	L0=98 / L1=99	L0=66 / L1=99	L1=64 / L1=95
r=20%	L0=100 / L1=10	L0=100 / L1=10	L0=100 / L1=10	L0=100 / L1=10	L0=98 / L1=100	L0=65 / L1=100	L1=66 / L1=95
r=25%	L0=100 / L1=10	L0=98 / L1=100	L0=99 / L1=100	L0=98 / L1=100	L0=97 / L1=100	L0=66 / L1=97	L1=65 / L1=95
r=30%	L0=100 / L1=10	L0=98 / L1=100	L0=99 / L1=100	L0=99 / L1=100	L0=97 / L1=100	L0=66 / L1=99	L1=65 / L1=97

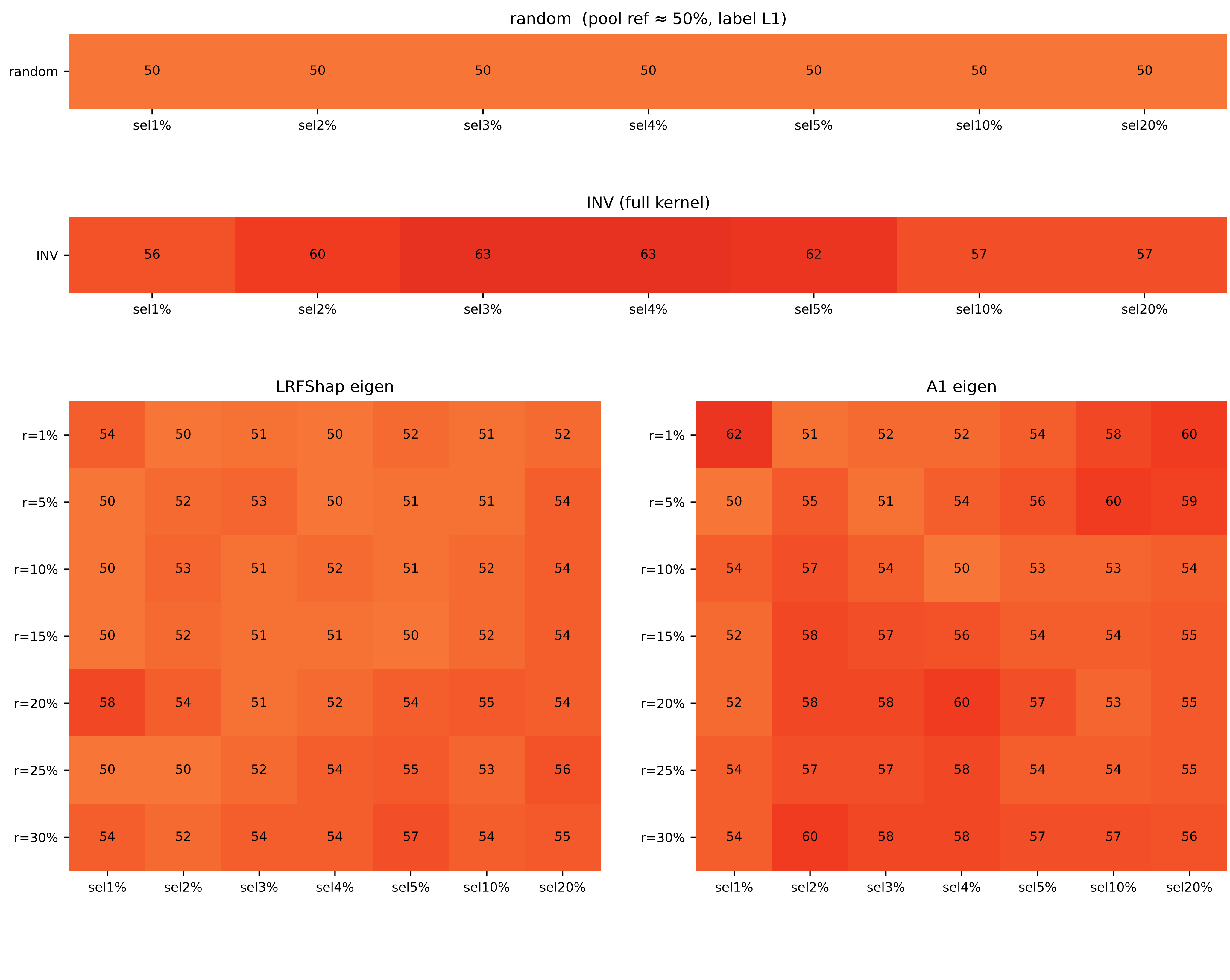


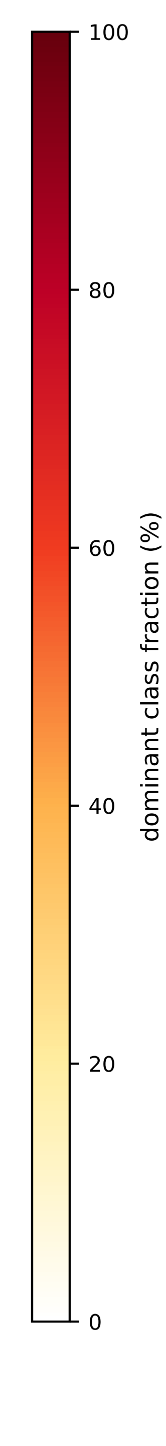

```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

sst2 pos50 valbal — top-Shapley class composition

SST-2 forced 50/50 (n=5000, valbal=856)

	sel 1%	**sel 2%**	**sel 3%**	**sel 4%**	**sel 5%**	**sel 10%**	**sel 20%**
INV / random	L0=56 / L1=50	L0=60 / L1=50	L0=63 / L1=50	L0=63 / L1=50	L0=62 / L1=50	L0=57 / L1=50	L0=57 / L1=50
LR / A1							
r=1%	L1=54 / L1=62	L1=50 / L0=51	L1=51 / L0=52	L1=50 / L0=52	L1=52 / L0=54	L0=51 / L0=58	L0=52 / L0=60
r=5%	L1=50 / L1=50	L1=52 / L0=55	L1=53 / L0=51	L0=50 / L0=54	L0=51 / L0=56	L0=51 / L0=60	L0=54 / L0=59
r=10%	L1=50 / L0=54	L1=53 / L0=57	L1=51 / L0=54	L0=52 / L0=50	L0=51 / L0=53	L0=52 / L0=53	L0=54 / L0=54
r=15%	L1=50 / L0=52	L1=52 / L0=58	L0=51 / L0=57	L0=51 / L0=56	L1=50 / L0=54	L0=52 / L0=54	L0=54 / L0=55
r=20%	L1=58 / L0=52	L1=54 / L0=58	L1=51 / L0=58	L0=52 / L0=60	L0=54 / L0=57	L0=55 / L0=53	L0=54 / L0=55
r=25%	L1=50 / L0=54	L1=50 / L0=57	L0=52 / L0=57	L0=54 / L0=58	L0=55 / L0=54	L0=53 / L0=54	L0=56 / L0=55
r=30%	L1=54 / L0=54	L1=52 / L0=60	L0=54 / L0=58	L0=54 / L0=58	L0=57 / L0=57	L0=54 / L0=57	L0=55 / L0=56



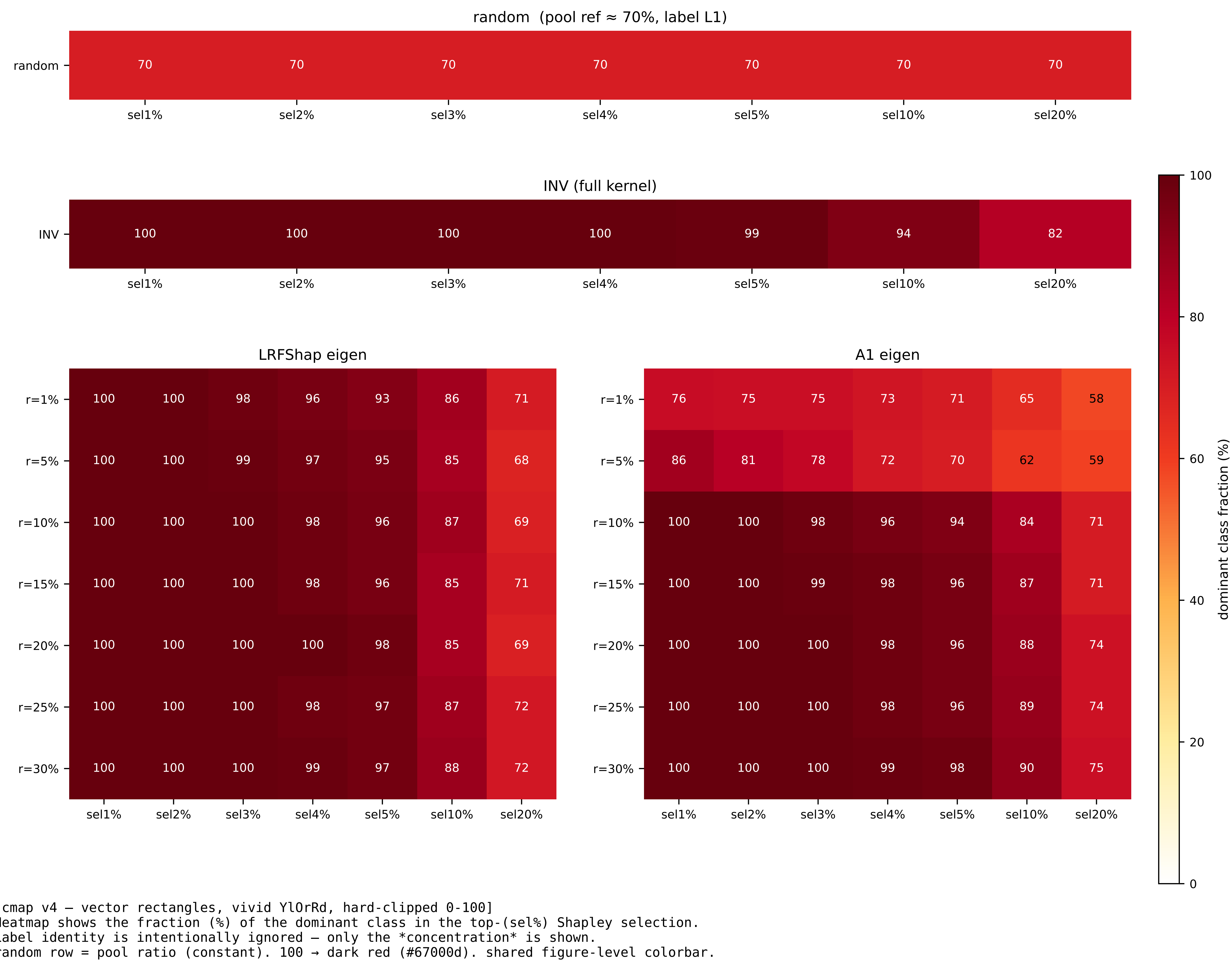



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

sst2 pos70 valbal — top-Shapley class composition

SST-2 forced 70/30 label1-majority (n=5000, valbal=856)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L0=100 / L1=70	L0=100 / L1=70	L0=100 / L1=70	L0=100 / L1=70	L0=99 / L1=70	L0=94 / L1=70	L0=82 / L1=70
LR / A1							
r=1%	L0=100 / L1=76	L0=100 / L1=75	L0=98 / L1=75	L0=96 / L1=73	L0=93 / L1=71	L0=86 / L1=65	L0=71 / L1=58
r=5%	L0=100 / L1=86	L0=100 / L1=81	L0=99 / L1=78	L0=97 / L1=72	L0=95 / L1=70	L0=85 / L1=62	L0=68 / L1=59
r=10%	L0=100 / L0=10	0=100 / L0=10	L0=100 / L0=98	L0=98 / L0=96	L0=96 / L0=94	L0=87 / L0=84	L0=69 / L0=71
r=15%	L0=100 / L0=10	0=100 / L0=10	L0=100 / L0=99	L0=98 / L0=98	L0=96 / L0=96	L0=85 / L0=87	L0=71 / L0=71
r=20%	L0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	L0=100 / L0=98	L0=98 / L0=96	L0=85 / L0=88	L0=69 / L0=74
r=25%	L0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	L0=98 / L0=98	L0=97 / L0=96	L0=87 / L0=89	L0=72 / L0=74
r=30%	L0=100 / L0=10	0=100 / L0=10	0=100 / L0=10	L0=99 / L0=99	L0=97 / L0=98	L0=88 / L0=90	L0=72 / L0=75

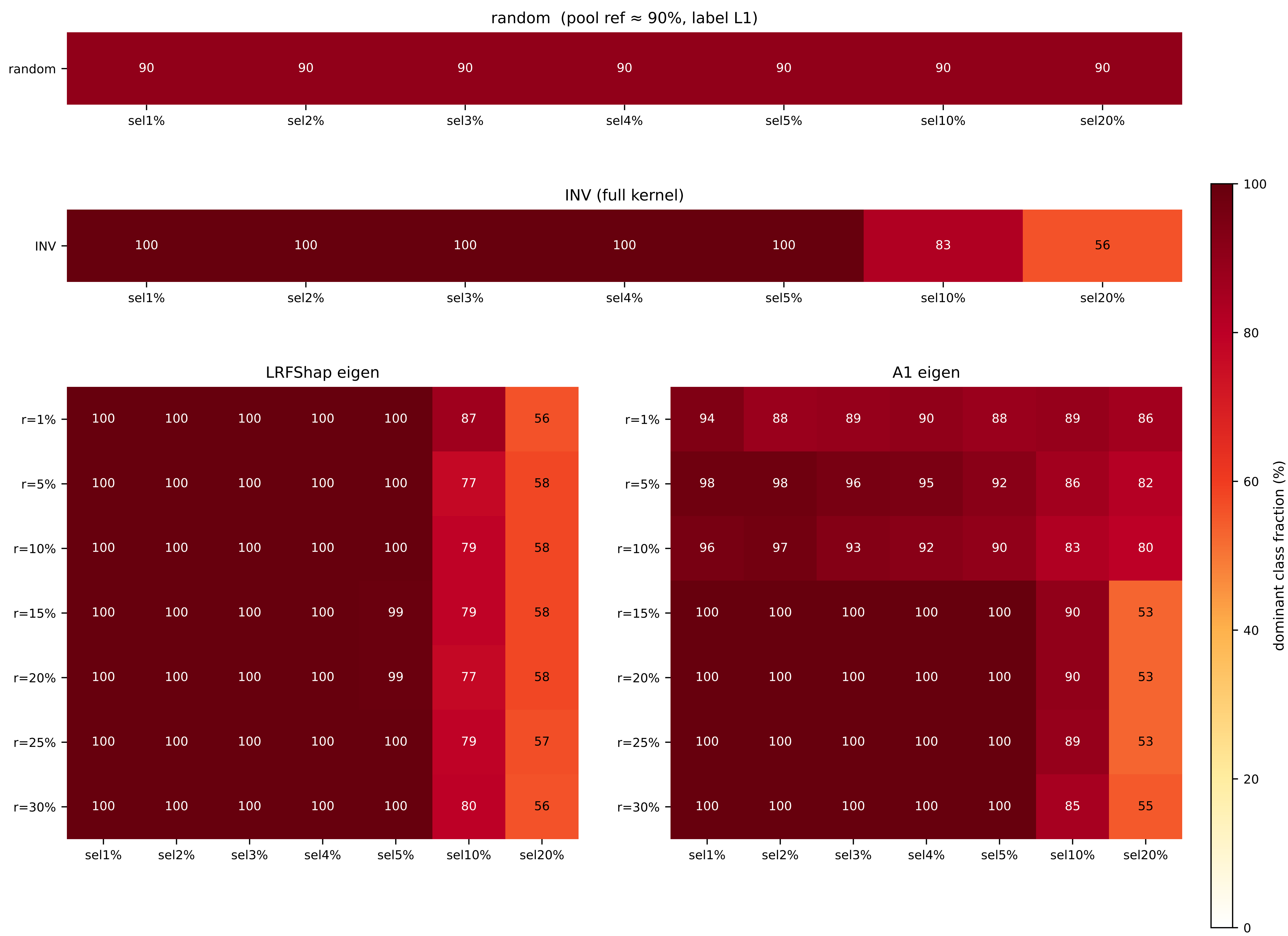



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

sst2 pos90 valbal — top-Shapley class composition

SST-2 forced 90/10 label1-majority (n=5000, valbal=856)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L0=100 / L1=90	L0=100 / L1=90	L0=100 / L1=90	L0=100 / L1=90	L0=100 / L1=90	L0=83 / L1=90	L1=56 / L1=90
LR / A1							
r=1%	L0=100 / L1=94	L0=100 / L1=88	L0=100 / L1=89	L0=100 / L1=90	L0=100 / L1=88	L0=87 / L1=89	L1=56 / L1=86
r=5%	L0=100 / L1=98	L0=100 / L1=98	L0=100 / L1=96	L0=100 / L1=95	L0=100 / L1=92	L0=77 / L1=86	L1=58 / L1=82
r=10%	L0=100 / L1=96	L0=100 / L1=97	L0=100 / L1=93	L0=100 / L1=92	L0=100 / L1=90	L0=79 / L1=83	L1=58 / L1=80
r=15%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=99 / L0=100	L0=79 / L0=90	L1=58 / L1=53
r=20%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=99 / L0=100	L0=77 / L0=90	L1=58 / L1=53
r=25%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=79 / L0=89	L1=57 / L1=53
r=30%	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=100 / L0=100	L0=80 / L0=85	L1=56 / L1=55

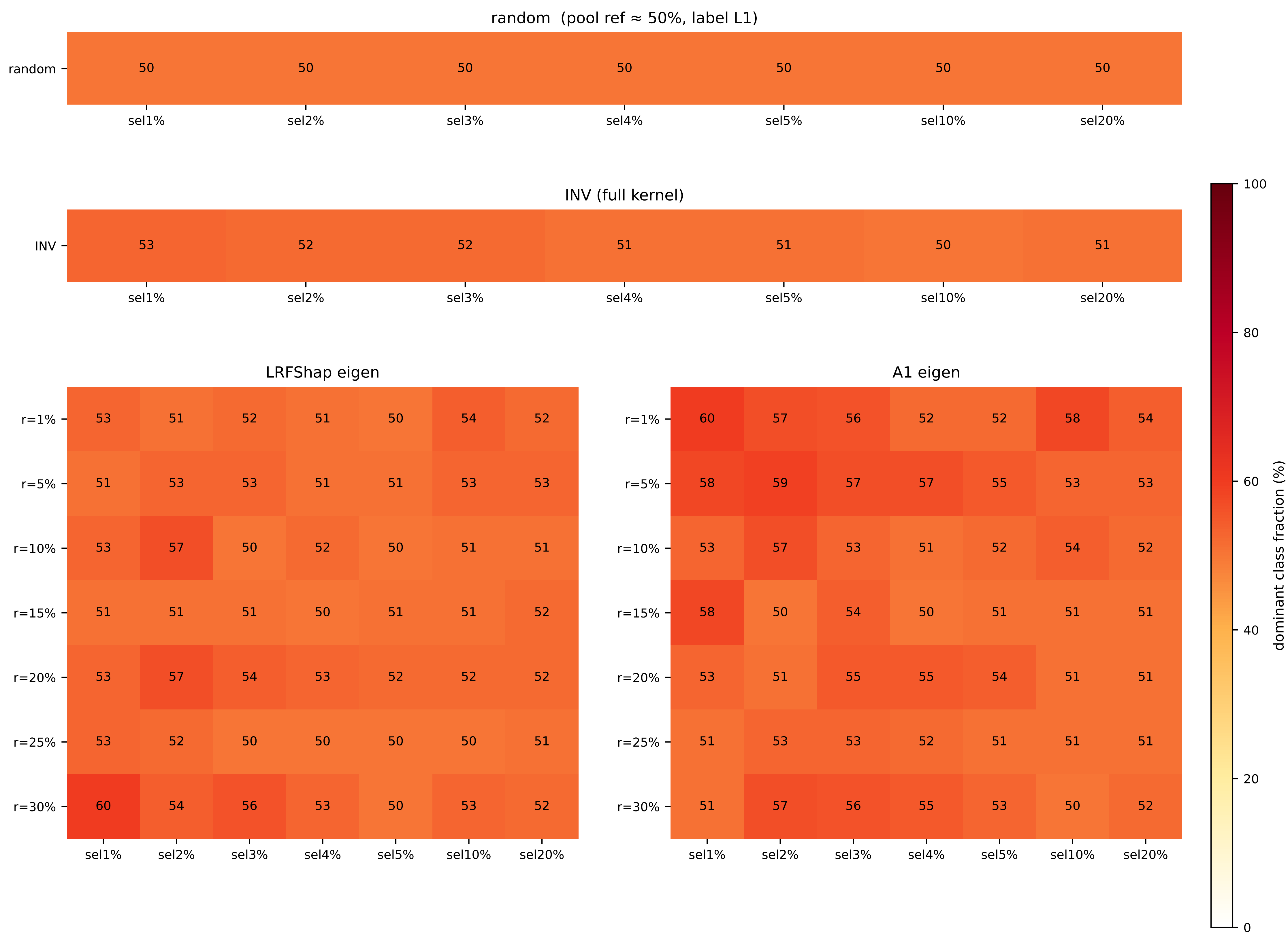



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

mr pos50 valbal — top-Shapley class composition

MR forced 50/50 (n=4500, valbal=1000)

	sel 1%	**sel 2%**	**sel 3%**	**sel 4%**	**sel 5%**	**sel 10%**	**sel 20%**
INV / random	L1=53 / L1=50	L1=52 / L1=50	L0=52 / L1=50	L1=51 / L1=50	L1=51 / L1=50	L1=50 / L1=50	L1=51 / L1=50
LR / A1							
r=1%	L0=53 / L0=60	L0=51 / L0=57	L0=52 / L0=56	L1=51 / L0=52	L1=50 / L0=52	L1=54 / L0=58	L1=52 / L0=54
r=5%	L1=51 / L0=58	L1=53 / L0=59	L1=53 / L0=57	L1=51 / L0=57	L1=51 / L0=55	L1=53 / L0=53	L1=53 / L0=53
r=10%	L0=53 / L0=53	L0=57 / L0=57	L0=50 / L0=53	L0=52 / L1=51	L1=50 / L1=52	L0=51 / L1=54	L1=51 / L1=52
r=15%	L0=51 / L1=58	L0=51 / L0=50	L1=51 / L1=54	L0=50 / L0=50	L0=51 / L1=51	L1=51 / L1=51	L1=52 / L1=51
r=20%	L0=53 / L1=53	L1=57 / L0=51	L1=54 / L1=55	L1=53 / L1=55	L1=52 / L1=54	L1=52 / L1=51	L1=52 / L1=51
r=25%	L0=53 / L0=51	L0=52 / L1=53	L1=50 / L1=53	L1=50 / L1=52	L0=50 / L1=51	L1=50 / L1=51	L1=51 / L1=51
r=30%	L0=60 / L0=51	L0=54 / L1=57	L0=56 / L1=56	L0=53 / L1=55	L1=50 / L1=53	L1=53 / L0=50	L1=52 / L1=52

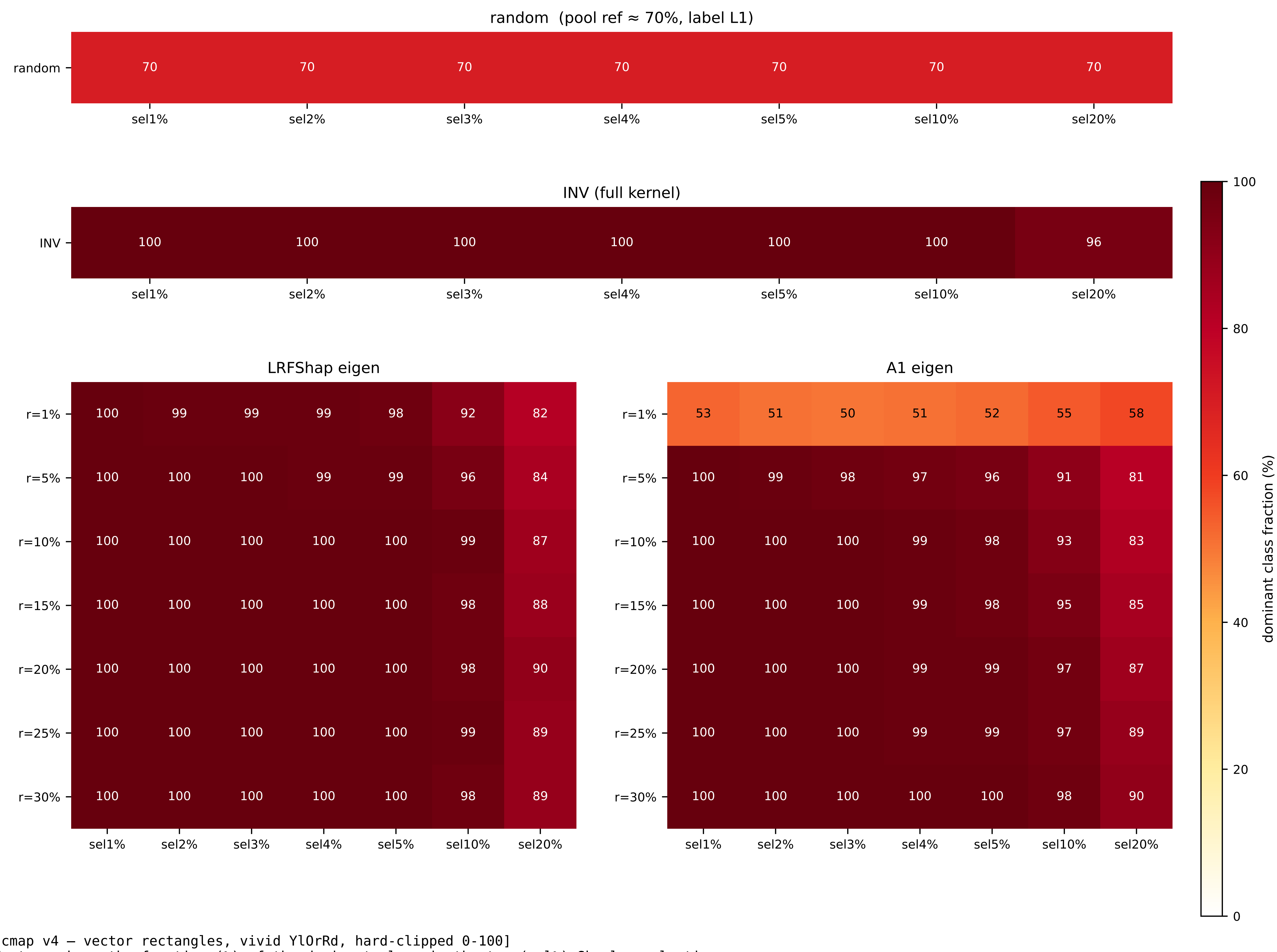


[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.

mr pos70 valbal — top-Shapley class composition

MR forced 70/30 label1-majority (n=4500, valbal=1000)

	sel 1%	**sel 2%**	**sel 3%**	**sel 4%**	**sel 5%**	**sel 10%**	**sel 20%**
INV / random	L0=100 / L1=7	L0=100 / L1=7	L0=100 / L1=7	L0=100 / L1=7	L0=100 / L1=7	L0=100 / L1=7	L0=96 / L1=70
LR / A1							
r=1%	L0=100 / L1=5	L0=99 / L0=51	L0=99 / L1=50	L0=99 / L1=51	L0=98 / L1=52	L0=92 / L1=55	L0=82 / L1=58
r=5%	L0=100 / L0=10	L0=100 / L0=9	L0=100 / L0=98	L0=99 / L0=97	L0=99 / L0=96	L0=96 / L0=91	L0=84 / L0=81
r=10%	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=9	L0=100 / L0=98	L0=99 / L0=93	L0=87 / L0=83
r=15%	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=9	L0=100 / L0=98	L0=98 / L0=95	L0=88 / L0=85
r=20%	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=9	L0=100 / L0=9	L0=98 / L0=97	L0=90 / L0=87
r=25%	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=9	L0=100 / L0=9	L0=99 / L0=97	L0=89 / L0=89
r=30%	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=98 / L0=98	L0=89 / L0=90

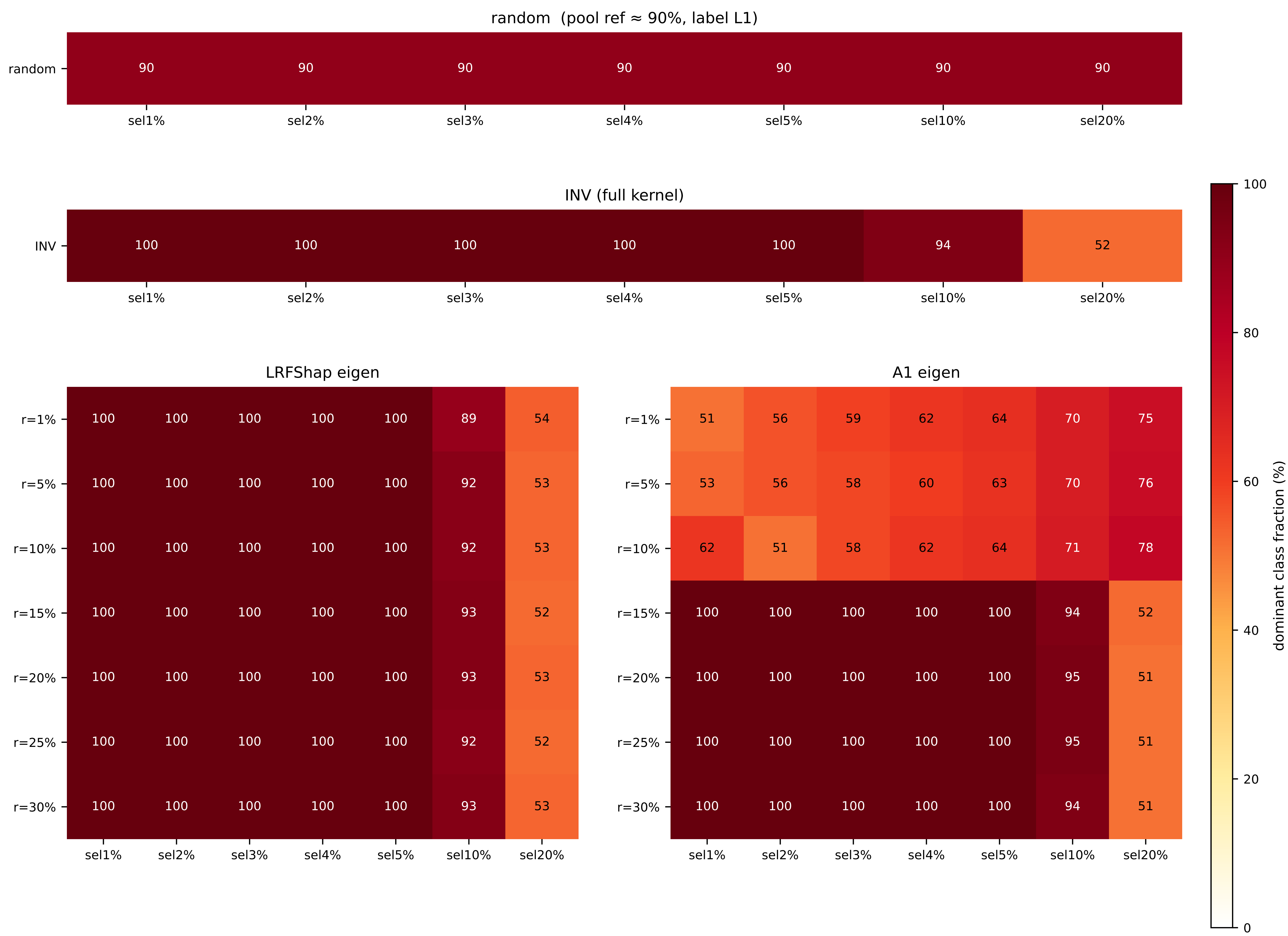



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

mr pos90 valbal — top-Shapley class composition

MR forced 90/10 label1-majority (n=4500, valbal=1000)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L0=100 / L1=90	L0=100 / L1=9	L0=100 / L1=9	L0=100 / L1=9	L0=100 / L1=9	L0=94 / L1=90	L1=52 / L1=90
LR / A1							
r=1%	L0=100 / L1=5	L0=100 / L1=5	L0=100 / L1=5	L0=100 / L1=6	L0=100 / L1=64	L0=89 / L1=70	L1=54 / L1=75
r=5%	L0=100 / L1=5	L0=100 / L1=5	L0=100 / L1=58	L0=100 / L1=60	L0=100 / L1=63	L0=92 / L1=70	L1=53 / L1=76
r=10%	L0=100 / L0=6	L0=100 / L1=5	L0=100 / L1=58	L0=100 / L1=6	L0=100 / L1=64	L0=92 / L1=71	L1=53 / L1=78
r=15%	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=93 / L0=94	L1=52 / L1=52
r=20%	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=93 / L0=95	L1=53 / L1=51
r=25%	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=92 / L0=95	L1=52 / L1=51
r=30%	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=100 / L0=10	L0=93 / L0=94	L1=53 / L1=51

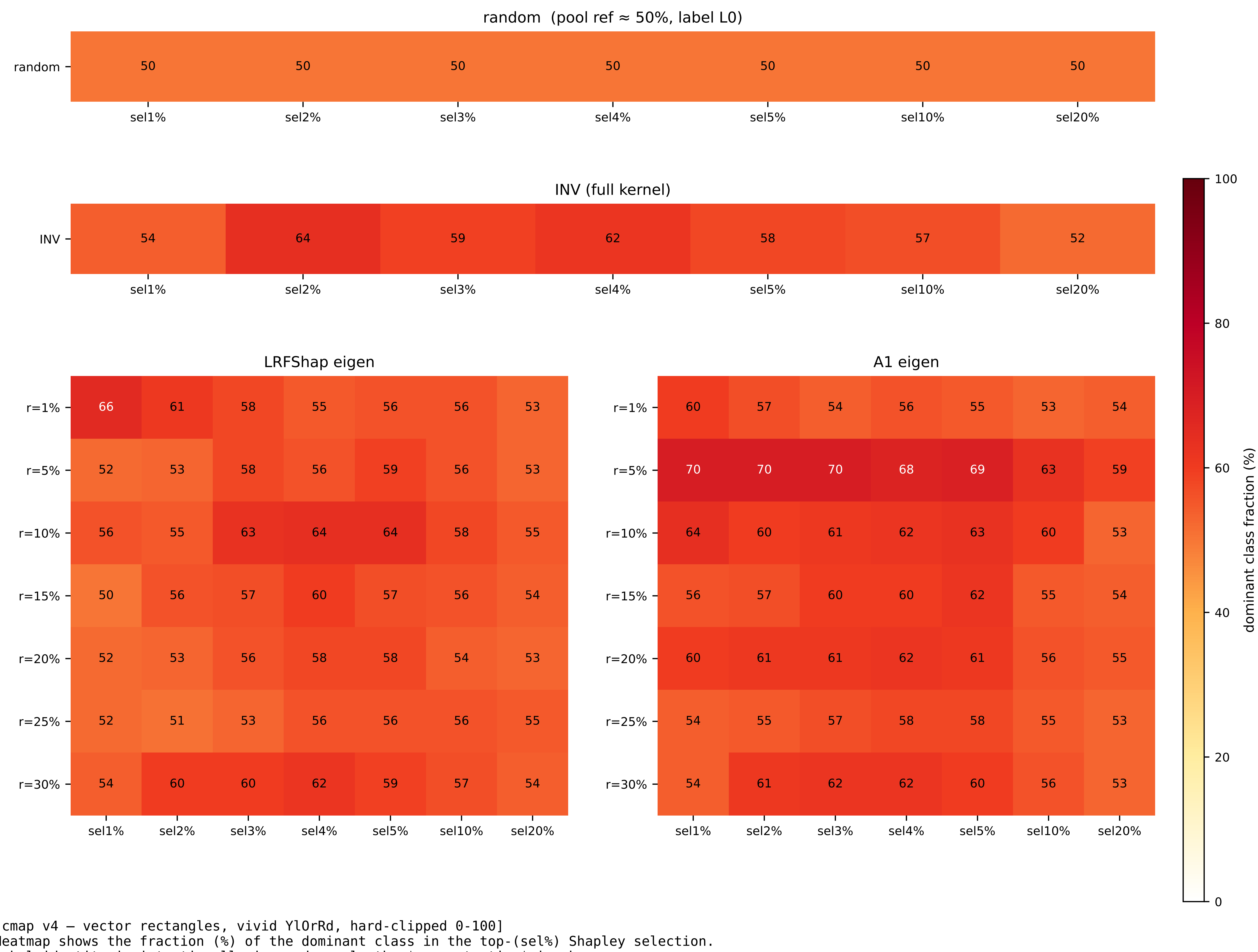



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

qqp pos50 valbal — top-Shapley class composition

QQP forced 50/50 (n=5000, valbal=1000)

	sel 1%	**sel 2%**	**sel 3%**	**sel 4%**	**sel 5%**	**sel 10%**	**sel 20%**
INV / random	L1=54 / L0=50	L1=64 / L0=50	L1=59 / L0=50	L1=62 / L0=50	L1=58 / L0=50	L1=57 / L0=50	L1=52 / L0=50
LR / A1							
r=1%	L1=66 / L1=60	L1=61 / L1=57	L1=58 / L1=54	L1=55 / L1=56	L1=56 / L1=55	L1=56 / L1=53	L1=53 / L1=54
r=5%	L1=52 / L1=70	L1=53 / L1=70	L1=58 / L1=70	L1=56 / L1=68	L1=59 / L1=69	L1=56 / L1=63	L1=53 / L1=59
r=10%	L1=56 / L1=64	L1=55 / L1=60	L1=63 / L1=61	L1=64 / L1=62	L1=64 / L1=63	L1=58 / L1=60	L1=55 / L1=53
r=15%	L0=50 / L1=56	L1=56 / L1=57	L1=57 / L1=60	L1=60 / L1=60	L1=57 / L1=62	L1=56 / L1=55	L1=54 / L1=54
r=20%	L0=52 / L1=60	L1=53 / L1=61	L1=56 / L1=61	L1=58 / L1=62	L1=58 / L1=61	L1=54 / L1=56	L1=53 / L1=55
r=25%	L1=52 / L0=54	L1=51 / L1=55	L1=53 / L1=57	L1=56 / L1=58	L1=56 / L1=58	L1=56 / L1=55	L1=55 / L1=53
r=30%	L1=54 / L1=54	L1=60 / L1=61	L1=60 / L1=62	L1=62 / L1=62	L1=59 / L1=60	L1=57 / L1=56	L1=54 / L1=53

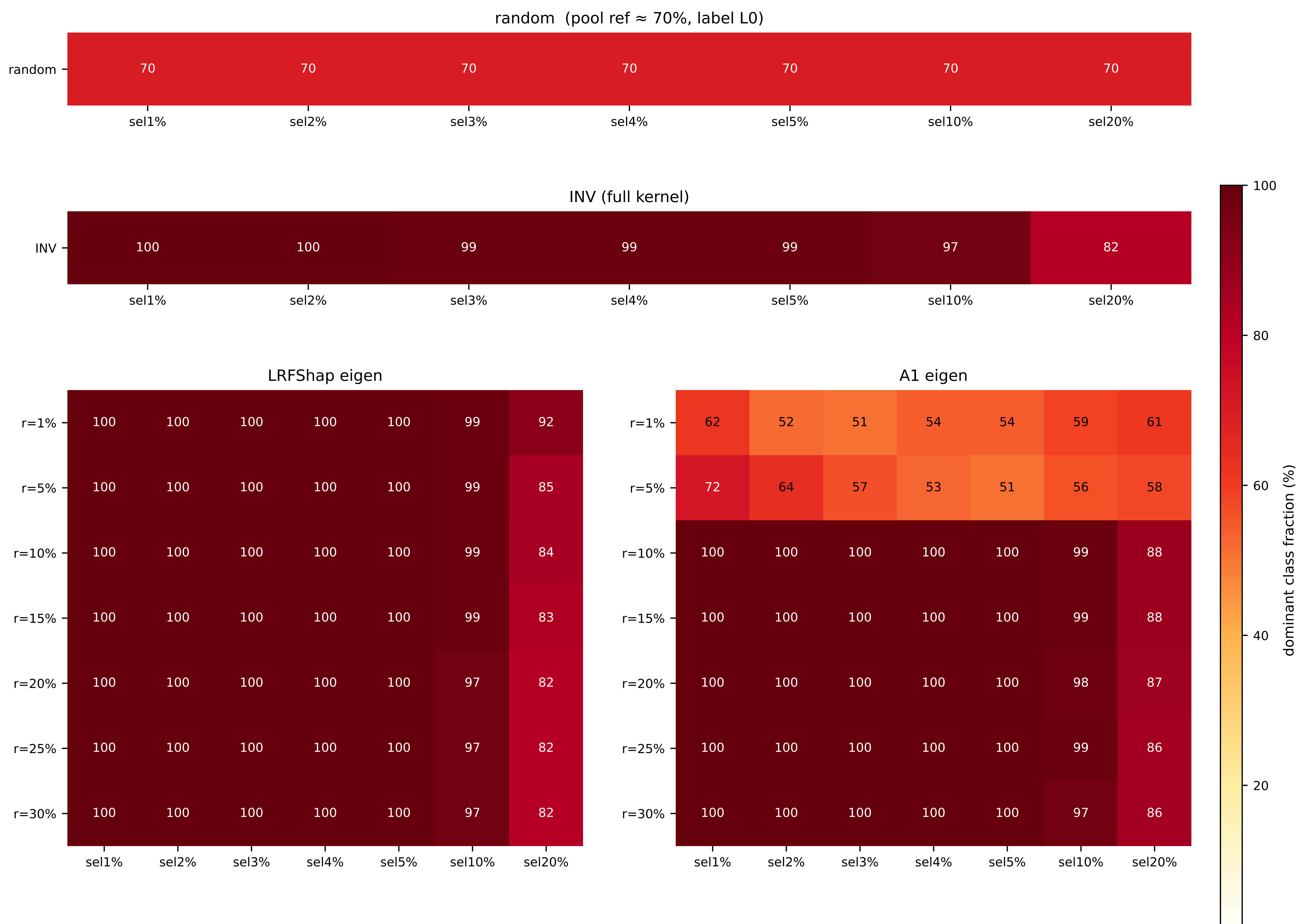



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

qqp pos30 valbal — top-Shapley class composition

QQP forced 70/30 label0-majority (n=5000, valbal=1000)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L1=100 / L0=70	L1=100 / L0=70	L1=99 / L0=70	L1=99 / L0=70	L1=99 / L0=70	L1=97 / L0=70	L1=82 / L0=70
LR / A1							
r=1%	L1=100 / L1=6	L1=100 / L0=5	L1=100 / L0=5	L1=100 / L0=5	L1=100 / L0=54	L1=99 / L0=59	L1=92 / L0=61
r=5%	L1=100 / L1=7	L1=100 / L1=6	L1=100 / L1=5	L1=100 / L1=5	L1=100 / L0=51	L1=99 / L0=56	L1=85 / L0=58
r=10%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=99 / L1=99	L1=84 / L1=88
r=15%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=99 / L1=99	L1=83 / L1=88
r=20%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=97 / L1=98	L1=82 / L1=87
r=25%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=97 / L1=99	L1=82 / L1=86
r=30%	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=100 / L1=10	L1=97 / L1=97	L1=82 / L1=86

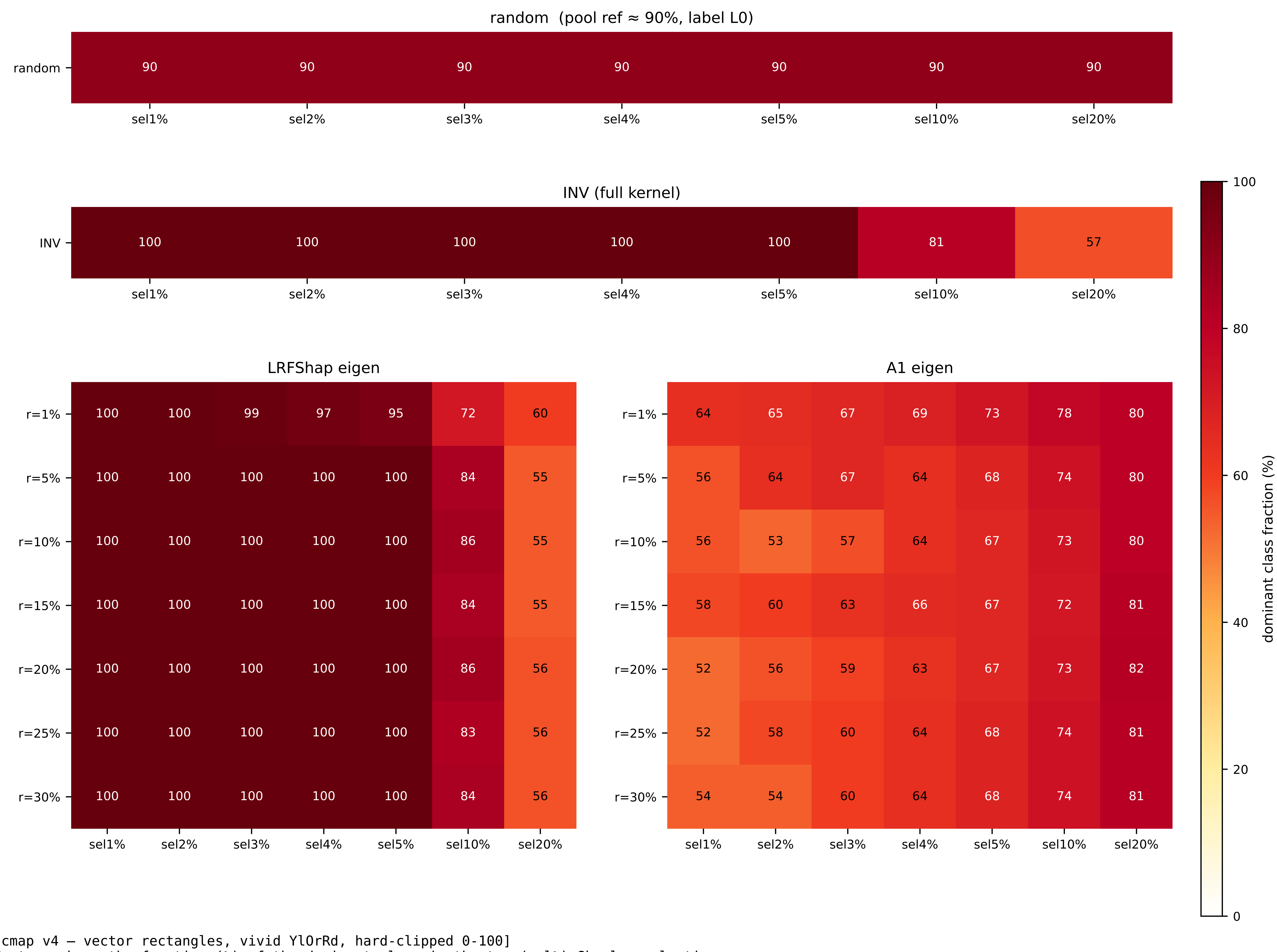



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

qqp pos10 valbal — top-Shapley class composition

QQP forced 90/10 label0-majority (n=5000, valbal=1000)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L1=100 / L0=90	L1=100 / L0=90	L1=100 / L0=90	L1=100 / L0=90	L1=100 / L0=90	L1=81 / L0=90	L0=57 / L0=90
LR / A1							
r=1%	L1=100 / L0=64	L1=100 / L0=65	L1=99 / L0=67	L1=97 / L0=69	L1=95 / L0=73	L1=72 / L0=78	L0=60 / L0=80
r=5%	L1=100 / L0=56	L1=100 / L0=64	L1=100 / L0=67	L1=100 / L0=64	L1=100 / L0=68	L1=84 / L0=74	L0=55 / L0=80
r=10%	L1=100 / L1=56	L1=100 / L0=53	L1=100 / L0=57	L1=100 / L0=64	L1=100 / L0=67	L1=86 / L0=73	L0=55 / L0=80
r=15%	L1=100 / L0=58	L1=100 / L0=60	L1=100 / L0=63	L1=100 / L0=66	L1=100 / L0=67	L1=84 / L0=72	L0=55 / L0=81
r=20%	L1=100 / L0=52	L1=100 / L0=56	L1=100 / L0=59	L1=100 / L0=63	L1=100 / L0=67	L1=86 / L0=73	L0=56 / L0=82
r=25%	L1=100 / L1=52	L1=100 / L0=58	L1=100 / L0=60	L1=100 / L0=64	L1=100 / L0=68	L1=83 / L0=74	L0=56 / L0=81
r=30%	L1=100 / L1=54	L1=100 / L0=54	L1=100 / L0=60	L1=100 / L0=64	L1=100 / L0=68	L1=84 / L0=74	L0=56 / L0=81

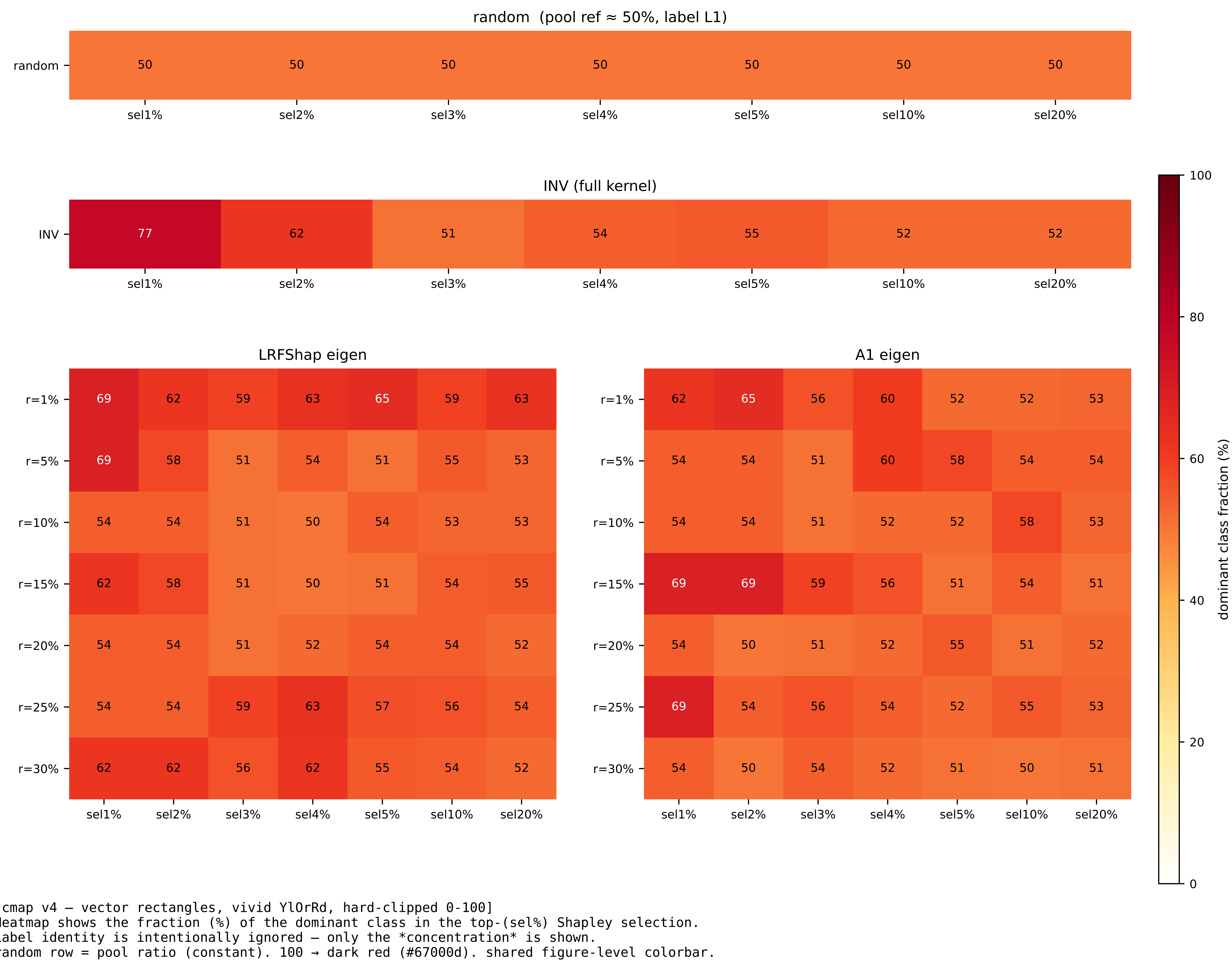



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

rte pos50 valbal — top-Shapley class composition

RTE forced 50/50 (n=1300, valbal=262)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L0=77 / L1=50	L0=62 / L1=50	L1=51 / L1=50	L0=54 / L1=50	L0=55 / L1=50	L0=52 / L1=50	L0=52 / L1=50
LR / A1							
r=1%	L0=69 / L0=62	L0=62 / L0=65	L0=59 / L0=56	L0=63 / L0=60	L0=65 / L0=52	L0=59 / L1=52	L0=63 / L1=53
r=5%	L1=69 / L0=54	L1=58 / L0=54	L0=51 / L1=51	L1=54 / L1=60	L1=51 / L1=58	L0=55 / L1=54	L0=53 / L1=54
r=10%	L0=54 / L0=54	L0=54 / L0=54	L0=51 / L0=51	L0=50 / L1=52	L1=54 / L0=52	L0=53 / L1=58	L0=53 / L1=53
r=15%	L0=62 / L1=69	L0=58 / L1=69	L0=51 / L1=59	L0=50 / L1=56	L1=51 / L0=51	L0=54 / L1=54	L0=55 / L1=51
r=20%	L0=54 / L1=54	L0=54 / L1=50	L1=51 / L0=51	L0=52 / L1=52	L0=54 / L1=55	L0=54 / L1=51	L0=52 / L1=52
r=25%	L1=54 / L1=69	L0=54 / L1=54	L0=59 / L1=56	L0=63 / L1=54	L0=57 / L1=52	L0=56 / L1=55	L0=54 / L1=53
r=30%	L0=62 / L0=54	L0=62 / L0=50	L0=56 / L1=54	L0=62 / L1=52	L0=55 / L1=51	L0=54 / L0=50	L0=52 / L1=51

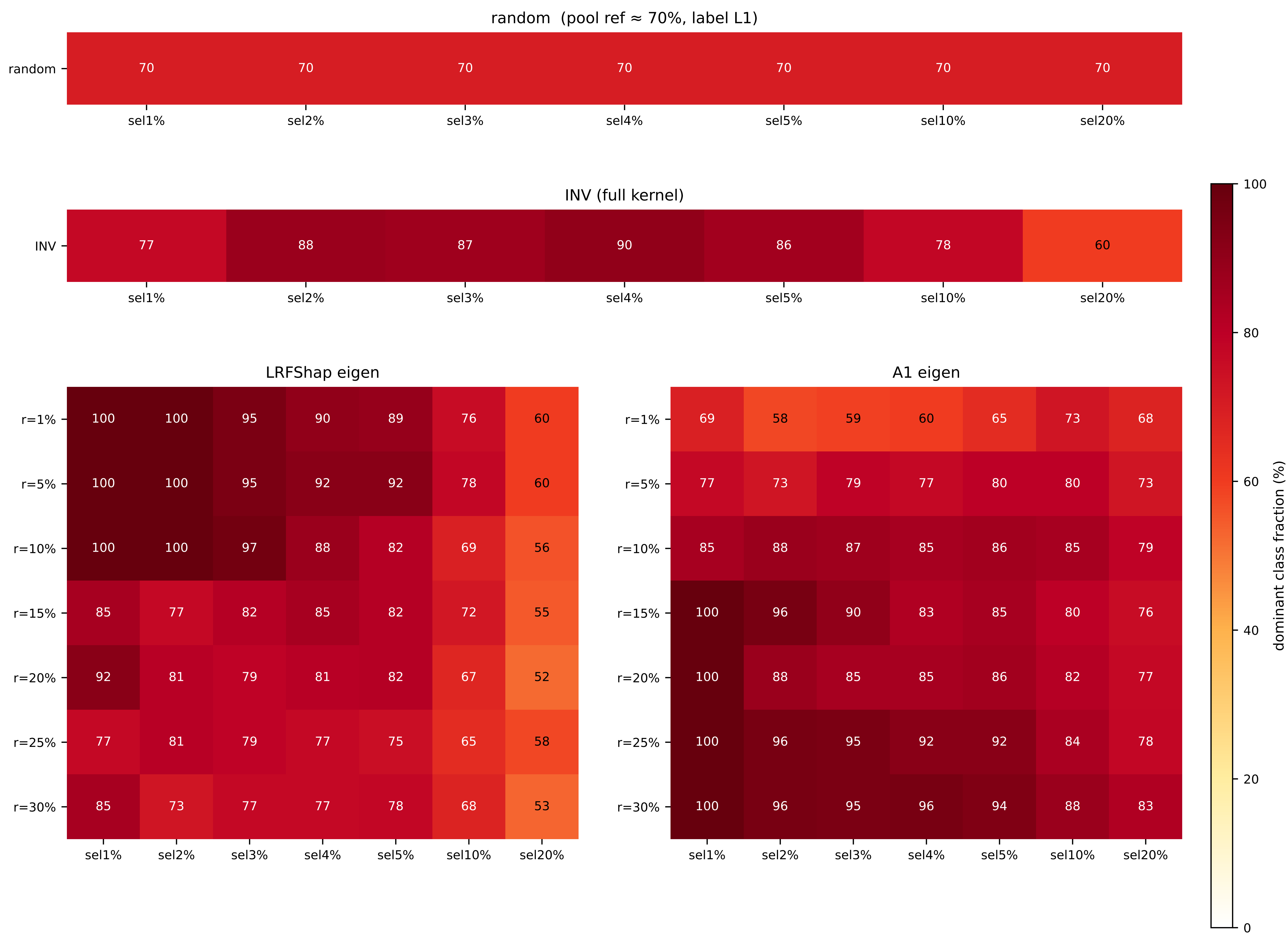


[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.

rte pos70 valbal — top-Shapley class composition

RTE forced 70/30 label1-majority (n=1300, valbal=262)

	sel 1%	**sel 2%**	**sel 3%**	**sel 4%**	**sel 5%**	**sel 10%**	**sel 20%**
INV / random	L0=77 / L1=70	L0=88 / L1=70	L0=87 / L1=70	L0=90 / L1=70	L0=86 / L1=70	L0=78 / L1=70	L0=60 / L1=70
LR / A1							
r=1%	L0=100 / L1=69	L0=100 / L1=58	L0=95 / L1=59	L0=90 / L1=60	L0=89 / L1=65	L0=76 / L1=73	L0=60 / L1=68
r=5%	L0=100 / L1=77	L0=100 / L1=73	L0=95 / L1=79	L0=92 / L1=77	L0=92 / L1=80	L0=78 / L1=80	L0=60 / L1=73
r=10%	L0=100 / L1=85	L0=100 / L1=88	L0=97 / L1=87	L0=88 / L1=85	L0=82 / L1=86	L0=69 / L1=85	L0=56 / L1=79
r=15%	L0=85 / L1=100	L0=77 / L1=96	L0=82 / L1=90	L0=85 / L1=83	L0=82 / L1=85	L0=72 / L1=80	L0=55 / L1=76
r=20%	L0=92 / L1=100	L0=81 / L1=88	L0=79 / L1=85	L0=81 / L1=85	L0=82 / L1=86	L0=67 / L1=82	L0=52 / L1=77
r=25%	L0=77 / L1=100	L0=81 / L1=96	L0=79 / L1=95	L0=77 / L1=92	L0=75 / L1=92	L0=65 / L1=84	L0=58 / L1=78
r=30%	L0=85 / L1=100	L0=73 / L1=96	L0=77 / L1=95	L0=77 / L1=96	L0=78 / L1=94	L0=68 / L1=88	L0=53 / L1=83

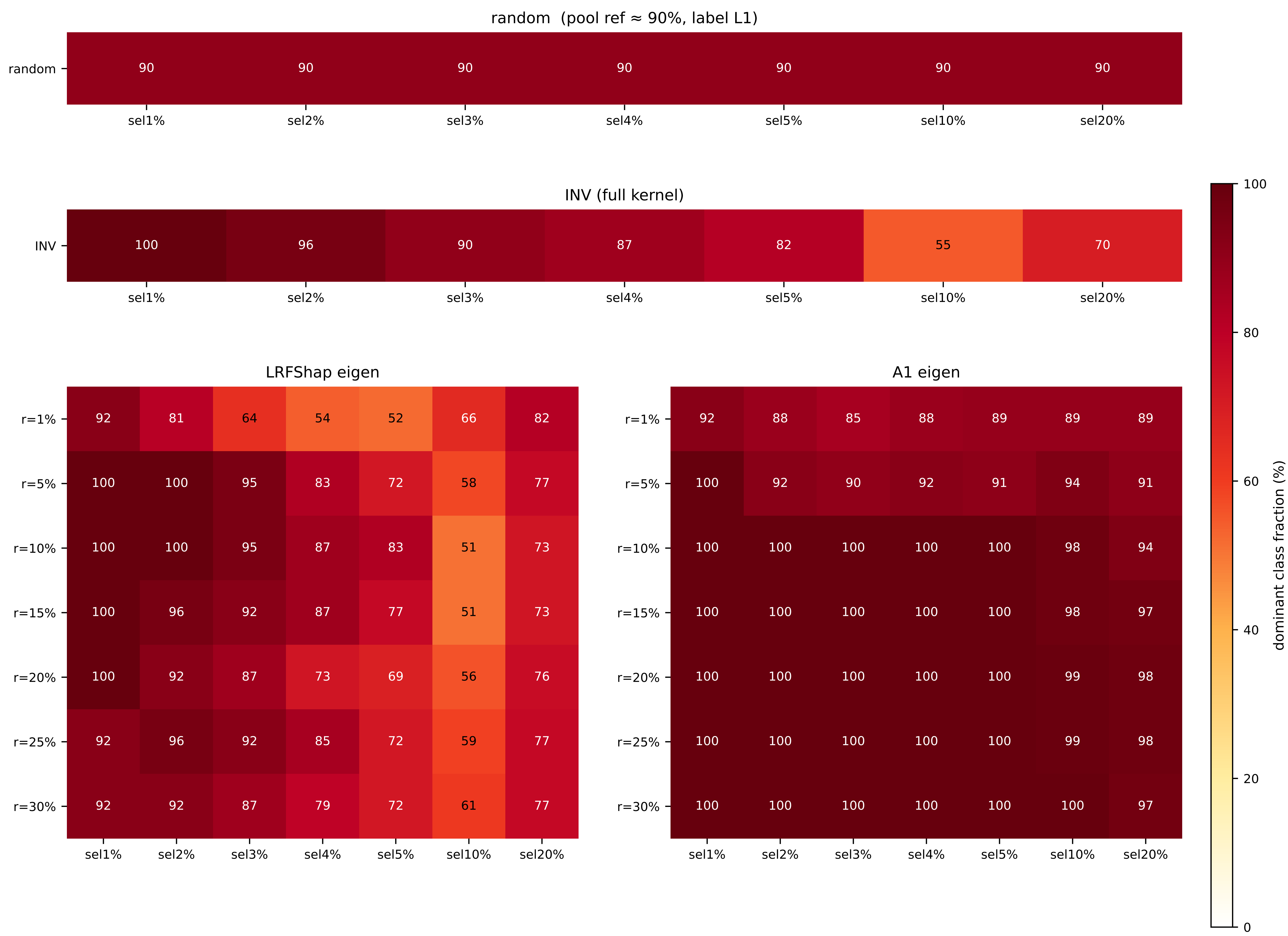



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

rte pos90 valbal — top-Shapley class composition

RTE forced 90/10 label1-majority (n=1300, valbal=262)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L0=100 / L1=90	L0=96 / L1=90	L0=90 / L1=90	L0=87 / L1=90	L0=82 / L1=90	L0=55 / L1=90	L1=70 / L1=90
LR / A1							
r=1%	L0=92 / L1=92	L0=81 / L1=88	L0=64 / L1=85	L0=54 / L1=88	L1=52 / L1=89	L1=66 / L1=89	L1=82 / L1=89
r=5%	L0=100 / L1=10	L0=100 / L1=92	L0=95 / L1=90	L0=83 / L1=92	L0=72 / L1=91	L1=58 / L1=94	L1=77 / L1=91
r=10%	L0=100 / L1=10	L0=100 / L1=10	L0=95 / L1=100	L0=87 / L1=100	L0=83 / L1=100	L0=51 / L1=98	L1=73 / L1=94
r=15%	L0=100 / L1=10	L0=96 / L1=100	L0=92 / L1=100	L0=87 / L1=100	L0=77 / L1=100	L1=51 / L1=98	L1=73 / L1=97
r=20%	L0=100 / L1=10	L0=92 / L1=100	L0=87 / L1=100	L0=73 / L1=100	L0=69 / L1=100	L1=56 / L1=99	L1=76 / L1=98
r=25%	L0=92 / L1=100	L0=96 / L1=100	L0=92 / L1=100	L0=85 / L1=100	L0=72 / L1=100	L1=59 / L1=99	L1=77 / L1=98
r=30%	L0=92 / L1=100	L0=92 / L1=100	L0=87 / L1=100	L0=79 / L1=100	L0=72 / L1=100	L1=61 / L1=100	L1=77 / L1=97

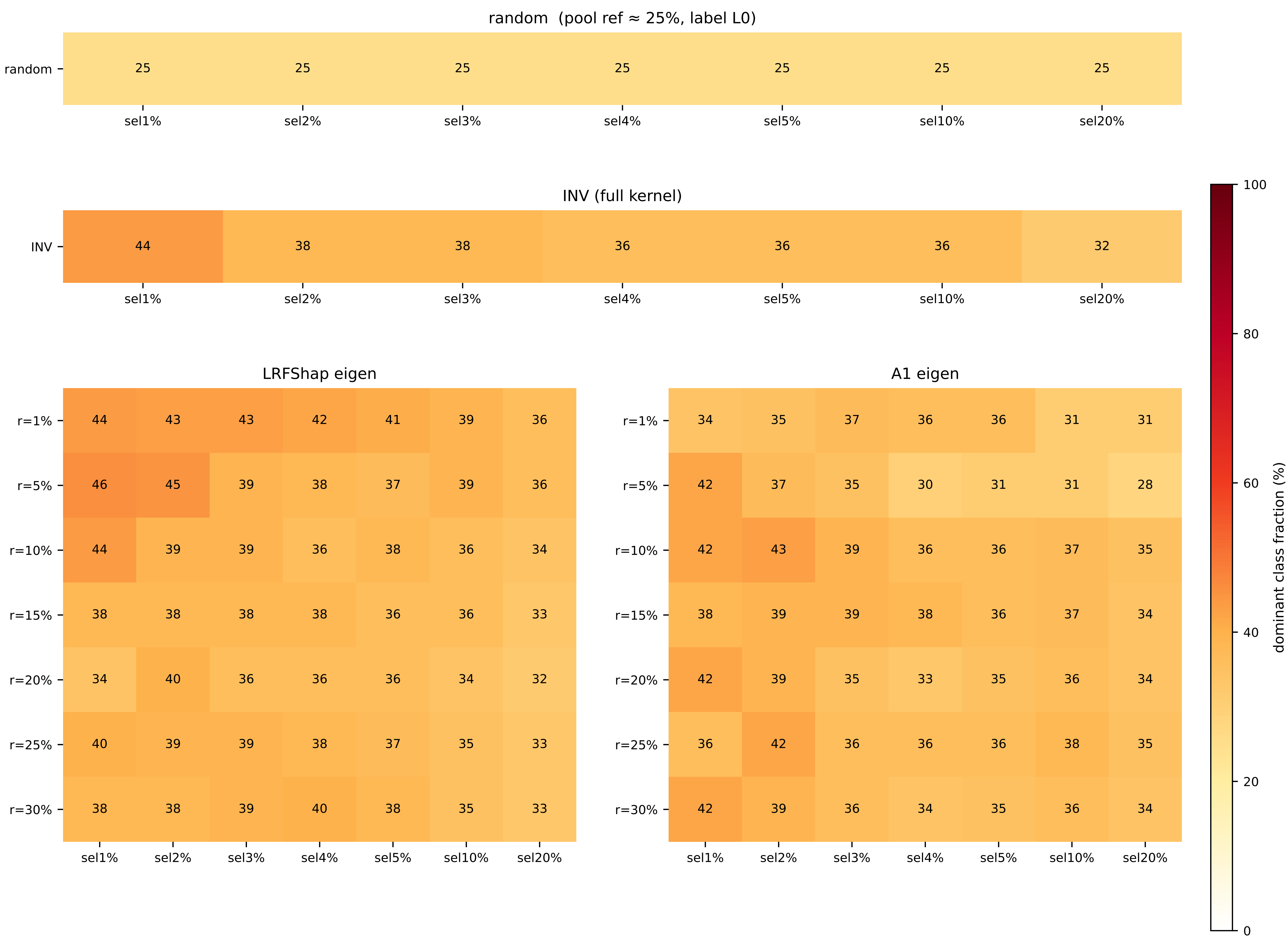



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

ag_news cls25_25_25_25 valbal — top-Shapley class composition

AG News balanced 25/25/25/25 (n=5000, valbal=1000)

	sel 1%	**sel 2%**	**sel 3%**	**sel 4%**	**sel 5%**	**sel 10%**	**sel 20%**
INV / random	L1=44 / L0=25	L1=38 / L0=25	L1=38 / L0=25	L1=36 / L0=25	L1=36 / L0=25	L1=36 / L0=25	L1=32 / L0=25
LR / A1							
r=1%	L1=44 / L0=34	L1=43 / L0=35	L1=43 / L0=37	L1=42 / L0=36	L1=41 / L0=36	L1=39 / L1=31	L1=36 / L1=31
r=5%	L1=46 / L0=42	L1=45 / L0=37	L1=39 / L0=35	L1=38 / L0=30	L1=37 / L0=31	L1=39 / L0=31	L1=36 / L1=28
r=10%	L1=44 / L1=42	L1=39 / L1=43	L1=39 / L1=39	L1=36 / L1=36	L1=38 / L1=36	L1=36 / L1=37	L1=34 / L1=35
r=15%	L1=38 / L1=38	L1=38 / L1=39	L1=38 / L1=39	L1=38 / L1=38	L1=36 / L1=36	L1=36 / L1=37	L1=33 / L1=34
r=20%	L1=34 / L1=42	L1=40 / L1=39	L1=36 / L1=35	L1=36 / L1=33	L1=36 / L1=35	L1=34 / L1=36	L1=32 / L1=34
r=25%	L1=40 / L1=36	L1=39 / L1=42	L1=39 / L1=36	L1=38 / L1=36	L1=37 / L1=36	L1=35 / L1=38	L1=33 / L1=35
r=30%	L1=38 / L1=42	L1=38 / L1=39	L1=39 / L1=36	L1=40 / L1=34	L1=38 / L1=35	L1=35 / L1=36	L1=33 / L1=34

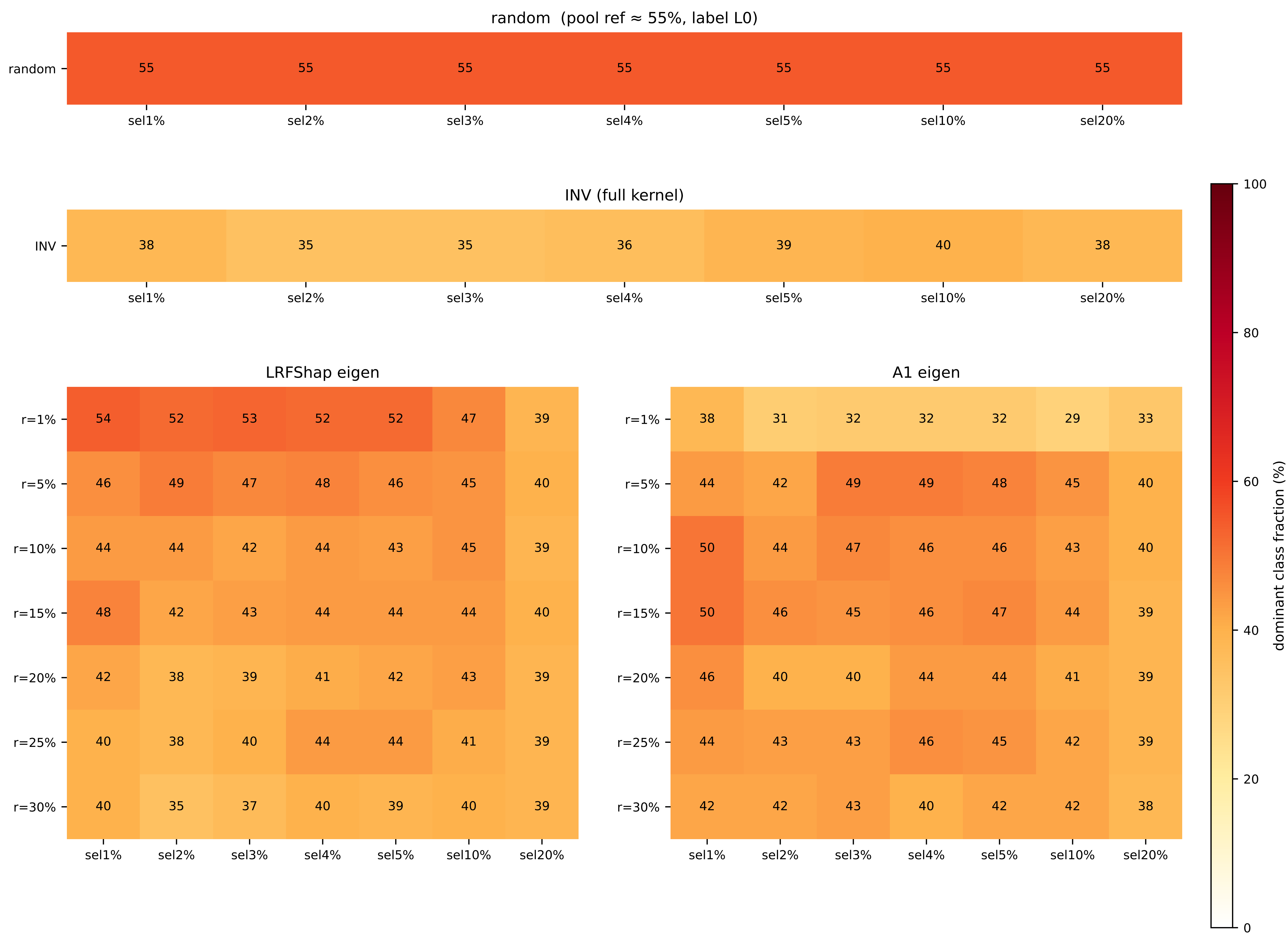



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

ag_news cls55_15_15_15 valbal — top-Shapley class composition

AG News forced 55/15/15/15 label0-majority (n=5000, valbal=1000)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L1=38 / L0=55	L1=35 / L0=55	L1=35 / L0=55	L1=36 / L0=55	L1=39 / L0=55	L1=40 / L0=55	L1=38 / L0=55
LR / A1							
r=1%	L1=54 / L1=38	L1=52 / L1=31	L1=53 / L1=32	L1=52 / L1=32	L1=52 / L1=32	L1=47 / L1=29	L1=39 / L0=33
r=5%	L1=46 / L1=44	L1=49 / L1=42	L1=47 / L1=49	L1=48 / L1=49	L1=46 / L1=48	L1=45 / L1=45	L1=40 / L1=40
r=10%	L1=44 / L1=50	L1=44 / L1=44	L1=42 / L1=47	L1=44 / L1=46	L1=43 / L1=46	L1=45 / L1=43	L1=39 / L1=40
r=15%	L1=48 / L1=50	L1=42 / L1=46	L1=43 / L1=45	L1=44 / L1=46	L1=44 / L1=47	L1=44 / L1=44	L1=40 / L1=39
r=20%	L1=42 / L1=46	L1=38 / L1=40	L1=39 / L1=40	L1=41 / L1=44	L1=42 / L1=44	L1=43 / L1=41	L1=39 / L1=39
r=25%	L1=40 / L1=44	L2=38 / L1=43	L1=40 / L1=43	L1=44 / L1=46	L1=44 / L1=45	L1=41 / L1=42	L1=39 / L1=39
r=30%	L2=40 / L1=42	L2=35 / L1=42	L1=37 / L1=43	L1=40 / L1=40	L1=39 / L1=42	L1=40 / L1=42	L1=39 / L1=38

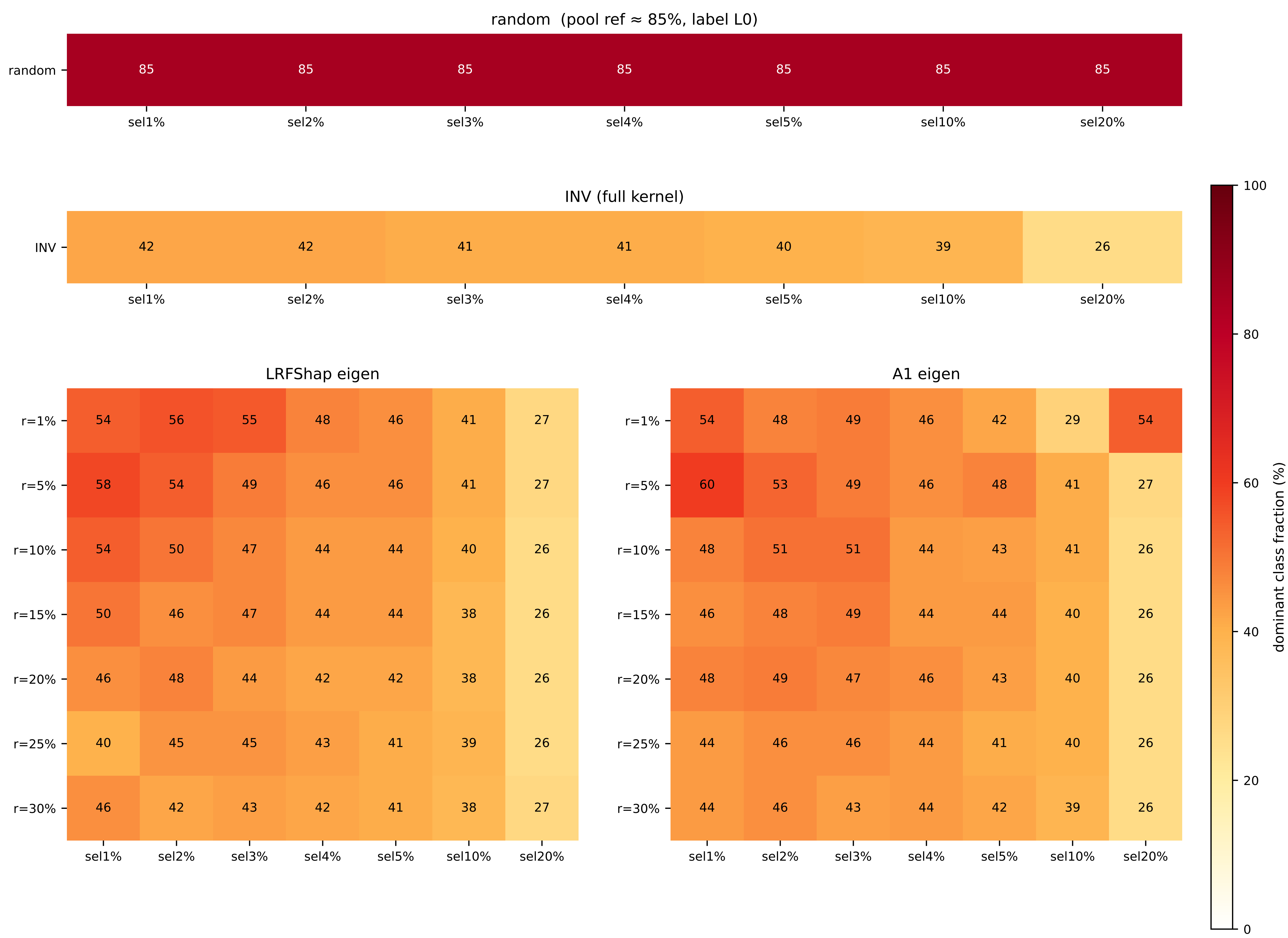



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

ag_news cls85_05_05_05 valbal — top-Shapley class composition

AG News forced 85/5/5/5 label0-majority (n=5000, valbal=1000)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L1=42 / L0=85	L1=42 / L0=85	L1=41 / L0=85	L2=41 / L0=85	L2=40 / L0=85	L1=39 / L0=85	L0=26 / L0=85
LR / A1							
r=1%	L1=54 / L2=54	L1=56 / L2=48	L1=55 / L2=49	L1=48 / L2=46	L1=46 / L2=42	L1=41 / L2=29	L0=27 / L0=54
r=5%	L1=58 / L1=60	L1=54 / L1=53	L1=49 / L1=49	L1=46 / L1=46	L1=46 / L1=48	L1=41 / L1=41	L0=27 / L0=27
r=10%	L1=54 / L1=48	L1=50 / L1=51	L1=47 / L1=51	L1=44 / L1=44	L1=44 / L1=43	L1=40 / L1=41	L0=26 / L0=26
r=15%	L1=50 / L1=46	L1=46 / L1=48	L1=47 / L1=49	L1=44 / L1=44	L1=44 / L1=44	L1=38 / L1=40	L0=26 / L0=26
r=20%	L1=46 / L1=48	L1=48 / L1=49	L1=44 / L1=47	L1=42 / L1=46	L1=42 / L1=43	L1=38 / L1=40	L0=26 / L0=26
r=25%	L1=40 / L1=44	L1=45 / L1=46	L1=45 / L1=46	L1=43 / L1=44	L1=41 / L1=41	L1=39 / L1=40	L0=26 / L0=26
r=30%	L1=46 / L2=44	L1=42 / L1=46	L1=43 / L1=43	L1=42 / L1=44	L1=41 / L1=42	L1=38 / L1=39	L0=27 / L0=26

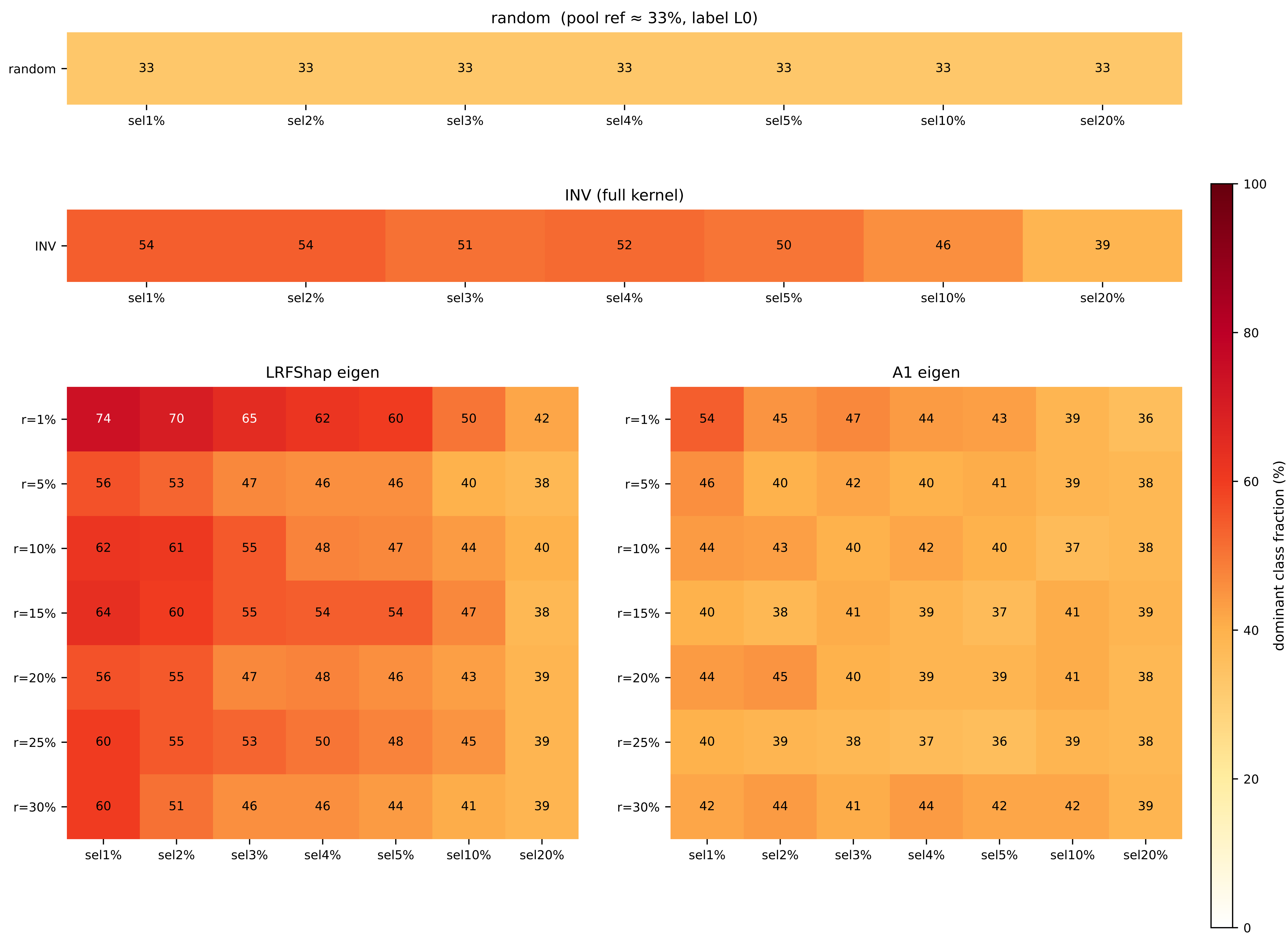



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

mnli cls33_33_33 valbal — top-Shapley class composition

MNLI balanced 33/33/33 (n=5000, valbal=1000)

	sel 1%	**sel 2%**	**sel 3%**	**sel 4%**	**sel 5%**	**sel 10%**	**sel 20%**
INV / random	L0=54 / L0=33	L0=54 / L0=33	L0=51 / L0=33	L0=52 / L0=33	L0=50 / L0=33	L0=46 / L0=33	L0=39 / L0=33
LR / A1							
r=1%	L0=74 / L0=54	L0=70 / L0=45	L0=65 / L0=47	L0=62 / L0=44	L0=60 / L0=43	L0=50 / L0=39	L0=42 / L2=36
r=5%	L0=56 / L2=46	L0=53 / L2=40	L0=47 / L2=42	L0=46 / L2=40	L0=46 / L2=41	L0=40 / L2=39	L0=38 / L2=38
r=10%	L0=62 / L0=44	L0=61 / L0=43	L0=55 / L0=40	L0=48 / L0=42	L0=47 / L0=40	L0=44 / L2=37	L0=40 / L2=38
r=15%	L0=64 / L0=40	L0=60 / L0=38	L0=55 / L0=41	L0=54 / L0=39	L0=54 / L0=37	L0=47 / L2=41	L0=38 / L2=39
r=20%	L0=56 / L0=44	L0=55 / L0=45	L0=47 / L0=40	L0=48 / L2=39	L0=46 / L2=39	L0=43 / L2=41	L0=39 / L2=38
r=25%	L0=60 / L0=40	L0=55 / L2=39	L0=53 / L2=38	L0=50 / L2=37	L0=48 / L2=36	L0=45 / L2=39	L0=39 / L2=38
r=30%	L0=60 / L2=42	L0=51 / L2=44	L0=46 / L2=41	L0=46 / L2=44	L0=44 / L2=42	L0=41 / L2=42	L0=39 / L2=39

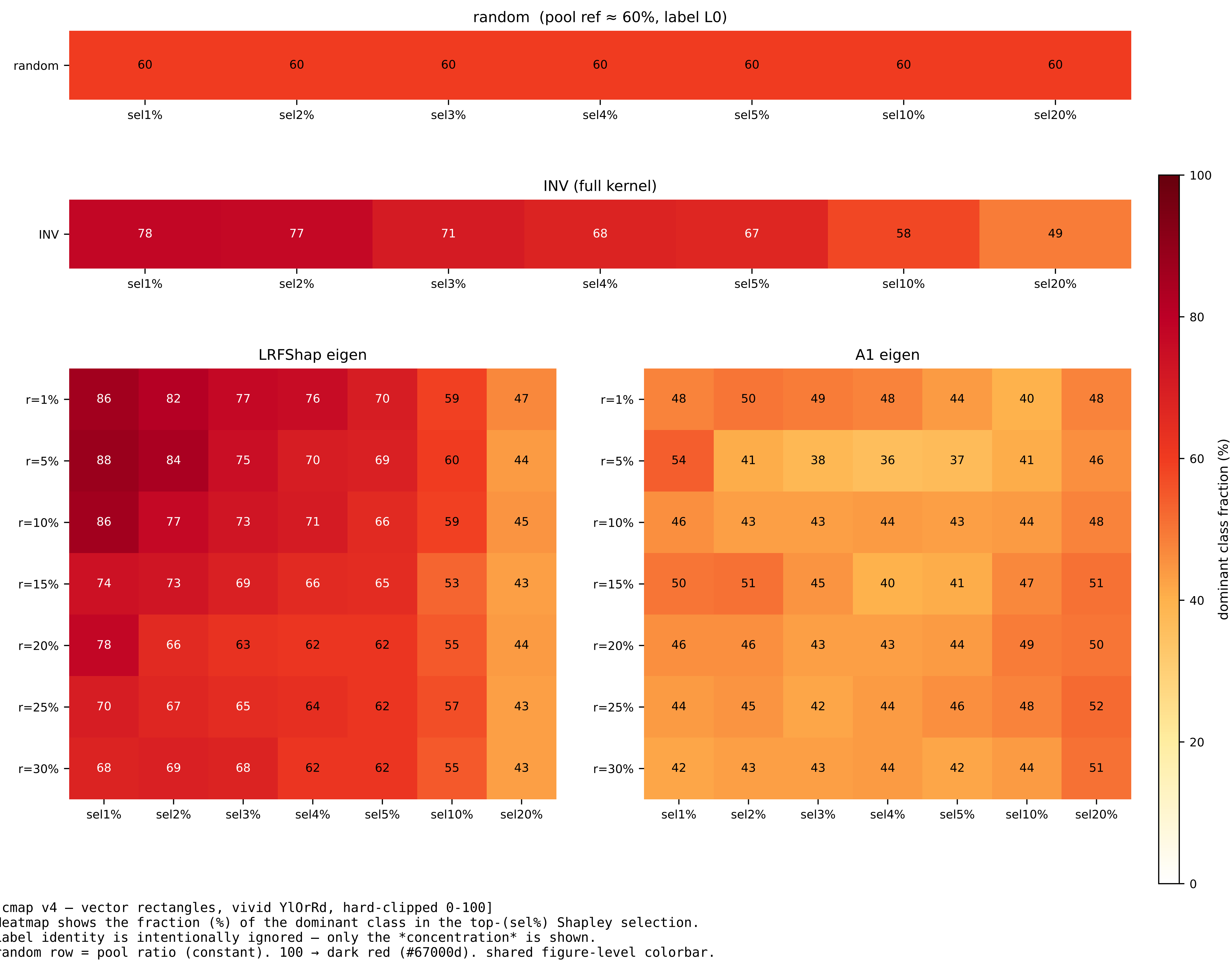



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

mnli cls60_20_20 valbal — top-Shapley class composition

MNLI forced 60/20/20 label0-majority (n=5000, valbal=1000)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L2=78 / L0=60	L2=77 / L0=60	L2=71 / L0=60	L2=68 / L0=60	L2=67 / L0=60	L2=58 / L0=60	L2=49 / L0=60
LR / A1							
r=1%	L2=86 / L2=48	L2=82 / L2=50	L2=77 / L2=49	L2=76 / L2=48	L2=70 / L2=44	L2=59 / L0=40	L2=47 / L0=48
r=5%	L2=88 / L2=54	L2=84 / L2=41	L2=75 / L2=38	L2=70 / L0=36	L2=69 / L0=37	L2=60 / L0=41	L2=44 / L0=46
r=10%	L2=86 / L2=46	L2=77 / L2=43	L2=73 / L2=43	L2=71 / L2=44	L2=66 / L2=43	L2=59 / L0=44	L2=45 / L0=48
r=15%	L2=74 / L2=50	L2=73 / L2=51	L2=69 / L2=45	L2=66 / L2=40	L2=65 / L0=41	L2=53 / L0=47	L2=43 / L0=51
r=20%	L2=78 / L0=46	L2=66 / L2=46	L2=63 / L2=43	L2=62 / L0=43	L2=62 / L0=44	L2=55 / L0=49	L2=44 / L0=50
r=25%	L2=70 / L2=44	L2=67 / L0=45	L2=65 / L0=42	L2=64 / L0=44	L2=62 / L0=46	L2=57 / L0=48	L2=43 / L0=52
r=30%	L2=68 / L0=42	L2=69 / L2=43	L2=68 / L2=43	L2=62 / L2=44	L2=62 / L2=42	L2=55 / L0=44	L2=43 / L0=51

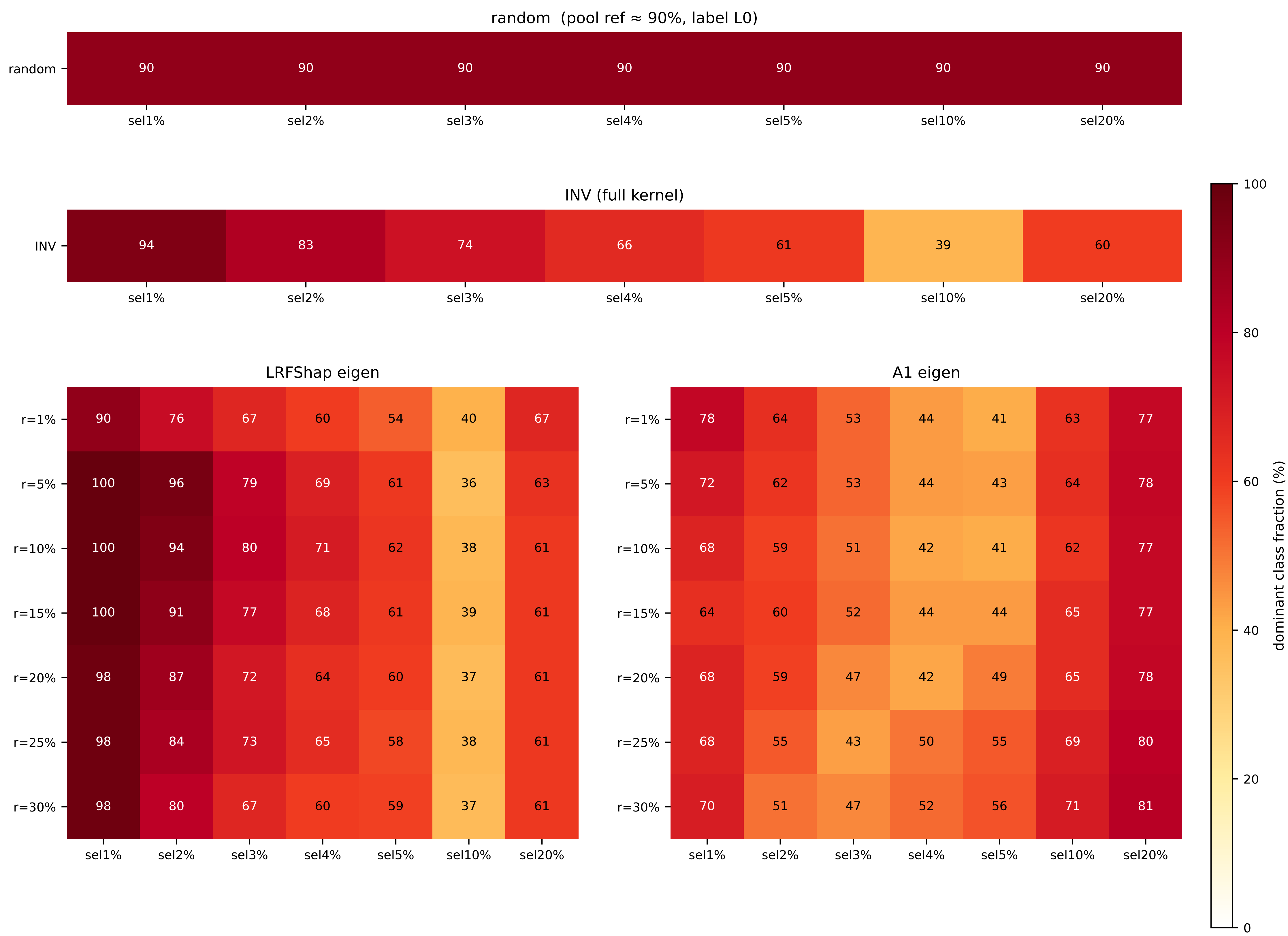



```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

mnli cls90_05_05 valbal — top-Shapley class composition

MNLI forced 90/5/5 label0-majority (n=5000, valbal=1000)

	sel 1%	sel 2%	sel 3%	sel 4%	sel 5%	sel 10%	sel 20%
INV / random	L2=94 / L0=90	L2=83 / L0=90	L2=74 / L0=90	L2=66 / L0=90	L2=61 / L0=90	L2=39 / L0=90	L0=60 / L0=90
LR / A1							
r=1%	L2=90 / L2=78	L2=76 / L2=64	L2=67 / L2=53	L2=60 / L2=44	L2=54 / L0=41	L0=40 / L0=63	L0=67 / L0=77
r=5%	L2=100 / L2=72	L2=96 / L2=62	L2=79 / L2=53	L2=69 / L2=44	L2=61 / L0=43	L2=36 / L0=64	L0=63 / L0=78
r=10%	L2=100 / L2=68	L2=94 / L2=59	L2=80 / L2=51	L2=71 / L2=42	L2=62 / L0=41	L2=38 / L0=62	L0=61 / L0=77
r=15%	L2=100 / L2=64	L2=91 / L2=60	L2=77 / L2=52	L2=68 / L2=44	L2=61 / L0=44	L2=39 / L0=65	L0=61 / L0=77
r=20%	L2=98 / L2=68	L2=87 / L2=59	L2=72 / L2=47	L2=64 / L0=42	L2=60 / L0=49	L2=37 / L0=65	L0=61 / L0=78
r=25%	L2=98 / L2=68	L2=84 / L2=55	L2=73 / L2=43	L2=65 / L0=50	L2=58 / L0=55	L2=38 / L0=69	L0=61 / L0=80
r=30%	L2=98 / L2=70	L2=80 / L2=51	L2=67 / L0=47	L2=60 / L0=52	L2=59 / L0=56	L2=37 / L0=71	L0=61 / L0=81




```
[cmap v4 — vector rectangles, vivid YlOrRd, hard-clipped 0-100]
Heatmap shows the fraction (%) of the dominant class in the top-(sel%) Shapley selection.
Label identity is intentionally ignored — only the *concentration* is shown.
random row = pool ratio (constant). 100 → dark red (#67000d). shared figure-level colorbar.
```

Shapley wall-clock time and speedup (= INV time / method time)

setting (node, GPU)	INV	LR r=1%	LR r=5%	LR r=10%	LR r=15%	LR r=20%	LR r=25%	LR r=30%	A1 r=1%	A1 r=5%	A1 r=10%	A1 r=15%	A1 r=20%	A1 r=25%	A1 r=30%
mrpc/pos50 (node01, RTX 2080 Ti)	3.2 h 1.0x	5.2 min 36.1x	13.9 min 13.6x	19.8 min 9.6x	21.2 min 9.0x	23.4 min 8.1x	24.8 min 7.7x	27.1 min 7.0x	5.8 min 32.5x	8.1 min 23.5x	8.8 min 21.4x	10.7 min 17.8x	9.9 min 19.2x	10.1 min 18.7x	13.7 min 13.9x
mrpc/pos70 (node01, RTX 2080 Ti)	4.0 h 1.0x	8.6 min 28.0x	14.9 min 16.2x	21.1 min 11.4x	20.6 min 11.7x	24.0 min 10.0x	27.2 min 8.9x	32.2 min 7.5x	6.7 min 35.8x	9.6 min 25.1x	11.7 min 20.6x	13.9 min 17.4x	16.6 min 14.5x	15.8 min 15.3x	18.5 min 13.1x
mrpc/pos90 (node01, RTX 2080 Ti)	3.3 h 1.0x	5.7 min 35.1x	20.1 min 9.9x	24.2 min 8.2x	29.4 min 6.8x	28.5 min 7.0x	30.7 min 6.5x	36.8 min 5.4x	12.3 min 16.2x	14.7 min 13.6x	22.8 min 8.8x	34.2 min 5.8x	31.0 min 6.4x	23.8 min 8.4x	36.4 min 5.5x
sst2/pos50 (node04, RTX 3090)	3.5 h 1.0x	6.6 min 31.5x	21.8 min 9.5x	30.6 min 6.8x	38.8 min 5.4x	1.0 h 3.3x	1.1 h 3.3x	1.4 h 2.5x	4.3 min 48.3x	11.8 min 17.6x	13.7 min 15.2x	18.8 min 11.1x	19.8 min 10.5x	21.7 min 9.6x	28.4 min 7.3x
sst2/pos70 (node04, RTX 3090)	4.6 h 1.0x	10.6 min 26.0x	25.2 min 10.9x	25.4 min 10.9x	35.7 min 7.7x	1.0 h 4.4x	57.5 min 4.8x	1.4 h 3.3x	10.3 min 26.8x	24.2 min 11.4x	12.6 min 21.8x	14.8 min 18.6x	19.0 min 14.5x	22.0 min 12.5x	25.5 min 10.8x
sst2/pos90 (node04, RTX 3090)	16.0 h 1.0x	1.7 h 9.3x	45.8 min 20.9x	49.5 min 19.3x	53.9 min 17.8x	1.1 h 14.6x	1.2 h 13.8x	1.3 h 12.6x	50.5 min 19.0x	1.6 h 10.2x	1.4 h 11.6x	24.6 min 39.0x	29.1 min 33.0x	34.2 min 28.0x	46.2 min 20.7x
mr/pos50 (node05, RTX A6000)	6.8 h 1.0x	9.4 min 43.2x	23.7 min 17.2x	35.5 min 11.5x	55.6 min 7.3x	1.0 h 6.8x	1.3 h 5.2x	1.6 h 4.4x	9.4 min 43.3x	15.1 min 27.1x	16.8 min 24.2x	20.5 min 19.8x	23.4 min 17.4x	26.4 min 15.5x	28.4 min 14.4x
mr/pos70 (node05, RTX A6000) (partial — some cells —)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
mr/pos90 (node05, RTX A6000)	11.7 h 1.0x	33.0 min 21.2x	38.0 min 18.4x	38.4 min 18.2x	42.4 min 16.5x	55.4 min 12.6x	1.2 h 10.1x	1.1 h 10.4x	20.2 min 34.5x	53.1 min 13.2x	1.1 h 10.5x	20.6 min 34.0x	21.4 min 32.7x	26.8 min 26.1x	34.0 min 20.6x
qqp/pos50 (node05, RTX A6000)	19.2 h 1.0x	22.2 min 51.7x	32.2 min 35.6x	51.1 min 22.5x	1.2 h 16.0x	1.8 h 10.8x	1.9 h 10.3x	1.9 h 9.9x	15.3 min 75.3x	20.4 min 56.2x	29.3 min 39.3x	33.5 min 34.3x	41.2 min 27.9x	50.0 min 23.0x	57.7 min 19.9x
qqp/pos30 (node01, RTX 2080 Ti)	38.5 h 1.0x	18.5 min 124.6x	48.4 min 47.8x	1.7 h 22.2x	1.7 h 23.0x	2.1 h 18.6x	2.9 h 13.4x	2.8 h 13.6x	19.2 min 120.4x	46.4 min 49.8x	49.2 min 47.0x	44.6 min 51.8x	58.2 min 39.7x	1.3 h 29.6x	1.4 h 27.2x
qqp/pos10 (node05, RTX A6000) (partial — some cells —)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
rte/pos50 (node04, RTX 3090)	1.0 h 1.0x	8.4 min 7.3x	7.9 min 7.9x	10.3 min 6.0x	12.1 min 5.1x	11.6 min 5.4x	14.7 min 4.2x	13.7 min 4.5x	10.9 min 5.7x	6.9 min 9.0x	9.6 min 6.5x	8.6 min 7.2x	12.3 min 5.0x	10.4 min 6.0x	12.9 min 4.8x
rte/pos70 (node04, RTX 3090)	38.9 min 1.0x	6.5 min 6.0x	11.0 min 3.5x	14.9 min 2.6x	17.4 min 2.2x	15.0 min 2.6x	15.4 min 2.5x	15.8 min 2.5x	7.9 min 4.9x	13.7 min 2.8x	15.3 min 2.5x	12.7 min 3.1x	13.7 min 2.8x	13.7 min 2.8x	16.1 min 2.4x
rte/pos90 (node04, RTX 3090)	39.2 min 1.0x	2.0 min 19.4x	3.5 min 11.1x	7.1 min 5.5x	7.0 min 5.6x	9.2 min 4.3x	8.3 min 4.7x	8.9 min 4.4x	11.6 min 3.4x	11.8 min 3.3x	12.9 min 3.0x	15.1 min 2.6x	16.0 min 2.4x	15.8 min 2.5x	15.8 min 2.5x
ag_news/cls25_25_25_25 (node03, RTX 3090)	6.9 h 1.0x	4.2 min 97.9x	13.9 min 29.7x	21.4 min 19.3x	32.0 min 12.9x	38.9 min 10.6x	41.7 min 9.9x	47.6 min 8.7x	4.2 min 97.9x	10.4 min 39.8x	14.5 min 28.4x	15.2 min 27.1x	18.9 min 21.8x	21.7 min 19.0x	22.5 min 18.4x
ag_news/cls55_15_15_15 (node03, RTX 3090)	11.0 h 1.0x	5.3 min 124.9x	15.3 min 43.1x	21.7 min 30.4x	32.6 min 20.3x	42.9 min 15.4x	50.0 min 13.2x	59.6 min 11.1x	6.3 min 105.0x	12.2 min 54.1x	14.0 min 47.2x	20.4 min 32.3x	25.0 min 26.4x	27.2 min 24.3x	30.8 min 21.4x
ag_news/cls85_05_05_05 (node03, RTX 3090)	38.6 h 1.0x	7.1 min 327.1x	12.8 min 180.3x	23.2 min 99.9x	30.8 min 75.1x	49.0 min 47.3x	59.6 min 38.9x	1.5 h 26.4x	18.3 min 126.6x	12.5 min 185.1x	20.8 min 111.6x	30.1 min 77.0x	36.6 min 63.3x	49.8 min 46.6x	53.0 min 43.7x
mnli/cls33_33_33 (multi, —) (cross-node, skipped)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
mnli/cls60_20_20 (narnia, H100 NVL)	44.6 h 1.0x	13.2 min 203.3x	27.0 min 99.0x	42.2 min 63.4x	43.5 min 61.5x	1.6 h 28.7x	1.7 h 25.5x	2.0 h 22.2x	13.6 min 197.3x	20.1 min 132.8x	38.0 min 70.5x	48.9 min 54.7x	1.1 h 41.1x	1.6 h 28.2x	1.4 h 31.7x
mnli/cls90_05_05 (multi, —) (cross-node, skipped)	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

```
Cell format: 'wall-clock time / speedup' (speedup = INV time / method time at same setting).
INV column is the FreeShap baseline (full kernel inverse); its speedup is 1.0x by definition.
Time is the elapsed portion of the final tqdm '500/500 [HH:MM:SS<00:00, ...]' line
captured for each banner-tagged sub-run in the experiment logs.
Settings tagged (cross-node, skipped) span multiple machines (mnli/cls33_33_33 split across
node01+node03; mnli/cls90_05_05 split across 3 nodes); per-row speedups would mix hardware
and are therefore omitted.
Hardware labels reflect the node where each ratio was actually run; speedups are NOT
comparable across rows because different GPUs were used.
```